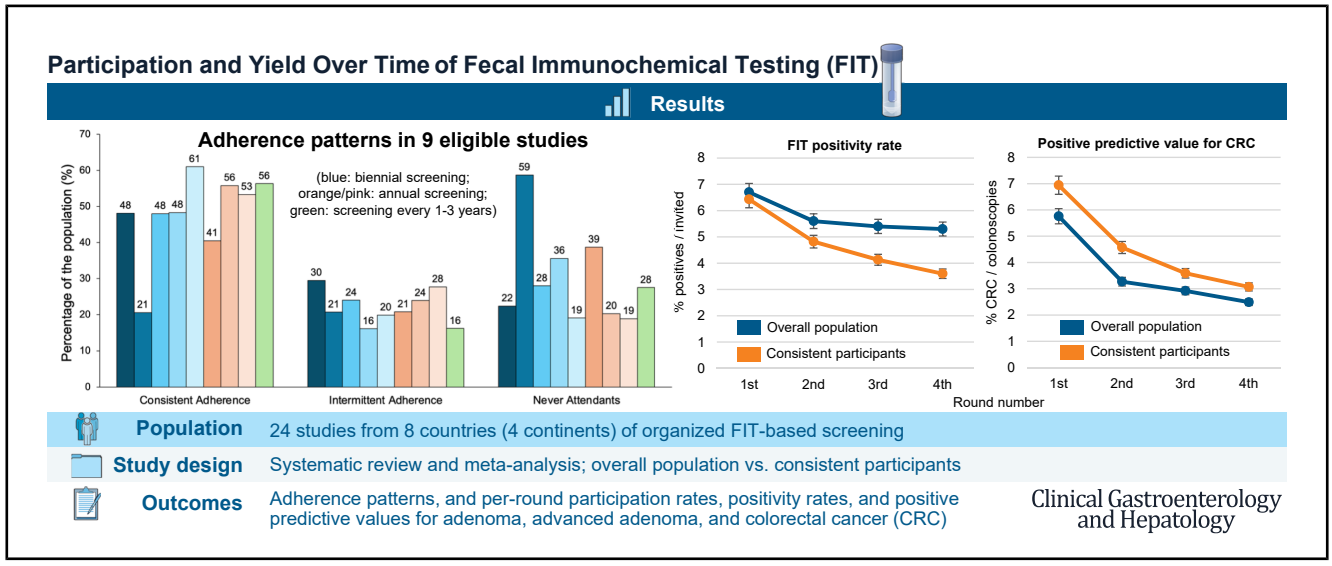
Systematic Review of Participation, Positivity, and Yield Over Time in Fecal Immunochemical Test-Based Organized Colorectal Cancer Screening Programs

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BACKGROUND & AIMS: **Fecal immunochemical testing (FIT), the most common colorectal cancer (CRC) screening modality worldwide, requires programmatic participation over decades. We performed a systematic review and meta-analysis to ascertain overall adherence patterns, per-round participation, FIT positivity, and neoplasia detection rates in organized FIT-based screening programs, contrasting initial vs subsequent screening rounds.**

METHODS: **We systematically searched the PubMed, Embase, and Cochrane databases. Adherence patterns were defined as consistent screeners (all rounds), intermittent screeners, and never attendants. Per-round participation, FIT positivity, and positive predictive values (PPVs) for CRC, advanced adenoma (AA), and adenoma were determined overall and for consistent screeners. Subgroup analyses were performed by screening periodicity and FIT positivity cutoff.**

*Authors share co-first authorship.

Abbreviations used in this paper: AA, advanced adenoma; AXIS, Appraisal of Cross-Sectional Studies; CI, confidence interval; CRC, colorectal cancer; DR, detection rate; FIT, fecal immunochemical testing; gFOBT, guaiac fecal occult blood testing; Hb, hemoglobin; PICO, patient, intervention, comparison, and outcome; PPV, positive predictive value; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; SD, standard deviation.

Most current article

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RESULTS: **Of 4414 candidate studies, 24 were included. Adherence patterns were: 47.1% (standard deviation [SD], 11.6%) consistent screeners, 22.7% (SD, 4.7%) intermittent screeners, and 30.3% (SD, 12.2%) never attendants. Overall pooled results from the first through fourth rounds showed per-round participation of 51.3% (95% confidence interval [CI], 37.4%–65.3%); 56.7% (95% CI, 41.3%–72.2%), 56.9% (95% CI, 38.7%–75.3%), and 56.5% (95% CI, 44%–68.6%); FIT positivity of 6.6% (95% CI, 5.9%–7.3%), 5.6% (95% CI, 4.6%–6.6%), 5.3% (95% CI, 4.0%–6.6%), and 5.3% (3.7%–6.9%); CRC PPV of 5.8% (95% CI, 4.1%–7.4%), 3.3% (95% CI, 2.3%–4.3%), 2.9% (95% CI, 2.2%–3.7%), and 2.5% (95% CI, 1.4%–3.5%); AA PPV of 32.8% (95% CI, 23.5%–42.0%), 28.3% (95% CI, 21.8%–34.7%), 22.7% (95% CI, 16.5%–28.9%), and 23% (95% CI, 10.1%–35.9%); and adenoma PPV of 48.6% (95% CI, 45.8%–51.4%), 46.2% (95% CI, 42.4%–50.1%), 48.7% (95% CI, 43.9%–53.6%), and 44.6% (95% CI, 38.2%–51.0%), respectively. Similar trends were observed for consistent screeners and for biennial FIT with 20 μg Hb/g feces positivity cutoff.**

CONCLUSIONS: **Organized FIT-based CRC screening programs achieve consistent participation in nearly one-half of persons and no participation in approximately one-third. FIT-positivity and yields for CRC, AA, and adenoma are highest in the first round and may stabilize at clinically relevant steady states at the third to fourth rounds.**

Keywords: Adherence; Colorectal Cancer; Fecal Immunochemical Test; Participation; Screening; Uptake.

Colorectal cancer (CRC) is the third most common cancer in the world.[1] Screening for CRC precursors including advanced adenomas (AAs) and for early-stage CRCs has proven effective in decreasing CRC incidence and CRC-related deaths.[2] CRC screening around the world commonly spans ages 50 to 74 years,[3] with a few countries including the United States recently lowering the screening initiation age to 45 years. Various CRC screening modalities exist, including stool-based tests and colonoscopy, each with its own set of barriers, invasiveness, and diagnostic accuracy.[4]

The fecal immunochemical test (FIT) is the preferred nonendoscopic test in programmatic screening worldwide. As with any test, participation rates are a key determinant of screening effectiveness, but it is especially relevant for FIT, in which the impact of screening is achieved over multiple rounds. Randomized controlled trials of guaiac fecal occult blood testing (gFOBT) demonstrated the effectiveness of gFOBT vs no screening.[5–8] In practice, FIT has overtaken gFOBT as the preferred stool-based CRC screening test, but we are unlikely to generate evidence for FIT vs no screening from randomized controlled trials with long-term follow-up.[9,10] As multiple FIT-based organized screening programs around the world mature, the possibility to analyze observational data through multiple screening rounds is now arising.

Most studies assessing FIT effectiveness have focused on short-term metrics, including one-time interventions.[11] Multiple ways to describe longitudinal adherence in FOBT programs have been developed, as summarized in a recent literature review by Doria-Rose et al.[12] These include adherence patterns such as consistent screeners vs inconsistent screeners vs never attendants, number of cycles attended, and detailed reporting of the possible permutations in the heterogeneous group of inconsistent screeners. There is no consensus on how to capture longitudinal adherence, and there has been no comprehensive assessment of participation, FIT positivity, and yield over time across multiple screening programs around the world.

Our aims were to characterize adherence patterns, and to evaluate FIT participation rates, positivity rates, and neoplasia detection yield over time in organized (non-opportunistic) multi-round FIT-based screening programs across the world. We performed a systematic review and meta-analysis of published studies. Our goal was to ascertain if there are generalizable conclusions regarding adherence patterns across different organized screening programs, and to explore how FIT positivity and neoplasia detection yield vary from initial through subsequent screening rounds, accounting for FIT interval and positivity threshold.

Methods

This systematic review and meta-analysis were performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Supplementary Table 1).[13] The protocol was registered prospectively at PROSPERO (CRD42023448999).

Search Strategy

We developed patient, intervention, comparison, and outcome (PICO) questions to capture our research questions (Supplementary Table 2). In consultation with a certified medical librarian (CS-A), a comprehensive search of the available electronic literature was performed to find studies describing overall adherence patterns, and per-round participation rates, FIT

positivity rates, and CRC, AA, or adenoma detection yield in FIT-based organized screening programs through multiple screening rounds.

We searched PubMed, EMBASE, and Cochrane databases from January 1, 2012, through July 31, 2023. The search strategy is shown in Supplementary Appendix 1. Language was restricted to English or Spanish. No publication status restrictions were imposed. References cited in related articles and meta-analyses were searched for additional eligible studies, referred to as cross-references. Before drafting of the final manuscript, an updated search through December 12, 2024, was performed.

Study Selection and Data Extraction

Two reviewers (SB-M and MP) independently screened all titles and abstracts, reviewed full-text manuscripts, and, after selection of articles fulfilling the eligibility criteria, carried out data extraction. A standardized template was developed to extract data from eligible studies. Extracted data comprised first author, year of publication, study design, study location, screening years, participants' age range, screening periodicity (annual, biennial, or triennial), FIT cutoff (in μg of Hb per g of feces), number of FIT samples, number of screening rounds, number of participants, positive FITs, follow-up colonoscopies, and the number of subjects identified as having CRC, AA, or adenoma. Data on CRC stage were extracted when available. Such data were scarce, and we therefore elected to report in the Supplementary Material. Disagreement among reviewers was resolved through discussion among all authors.

Study Type

We included cohort studies and clinical trials describing participation rates and patterns, and FIT results over time in FIT-based organized screening programs for average-risk populations.[14–37] Studies were excluded if they reported on opportunistic screening or screening targeting special groups, screening performed with methods other than FIT (eg, gFOBT, sigmoidoscopy, direct colonoscopy), screening programs offering participants the choice among different screening tests, screening programs in which FIT was used for triage after a positive primary screening test (eg, FIT following gFOBT), subjects with any high-risk condition for CRC (inflammatory bowel disease, hereditary CRC syndromes, personal history of CRC), or if data from only one round of screening were reported. Studies or trials with multiple arms comparing different screening methods were also excluded. When identical populations in overlapped study periods were included in several studies reporting the same outcomes, only the most comprehensive study was selected.

What You Need to Know

Background

Fecal immunochemical testing (FIT) is the preferred screening modality in organized colorectal cancer screening programs globally. Although FIT effectiveness relies on repeated participation over time, most studies have focused on single-round performance. There has been no comprehensive synthesis of evidence on longitudinal adherence patterns and yield across multiple FIT screening rounds.

Findings

This systematic review and meta-analysis, which identified 24 relevant studies, shows that only about one-half of individuals consistently participate in organized FIT screening, whereas one-third never participate. FIT positivity and positive predictive values for colorectal cancer and advanced neoplasia are highest at the first round and appear to stabilize at lower, but still clinically significant, levels by the third or fourth round. These trends are more marked in consistent screeners and closed cohorts.

Implications for patient care

At the level of screening programs, standardized reporting of individual longitudinal adherence trajectories and outcomes could optimize program evaluation and comparisons. At the level of individuals, screening trajectories and results could support personalized screening strategies, including extended intervals or early cessation in low-risk consistent screeners, which could improve programmatic cost-effectiveness and resource allocation.

Definitions and Outcomes

Subjects were classified into 3 overall adherence patterns across all rounds offered: consistent screeners who attended all rounds for which they were invited, intermittent screeners who attended some but not all rounds for which they were invited, and never attendants who did not attend any round for which they were invited. Intermittent screeners were further allocated to 2 subcategories when relevant data were available: drop out (participation in initial rounds followed by subsequent non-participation), and delayed entry (initial non-participation followed by participation in later rounds).[14]

The population of each study was defined as an open cohort when, in every screening round, new subjects with no prior screening round offers could be offered screening, and as a closed cohort when only patients invited at an initial round were followed through subsequent rounds.

For each screening round, participation rates, FIT positivity rates, follow-up colonoscopy completion rates, and positive predictive values (PPV) and detection rates

(DRs) for CRC, AA, and adenoma were calculated. Participation rate was calculated as the number of participants (individuals who returned a completed FIT kit) relative to all invitees, FIT positivity rate as the number of positive FIT relative to all participants, follow-up colonoscopy completion rate as the number of follow-up colonoscopies relative to all positive FIT, PPV as the number of persons with lesions detected relative to all persons undergoing colonoscopies following a positive FIT, all expressed as a percentage, and DR as the number of persons with lesions detected relative to all participants, all expressed as a percentage.

All study outcomes were ascertained for each program's screening round among the overall screening population in each study. The study outcomes were also ascertained for consistent screeners across all screening rounds offered when such data were reported.

Among the 11 included studies describing the yield of FIT for AA across rounds, 8 studies defined AA as those adenomas $\geq$10 mm, with villous component or high-grade dysplasia,[15,20,22,24,28,29,35,36] 2 studies defined AA as either an adenoma $\geq$10 mm, with villous component or high-grade dysplasia or 3 or more adenomas,[23,25] and finally, 1 study defined AA as those with villous histology without referring to the size or the presence of dysplasia.[21]

Risk of Bias Assessment

We assessed the quality of the included articles using the Appraisal of Cross-Sectional Studies (AXIS) tool, which consists of 20 items rated as "yes," "no," or "don't know" to evaluate the risk of bias (Supplementary Table 3). Because our review focused solely on programs utilizing FIT and colonoscopy after positive FIT (screening tools known for their high diagnostic accuracy), we excluded items 8 and 9, which address the validity and reliability of measurements.

Subgroup Analyses

In an attempt to minimize heterogeneity among studies, subgroup analyses were performed for different screening periodicities (ie, annual or biennial), for different FIT positivity cutoff values (ie, 20 or 10 μg Hb/g of feces), and for open and closed cohorts.

Statistical Analysis

For adherence patterns, means and standard deviations (SDs) were calculated across studies. For per-round participation rates, FIT positivity rates, PPVs, and DRs, studies were combined in separate meta-analyses to obtain pooled mean estimates and 95% confidence intervals (CIs). Due to the scarcity of studies reporting on 5 or more rounds, per-round pooled estimates were provided for the first through fourth rounds only. Random-effects meta-analyses using the generic inverse variance weighting method were performed. To mitigate potential bias arising from low-event data, studies reporting zero cases of the outcome of interest were excluded from the meta-analysis. Statistical heterogeneity among studies was assessed using the I^2 statistic, with an I^2 <25%, 25% to 50%, and >50% indicating low, medium, and high heterogeneity, respectively. The possibility of publication bias was assessed by inspection of funnel plots. The meta-analyses were performed using Review Manager 5.3 (The Nordic-Cochrane Center, The Cochrane Collaboration).

Results

Literature Search, Included Studies, Risk of Bias, and Publication Bias Assessment

The initial literature search yielded 7772 citations, of which 4414 remained after removal of duplicates. After applying the selection criteria, addition of cross-references, and a first screening of titles and abstracts, 119 studies were selected and reviewed in detail (Supplementary Appendix 2), leading to selection of 24 studies for inclusion in the final analysis (Figure 1).[14–23,25–27,28–37] These studies consist of 8 studies from the Netherlands, 5 from Italy, 3 from the United States, 3 from Australia, 2 from Spain, 1 from Sweden, 1 from Slovenia, and 1 from South Korea. Most of these studies provided real-world data, although some of them had an experimental setting. Also, most of the studies reported the initial rounds of the screening program, but there are some exceptions reporting on already stablished programs[17] or programs transitioning from gFOBT.[27,31] The age of participants ranged from 50 to 74 years. Most of the studies reported on 2 to 4 annual or biennial rounds, with 4 reporting on 5 to 7 rounds. Most studies (63%) reported on biennial programs. The most common FIT positivity cutoff values were 20 (38%), and 10 (38%) μg Hb/g feces. Table 1 provides an overview of the individual studies with their demographic data.

Supplementary Figure 1 shows the results of the risk of bias assessment. Supplementary Figure 2 shows funnel plots, which are quite symmetrical around the x-axis, suggesting the absence of publication bias.

Overall Adherence Patterns Across All Rounds Offered

Nine studies reported data on overall adherence patterns, allowing classification of participants as consistent screeners, intermittent screeners, and never attendants (Figure 2).[17–19,27,30,31,34,24,28] Across these 9 studies, the fraction of the population behaving as consistent screeners was on average 47.9% (SD, 11.6%; range, 20.6%–61.0%). The fraction of the population behaving as intermittent screeners was on average 22.1% (SD, 4.5%; range,

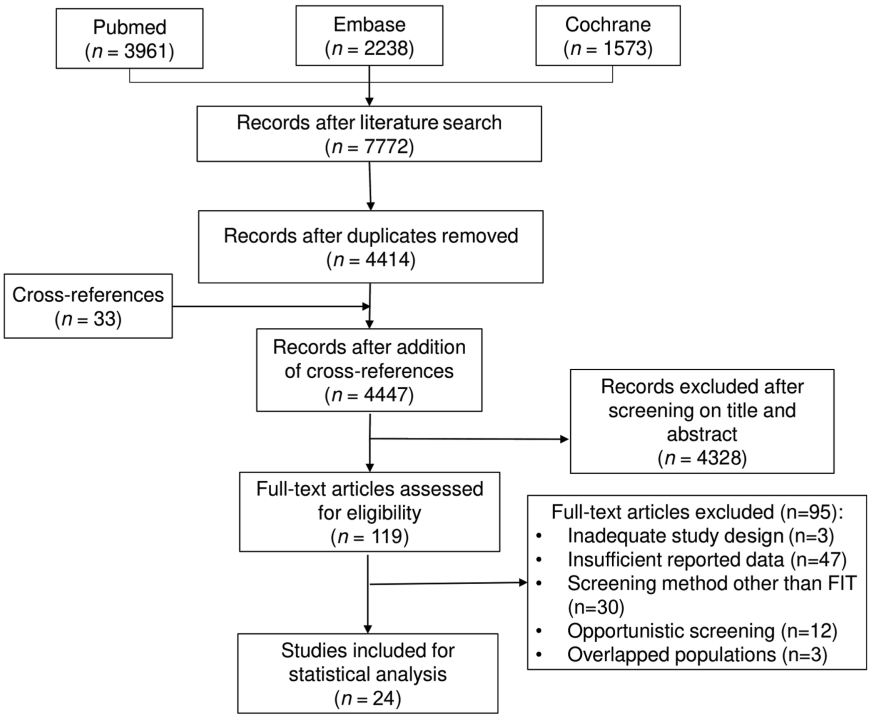


Figure 1. Flow diagram of study selection.

16.1%–29.5%). The fraction of the population behaving as never attendants was on average 29.9% (SD, 12.6%; range, 18.9%–58.7%). In studies that reported intermittent participation with some granularity,[14,27,30–32,34] the fractions of the population displaying late entry (on average, 9.9%; SD, 2.8%) or drop-out (on average, 9.5%; SD, 2.6%) appeared comparable.

Participation Rates per Round Over Time

Thirteen studies reported per-round participation rates across multiple screening rounds.[15,18,20–23,26,31,33,28,29,37] The pooled participation rates from the first through fourth rounds irrespective of FIT interval were 51.3%, 56.7%, 56.9%, and 56.5%, respectively (Figure 3, Supplementary Figures 3–5). Similar results for per-round participation rates were observed in annual (5 studies) vs biennial (8 studies) screening programs (Table 2, Figure 3, Supplementary Figures 3–5). In closed cohorts, per-round participation rates (61.4%–72.0%) were higher than in open cohorts (35.9%–43.9%) (Supplementary Figures 6 and 7).

Colonoscopy Follow-Up

Follow-up colonoscopy completion rate remained stable and over 87% through all rounds (Supplementary Figure 8).

FIT Positivity Rates per Round Over Time

Fourteen studies reported on FIT positivity rates across multiple screening rounds.[15,16,18,20–23,25,26,35,36,24,28,29] For the overall population, FIT positivity rates were highest at the first round and then decreased: 6.6%, 5.6%, 5.3%, and 5.3%, from the first through fourth rounds, respectively (Table 2, Figure 4, Supplementary Figures 9 and 10). In a subgroup analysis including only 4 studies reporting on biennial programs with a FIT positivity cutoff of 20 μg Hb/g feces, a similar pattern was observed but with lower positivity rates: 5.9%, 5.4%, 4.7%, and 4.1% from the first through fourth rounds, respectively (Supplementary Figure 11). For consistent screeners, FIT positivity across 5 studies from the first through fourth rounds was 6.4%, 4.8%, 4.1%, and 3.6%, respectively (Table 2, Supplementary Figures 9 and 12). In closed cohorts, the difference between first vs subsequent round results for FIT positivity was more marked compared with open cohorts (Supplementary Figures 13 and 14).

PPVs for CRC, AA, and Adenoma per Round Over Time

Twelve studies reported on detection yields across multiple screening rounds.[15,16,20–23,25,35,36,24,28,29] For the overall population (Table 2, Figure 4, Supplementary Figure 9), PPVs from the first through fourth rounds were: CRC 5.8%, 3.3%, 2.9%, and 2.5%, respectively (Supplementary Figure 15); AA 32.8%, 28.3%, 22.7%, and 23.0%, respectively (Supplementary Figure 16); and adenoma 48.6%, 46.2%, 48.7%, and 44.6%, respectively (Supplementary Figure 17).

In a subgroup analysis including only biennial programs with a FIT positivity cutoff of 20 μg Hb/g feces

Table 1. Characteristics of the Included Studies

Author, year of publication	Design; setting	Country	Patient inclusion	Periodicity	FIT cutoff, *μg Hb/g*	No. rounds	Age range, *y*	Type of cohort[a]	Population[b]	Analyses to be included in
Crotta, 2012	Cohort; real-world; program initiation	Italy	2001–2008	Biennial	20	4	50–74	Closed	Overall	Adherence patterns, participation rate, positivity rate, yields
Parente, 2013	Cohort; real-world; program initiation	Italy	2005–2009	Biennial	20	2	50–69	Open	Overall	Participation rate, positivity rate, yields
Portillo, 2018	Cohort; real-world; program initiation	Spain	2009–2014	Biennial	20	3	50–69	Closed	Consistent	Participation rate, positivity rate, yields
Zorzi, 2018	Cohort; real-world: program initiation	Italy	2002–2014	Biennial	20	6	50–69	Closed	Consistent	Participation rate, positivity rate, yields
Benito, 2019	Cohort; real-world; transition from gFOBT	Spain	2000–2017	Biennial	20	4	50–69	Open	Overall	Adherence patterns, participation rate
Baldacchini, 2020	Cohort; real-world; program initiation	Italy	2005–2016	Biennial	20	5	50–69	Open	Overall and consistent	Participation rate, positivity rates, yields
Tepes, 2020	Registry; real-world; program initiation	Slovenia	2008–2013	Biennial	20	3	50–69	Open	Overall	Participation rate, positivity rate, yields
Baldacchini, 2022	Cohort; real-world; program initiation	Italy	2005–2016	Biennial	20	7	50–69	Open	Overall	Adherence patterns
Denters, 2012	Cohort; real-world; program initiation	Netherlands	2006–2009	Biennial	10	2	50–74	Closed	Consistent	Participation rate, positivity rate, yields
Denters, 2013	Cohort; real-world; program initiation	Netherlands	2006–2009	Biennial	10	2	50–74	Open	Overall	Adherence patterns, participation rate
Stegeman, 2015	Cohort; experimental; program initiation	Netherlands	2007–2012	Biennial	10	3	50–75	Open	Overall	Participation rate, positivity rate, yields
Van der Vlugt, 2017	Cohort; experimental; program initiation	Netherlands	2006–2014	Biennial	10	4	50–74	Open	Overall	Adherence patterns
Schreuders, 2019	Cohort; real-world; program initiation	Netherlands	2006–2014	Biennial	10	3	50–74	Closed	Overall	Participation rate, positivity rate, yields
Heyman, 2024	Cohort; real-world; transition from gFOBT	Sweden	2015–2020	Biennial	40	3	60–70	Closed	Overall	Adherence patterns

Table 1. Continued

Author, year of publication	Design; setting	Country	Patient inclusion	Periodicity	FIT cutoff, *μg Hb/g*	No. rounds	Age range, *y*	Type of cohort[a]	Population[b]	Analyses to be included in
Breekveldt, 2022	Cohort; real-world; program initiation	Netherlands	2014–2017	Biennial	47	2	55–75	Closed	Consistent	Positivity rate, yields
Van Roon, 2013	RCT; experimental; program initiation	Netherlands	2006–2012	1–3 y	10	2	50–74	Close	Overall	Participation rate, positivity rate, yields
Kapidzic, 2014	Cohort; real-world; program initiation	Netherlands	2006–2016	1–3 y	10	3	50–74	Closed	Overall	Participation rate, positivity rate, yields
Jensen, 2016	Cohort; real-world: program initiation[c]	United States	2007–2010	Annual	20	4	50–70	Closed	Overall	Participation rate, positivity rate, yields
Choi, 2012	Cohort; real-world; program initiation	South Korea	2004–2008	Annual	NA[d]	5	>50	Open	Overall	Participation rate, positivity rate
Duncan, 2014	Cohort; experimental: program initiation[c]	Australia	2008–2010	Annual	NA[d]	3	50–75	Closed	Overall	Adherence patterns
Cole, 2013	Cohort; real-world; program initiation[c]	Australia	2003–2009	Annual	10	2	55–74	Closed	Overall	Adherence patterns
Baker, 2015	RCT; experimental; program initiation[c]	United States	2011–2012	Annual	10	2	51–75	Closed	Overall and consistent	Participation rate
Gordon, 2015	Cohort; real-world; established program	United States	2010–2012	Annual	NA[d]	3	50–75	Closed	Overall	Adherence patterns
Osborne, 2017	Cohort; experimental; program initiation[c]	Australia	2008–2011	Annual	NA[d]	4	50–74	Closed	Overall	Adherence patterns, participation rate positivity rate

FIT, fecal immunochemical test; gFOBT, guaiac fecal occult blood test; Hb, hemoglobin; NA, not available; RCT, randomized control trial.
[a]Open cohort: in every screening round new population with no prior rounds can come in. Closed cohort: only patients invited from the first round are invited across subsequent rounds (subsequent screening rounds not replenished with screening-naive individuals.)
[b]Overall population: all the invited population. Consistent population: only population who has participated in the screening every time they have been eligible.
[c]High opportunistic background.
[d]NA: not stated in publication and unable to verify via personal communication.

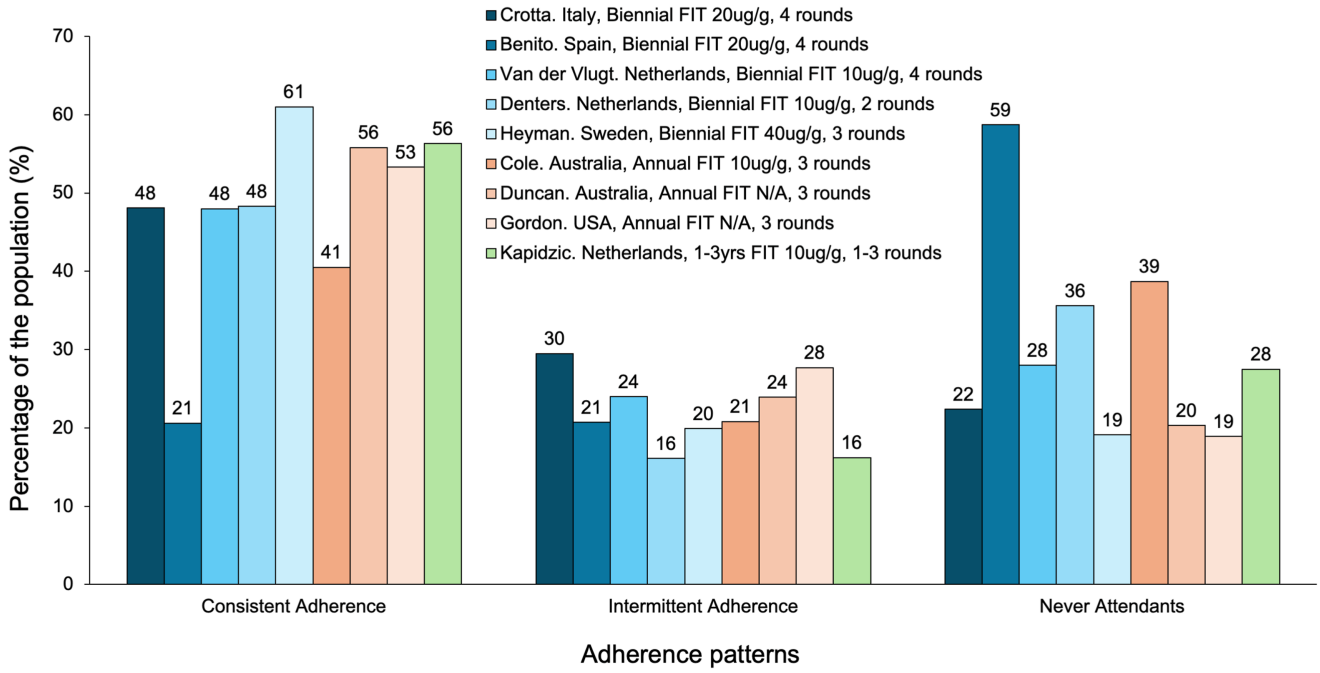


Figure 2. Adherence patterns allowing classification of participants as consistent screeners, intermittent screeners, and never attendants. Bars in shades of *blue* reflect biennial screening, bars in shades of *orange/pink* reflect annual screening, and the bar in *green* reflects screening every 1 to 3 years.

(Supplementary Figure 18), PPVs from the first through third rounds were: CRC 6.1%, 3.6%, and 3.4%, respectively (Supplementary Figure 19); AA 33.3%, 29.8%, and 27.3%, respectively (Supplementary Figure 20); and adenoma 46.1%, 45%, and 47.8%, respectively (Supplementary Figure 21).

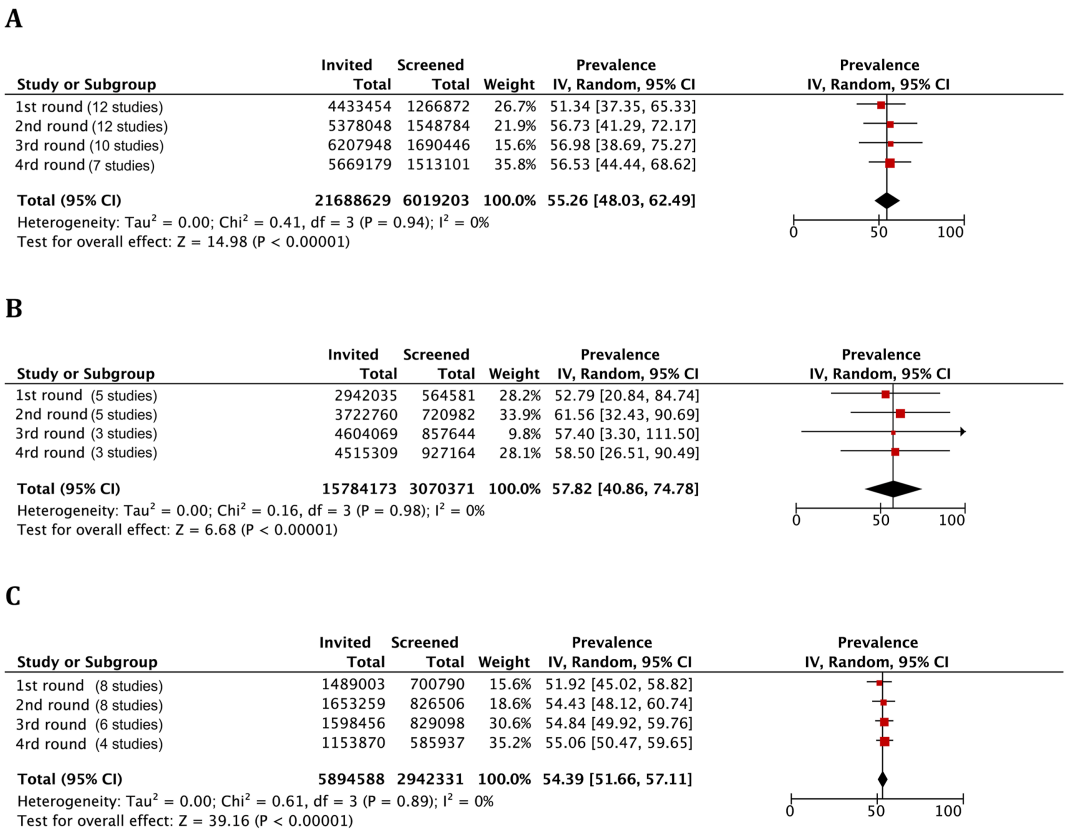


Figure 3. Participation rates across multiple screening rounds in FIT-based CRC organized screening programs for all programs (*A*), annual programs (*B*), and biennial programs (*C*). IV, inverse variance.

Table 2. Pooled Participation Rates, Positivity Rates, PPVs, and DRs for CRC, AA, and Adenoma, Over 4 Rounds of FIT Screening, Stratified by Program Interval and Adherence Pattern

	Round 1, % (95% CI)	Round 2, % (95% CI)	Round 3, % (95% CI)	Round 4, % (95% CI)
Participation				
Annual	52.8 (20.8–84.7)	61.6 (32.4–90.7)	57.4 (3.3–111.5)	58.5 (26.5–90.5)
Biennial	51.9 (45.0–58.8)	54.4 (48.1–60.8)	54.8 (49.9–59.8)	55.1 (50.5–59.7)
Positivity				
Overall	6.6 (5.9–7.3)	5.6 (4.6–6.7)	5.3 (4.0–6.6)	5.3 (3.7–6.9)
Consistent	6.4 (6.0–6.9)	4.8 (4.3–5.3)	4.1 (3.1–5.1)	3.6 (3.0–4.2)
PPV - CRC				
Overall	5.8 (4.1–7.4)	3.3 (2.3–4.3)	2.9 (2.2–3.7)	2.5 (1.5–3.6)
Consistent	6.9 (6.0–7.9)	4.6 (3.5–5.6)	3.6 (3.1–4.1)	3.1 (2.7–3.5)
PPV - AA				
Overall	32.8 (23.5–42.0)	28.3 (21.8–34.7)	22.7 (16.5–28.9)	23.0 (10.2–35.8)
Consistent	38.7 (32.9–44.6)	30.6 (24.2–37.0)	28.2 (20.7–35.8)	24.9 (21.2–28.6)
PPV - Adenoma				
Overall	48.6 (45.8–51.4)	46.2 (42.4–50.1)	48.7 (43.9–53.6)	44.6 (38.2–51.0)
Consistent	53.4 (47.5–59.4)	46.7 (42.1–51.3)	42.6 (37.6–47.6)	42.4 (39.0–45.9)
DR - CRC				
Overall	0.3 (0.2–0.4)	0.2 (0.1–0.2)	0.1 (0.1–0.2)	0.1 (0.1–0.1)
Consistent	0.4 (0.3–0.5)	0.2 (0.1–0.3)	0.1 (0.1–0.2)	0.1 (0.1–0.1)
DR - AA				
Overall	1.9 (1.2–2.7)	1.4 (0.9–2.0)	1.0 (0.5–1.5)	1.0 (0.5–1.5)
Consistent	2.3 (1.7–2.9)	1.4 (1.0–1.9)	1.1 (0.7–1.6)	0.8 (0.8–0.9)
DR - Adenoma				
Overall	2.5 (1.7–3.3)	2.1 (1.3–2.9)	2.0 (1.1–2.8)	1.1 (0.8–1.5)
Consistent	2.6 (2.3–2.9)	1.8 (1.5–2.1)	1.3 (1.3–1.4)	1.4 (1.4–1.5)

AA, advanced adenoma; CI, confidence interval; CRC, colorectal cancer; DR, detection rate; PPV, positive predictive value.

For consistent screeners (Table 2, Supplementary Figures 9 and 22), PPVs from the first through fourth rounds were: CRC 6.9%, 4.6%, 3.6%, and 3.1%, respectively (Supplementary Figure 23); AA 38.7%, 30.6%, 28.2%, and 24.9%, respectively (Supplementary Figure 24); and adenoma 53.4%, 46.7%, 42.6%, and 42.4%, respectively (Supplementary Figure 25).

In closed cohorts, the difference between first vs subsequent round results for CRC PPV appeared more marked compared with open cohorts (Supplementary Figures 26–31).

Data on DRs for CRC, AA, and adenoma for the overall population and the consistent screeners is described in Table 2 and Supplementary Figures 32–39.

CRC Stage per Round Over Time

Six studies provided information regarding CRC stage at diagnosis in organized screening programs. From those, we could extract information regarding CRC stage for the overall screening population from 5 studies,[20,22,23,25,29] and for the consistent screeners from 2 studies.[29,35] In the overall screening population, CRC stages I to II constituted 72.7%, 73.6%, and 71.3% of CRC's in the first through third rounds, respectively. In the consistent screeners, CRC stages I to II comprised 76.5%, 70.3% and 73.9% of CRCs in the first through third rounds, respectively. The available data on CRC stage are shown in Supplementary Table 4.

Discussion

Our synthesis of programmatic FIT-based screening data from around the world provides valuable insights on overall adherence patterns across multiple screening rounds, and on per-round participation rates, FIT positivity rates, and yields for neoplasia from the initial round through subsequent screening rounds. Overall, approximately 48% (range, 21%–61%) of screen-eligible individuals in organized programs consistently participate in screening, approximately 30% (range, 19%–59%) do not participate at all, and the remainder participate intermittently. Per-round participation rates remain relatively stable from the first through fourth screening rounds and are comparable between annual and biennial programs. Per-round FIT positivity rates and the PPV for CRC decline from the first through fourth rounds, with a more pronounced drop between the first and second rounds and an apparent later stabilization. In contrast, the per-round PPVs for AA and

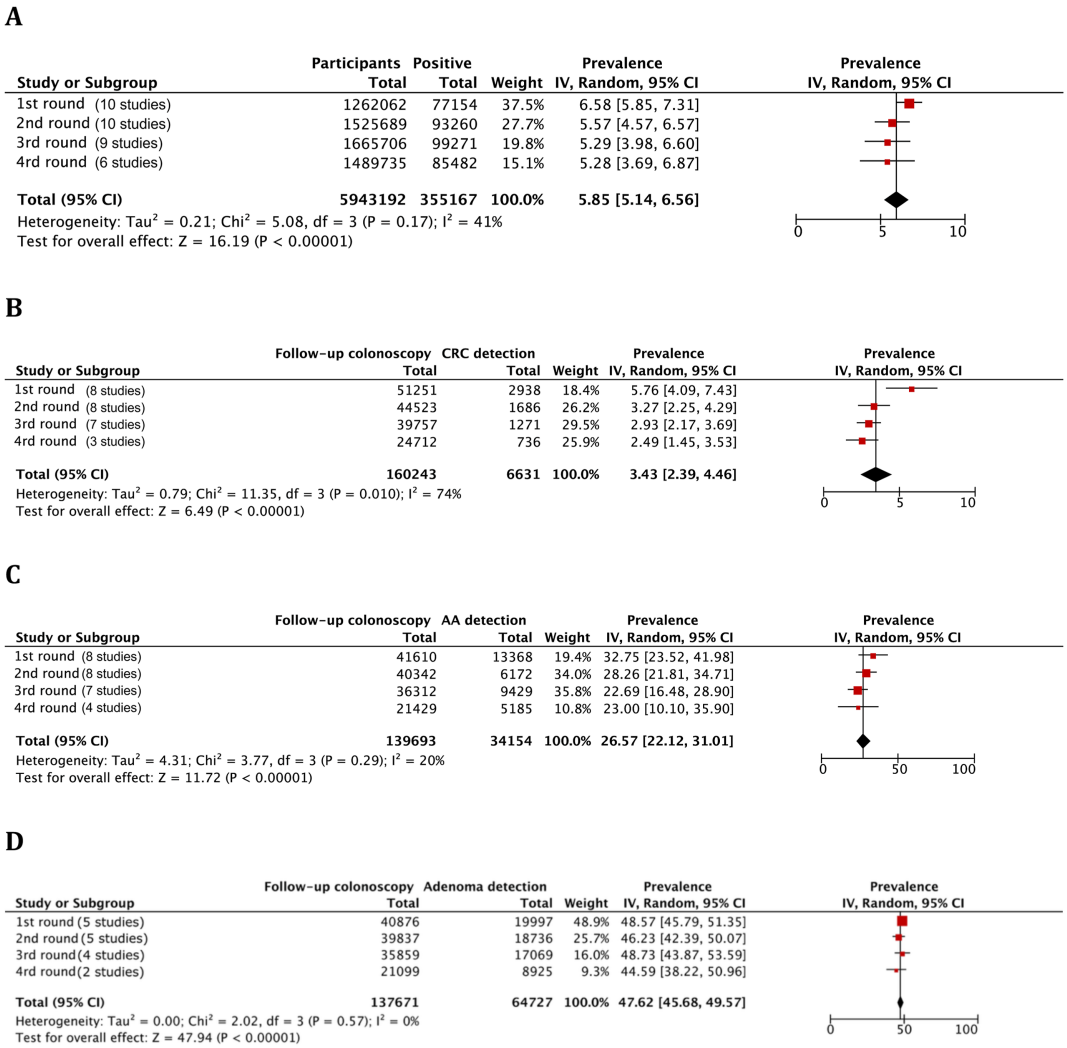


Figure 4. Positivity rate (*A*) and PPV for CRC (*B*), AA (*C*), and adenoma (*D*) of FIT per round in the overall screening population participating in FIT-based organized screening programs. IV, inverse variance.

adenomas decrease more gradually from the first through subsequent rounds. Among consistent screeners, the declines in FIT positivity rates seem to occur more rapidly than in the overall screening-eligible population. Although more subtle, this trend is also suggested for the PPVs.

Multiple clinical trials demonstrate that screening decreases CRC incidence and mortality.[5,6,10,38,39] Currently, the most widely adopted population-based screening strategies are colonoscopy every 10 years or FIT every 1 or 2 years, followed by colonoscopy if the result is positive. In a recent randomized controlled trial, invitation to biennial FIT screening was not inferior to invitation to a 1-time colonoscopy in reducing 10-year CRC incidence and CRC-related mortality.[10] Biennial FIT screening was better accepted by the population than colonoscopy, which enhanced the overall effectiveness of the FIT strategy.[40] Our results highlight the challenges and opportunities related to participation in organized FIT-based CRC screening programs: it is not clear how one might achieve participation at least once among those who remain unscreened after multiple screening round offers, or how one might sustain screening adherence over time among those who adhere only intermittently. A modeling study by Heisser et al[41] suggests that sporadic screening adherence among a large portion of the population prevents substantially more CRC cases and deaths than perfect, selective adherence within a small, highly compliant group, even at identical overall adherence levels. Public health interventions should prioritize maximizing the reach of screening to engage 'never-attendants,' rather than focusing resources on ensuring perfect repeat adherence among those already participating.

A major limitation of many reports of CRC screening programs' outcomes is that key metrics are reported according to the calendar year (ie, program's round).[21,24,28,29] This reporting approach fails to account for the individual screening round each participant is undergoing, or their screening behavior in previous rounds. Understanding longitudinal adherence within different subpopulations is essential. Evaluations of

closed cohorts and cohorts of consistent screenees provide the best assessment of changes in per-round results from initial to subsequent rounds, reflecting the balance between ongoing removal of neoplasia from the population and age-dependent development of new neoplasia. Such evaluations can define the maximum benefit that specific screening methods can offer and can identify when per-round results stabilize even as screenees age.

Several methods have been proposed to describe longitudinal adherence in CRC screening programs.[12] These include stratified single-round attendance (number attending among eligible per round); all possible adherence permutations; attendance stratified into consistent, inconsistent, and never; number of times attended; number of times attended with respect to offered rounds; overall program adherence; and proportion of time covered.[42,43] Each of these metrics has limitations, and no consensus has been reached on a standard method for reporting adherence across rounds. In our review, most studies only reported per-round stratified attendance; only 9 studies described participation in enough detail to allow for the extraction of overall adherence patterns at the level of individuals. In aggregate, total population per-round participation rates in the included studies were approximately in the 50% to 60% range and seemed to remain stable over multiple rounds, in contrast with reports of decline in uptake over subsequent rounds.[43] These participation rates exceed those of the landmark COLONPREV trial,[10] likely reflecting the difference between "real-world" established programs and an experimental randomized controlled trial, in which the investigational nature may have influenced uptake. European guidelines define a minimum screening uptake rate of 45% and a desirable rate of at least 65%, but do not currently provide recommendations for longitudinal adherence across multiple rounds.[44] Current outreach efforts appear sufficient for the subgroup of the population that is inclined to screen consistently. It remains to be determined if and how the approximately 30% who consistently do not screen at all could be encouraged to participate. Among intermittent screeners, we observed similar rates for dropouts and late adopters. These patterns probably reflect different underlying individual and system-level factors. The outlier results in the study by Benito et al[31] are likely attributable to a 'carry-over' effect from the program's initial rounds using gFOBT, a test known to have lower acceptability than FIT.

Our results highlight how FIT positivity rate and PPVs (specially for CRC) show a gradual decrease from the first through the fourth round, with the steepest drops between the first round ("prevalence round") and the second round. Robust data through further rounds might confirm a tendency towards stabilization at a steady state in per-round results. The few available studies reporting on fifth and sixth rounds show how adenoma and AA DRs seem to reach a steady state between the third and fifth rounds, whereas CRC detection maintains slight decreases through the fifth and sixth rounds.[36,29] Our results suggest that, in consistent screenees (vs the overall population) or closed (vs open) cohorts, FIT-positivity and neoplasia yield may reach a steady state within fewer rounds. These groups have minimal or no late-entry/intermittent screening at a wide range of age-related rates of prevalent neoplasia.

We are not aware of any previous systematic review and metanalysis that explores comprehensively the per-round participation and performance of population-based CRC screening programs using FIT. Previous systematic reviews focused on overall (no round-specific) performance of organized FIT-based programs,[45] repeat FIT or gFOBT prevalence in diverse settings,[46] gender differences in FIT uptake,[47] or FIT performance outside an organized setting.[48,49]

Our results reinforce the evidence that programmatic FIT screening identifies and removes a substantial cumulative number of CRCs and advanced precursor lesions from round to round. This could support the development of risk-based, personalized screening strategies based on an individual's screening history and results. Such individualized approaches could alleviate pressure on screening programs and endoscopy services. It remains to be determined, for instance, whether individuals who participate consistently in screening and have multiple consecutive negative results could be candidates for extended screening intervals, an increase in the FIT positivity cutoff value, or even early cessation of screening.

Our study has limitations. First, we observed substantial heterogeneity across all analyses, likely due to differences among studies in FIT cutoff thresholds, screening intervals, countries, recruitment periods, age ranges, underlying cancer prevalences, study design (experimental vs real world), and setting (initial, established, or gFOBT-transitioning programs; variable opportunistic background screening). Second, the number of individuals undergoing FIT decreased between the first and fourth rounds, with few studies reporting 5 or more rounds.[26,32,36,29] Third, a sex-based analysis was not feasible. Fourth, in some studies, FIT-based screening programs were preceded by gFOBT-based screening, potentially influencing early round positivity.[31] Fifth, it was beyond this study's scope to synthesize data on opportunistic screening, given that the absence of a dynamic list of screen-eligible persons in the opportunistic setting makes it challenging to determine screening results as a function of individual screening history. Adherence with fecal-based screening is superior in organized vs opportunistic screening.[50–52] Sixth, 1 study did not report colonoscopy completion rate, requiring the use of FIT positivity as the denominator to calculate PPVs.[16] Finally, a potential bias may arise from the reinvitation of individuals with a previous positive FIT and a negative colonoscopy, who may dilute positivity and detection rates in subsequent rounds.

Conclusion

In conclusion, our systematic review and meta-analysis provide a comprehensive overview of longitudinal adherence patterns, FIT positivity rates, and detection yields across successive rounds of organized FIT-based CRC screening programs worldwide. Per-round participation rates remain stable across rounds, with nearly one-half of the eligible population screening consistently, but a substantial proportion persistently not participating at all. Positivity rates and PPVs decline across rounds, more markedly among consistent participants. Our results underscore the need for standardized reporting of longitudinal participation, emphasizing individual-level per-round participation and results. This approach could enhance program evaluation, facilitate cross-country comparisons, promote research on personalized screening based on screening history, and ultimately optimize the effectiveness and cost-effectiveness of CRC screening and surveillance programs.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Clinical Gastroenterology and Hepatology* at www.cghjournal.org, and at https://doi.org/10.1016/j.cgh.2026.02.009.

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Conflicts of interest
These authors disclose the following: Rodrigo Jover has received research grants from MSD; and has participated as an advisor to MSD, Norgine, Alfa-Sigma, and GISupply. Uri Ladabaum has participated in the advisory board for UniversalDx, Lean Medical, Vivante, and Kohler Ventures; and as consultant for Medtronic, Clinical Genomics, Guardant Health, Freenome, and ChekCap.

Antoni Castells has participated in the advisory board for UniversalDx, Goodgut, and ProAlt; and received research grants from Eiken. The remaining authors disclose no conflicts.

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Data Availability
Data, analytic methods, and study materials will be made available to other researchers from the corresponding author upon reasonable request.

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Supplemental information

Systematic Review of Participation, Positivity, and Yield Over Time in Fecal Immunochemical Test-Based Organized Colorectal Cancer Screening Programs

Sandra Baile-Maxía, Michael Pimienta, Carmen Sánchez-Ardila, Antoni Castells, Rodrigo Jover, and Uri Ladabaum

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Appendix 1. Search strategies in Pubmed, EMBASE and Cochrane databases.

PUBMED (NLM)

Search: ((((("Early Detection of Cancer"[Mesh]) OR "Prospective Studies"[Mesh]) OR (((((("Clinical Study" [Publication Type] OR "Observational Study" [Publication Type] OR "Clinical Trial" [Publication Type]) OR (("Follow-Up Studies"[Mesh]) OR "Multivariate Analysis"[Mesh])) OR (("multivariate analysis"[Title]) OR ("multivariate analysis"[Other Term]))) OR (("observational study"[Title]) OR ("observational study"[Other Term]))) OR (("clinical trial"[Title]) OR (clinical trial[Other Term])))) AND (((((("Mass Screening"[Mesh]) OR ("Patient Compliance"[Mesh] OR "Treatment Adherence and Compliance"[Mesh])) OR ("mass screening"[Title] OR "screening"[Title] OR "round?"[Title] OR "MULTIPLE ROUND?"[Title] OR "PARTICIPATION"[Title] OR "ADHERENCE" [Title] OR "ATTENDANCE"[Title] OR "YIELD"[Title])) OR ("mass screening" [Other Term] OR "screening" [Other Term] OR "round?"[Other Term] OR "PARTICIPATION"[Other Term] OR "ADHERENCE" [Other Term] OR "ATTENDANCE"[Other Term] OR "YIELD"[Other Term])) OR (((((("patient compliance" [Title])) OR (patient compliance [Other Term])) OR (compliance [Other Term])) OR (compliance [Title]))))) AND (((((((((("Colonoscopy"[Mesh]) OR ("Occult Blood"[Mesh])) OR ("Endoscopy"[Mesh])) OR ("lmmunochemistry"[Mesh])) OR (("FAECAL IMMUNOCHEMICAL TEST"[Title]) OR ("FECAL IMMUNOCHEMICAL TEST"[Title]))) OR ((((((("faecal haemoglobin" [Title]) OR ("fecal hemoglobin"[Title])) OR ("faecal occult blood" [Title])) OR ("fecal occult blood"[Title])) OR ("haemoccult"[Title])) OR ("hemoccult"[Title])) OR (guaiac[Title])) OR ("immunochemical testing"[Title]))) OR ((((((("faecal haemoglobin"[Other Term]) OR ("fecal hemoglobin"[Other Term])) OR ("faecal occult blood"[Other Term])) OR ("fecal occult blood"[Other Term])) OR ("hemoccult" [Other Term])) OR (guaiac[Other Term])) OR ("immunochemical testing"[Other Term]))) OR ((((colonoscopy[Title]) OR (endoscopy[Title])) OR (occult blood[Title])) OR (immunochemistry[Title]))) OR ((((colonoscopy[Other Term]) OR (endoscopy[Other Term)) OR (occult blood[Other Term))) OR (immunochemistry[Other Term)))) AND ((((((((((((((((((((((("Colonic Neoplasms"[Mesh]) OR ("Colorectal Neoplasms"[Mesh))) OR ("Rectal Neoplasms"[Mesh])) OR ("Intestinal Neoplasms"[Mesh])) OR ("COLON TUMOR"[Other Term])) OR ("COLORECTAL TUMOR"[Other Term])) OR ("RECTUM TUMOR"[Other Term])) OR ("COLORECTAL CANCER"[Other Term])) OR ("INTESTINE TUMOR"[Other Term])) OR ("COLORECTAL ADENOMA"[Other Term))) OR ("BOWEL CANCER" [Other Term])) OR ("COLORECTAL CANCER"[Title])) OR ("BOWEL CANCER"[Title])) OR ("COLORECTAL ADENOMA"[Text Word])) OR ("COLON TUMOR"[Title])) OR ("COLORECTAL TUMOR"[Title])) OR ("RECTUM TUMOR"[Title])) OR ("INTESTINE TUMOR"[Title])) OR ("INTESTINAL TUMOR"[Title])) OR ("RECTAL TUMOR"[Title])) OR ("RECTAL TUMOR"[Title])) OR ("RECTAL TUMOR"[Other Term])) OR ("INTESTINAL TUMOR"[Other Term])))) Filters: Humans, from 2012/1/1 - 2023/7/31 Sort by: Publication Date

EMBASE (Elsevier)

#1 :id OR I609694575:id OR I605793786:id OR I601244335:id OR I603936239:id OR I53219484:id OR I600661696:id OR I605807315:id OR I52631115:id OR I51895856:id OR I52364983:id OR I51968849:id OR I364336486:id

#2 'colon tumor'/exp

#3 'colorectal tumor'/exp

#4 'rectum tumor'/exp

#5 'colorectal cancer'/exp

#6 'intestine tumor'/exp

#7 'intestine cancer'/exp

#8 'colorectal cancer cell line'/exp

#9 'colorectal cancer':ti OR 'colorectal cancer':kw

#10 'bowel cancer':ti OR 'bowel cancer':kw

#11 'colorectal tumor':ti OR 'colorectal tumor':kw

#12 'colorectal adenoma':ti OR 'colorectal adenoma':kw

#13 #2 OR#3 OR#4 OR#5 OR#6 OR#7 OR#8 OR#9 OR#10 OR#11 OR#12

#14 'faecal haemoglobin':ti,kw OR 'fecal hemoglobin':ti,kw

#15 (('faecal immunochemical test':ti,kw OR 'fecal immunochemical test':ti,kw OR 'faecal occult blood':ti,kw) AND 'fecal occult blood':ti,kw OR haemoccult:ti,kw OR hemoccult:ti,kw) AND guaiac:ti,kw OR 'immunochemical testing':ti,kw

#16 'occult blood test'/exp

#17 'colonoscopy'/exp

#18 'endoscopy'/exp

#19 'immunochemistry'

#20 'occult blood'

#21 #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20

#22 #13AND #21

#23 'mass screening':ti,kw OR screening:ti,kw OR round:ti,kw OR 'population-based screening':ti,kw OR 'multiple round*':ti,kw OR participation*:ti,kw OR adherence:ti,kw OR attendance:ti,kw OR yield:ti,kw

#24 'protocol compliance'/exp

#25 'patient compliance'/exp OR 'patient compliance'

#26 #23 OR #24 OR #25

#27 'major clinical study'/exp OR 'major clinical stud*'

#28 'follow up'/exp OR 'follow up'

#29 'multivariate analysis'/exp OR 'multivariate analysis'

#30 #27 OR #28 OR #29

#31 #22 AND #26 ANO #30

#32 #31 AND (2012:py OR 2013:py OR 2014:py OR 2015:py OR 2016:py OR 2017:py OR 2018:py OR 2019:py OR 2020:py OR 2021:py OR 2022:py OR 2023:py) AND ('clinical trial'/de OR 'human'/de OR 'major clinical study'/de OR 'observational study'/de OR 'randomized controlled trial'/de) AND ([aged]/lim OR [middle aged]/lim) AND 'Article'/it

#33 #31 AND (2012:py OR 2013:py OR 2014:py OR 2015:py OR 2016:py OR 2017:py OR 2018:py OR 2019:py OR 2020:py OR 2021:py OR 2022:py OR 2023:py) AND ('clinical trial'/de OR 'human'/de OR 'major clinical study'/de OR 'observational study'/de OR 'randomized controlled trial'/de) AND ([aged]/lim OR [middle aged]/lim) AND 'article'/it AND (english:la OR spanish:la)

Cochrane Library

ID	Search	Hits
#1	MeSH descriptor: [Colonic Neoplasms] explode all trees	2270
#2	(colorectal NEXT tumor*):ti,ab,kw	772

#3 (COLON* NEXT TUMOR*):ti,ab,kw 216

#4 (RECTUM NEXT TUMOR*):ti,ab,kw 145

#5 (RECTAL NEXT TUMOR*):ti,ab,kw 107

#6 (COLORECTAL NEXT CANCER):ti,ab,kw 15319

#7 (BOWEL NEXT CANCER):ti,ab,kw 332

#8 (COLORECTAL NEXT ADENOMA):ti,ab,kw 592

#9 #1 RO #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 16132

#10 (IMMUNOCHEMICAL NEXT TEST*):ti,ab,kw 481

#11 (FECAL OR FAECAL OR FECES):ti,ab,kw 17298

#12 (OCCULT NEXT BLOOD):ti,ab,kw 1565

#13 ("haemoccult" OR HEMOCCULT):ti,ab,kw 127

#14 (HAEMOGLOBIN OR HEMOGLOBIN):ti,ab,kw 44578

#15 (COLONOSCOP*):ti,ab,kw 8700

#16 (ENDOSCOP*):ti,ab,kw 33252

#17 (IMMUNOCHEM*):ti,ab,kw 1359

#18 #10 OR #11 OR #12 OR #13 OR #14 OR |5 OR #16 OR #17 967344

#19 #9 AND #18 10656

#20 (MASS NEXT SCREENING):ti,ab,kw 4978

#21 (SCREENING):ti,ab,kw 72823

#22 ("round"):ti,ab,kw 2978

#23 ("population"):ti,ab,kw 145441

#24 (ROUNDS):ti,ab,kw 1937

#25 (PARTICIPATION):ti,ab,kw 40757

#26 (ADHERENCE):ti,ab,kw 44242

#27 ("attendance"):ti,ab,kw 8352

#28 (YIELD):ti,ab,kw 8646

#29 ("patient compliance"):ti,ab,kw 22411

#30 #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 295976

#31 #19 AND #30 3528

#32 ("clinical studies" OR "clinical study"):ti,ab,kw 353504

#33 MeSH descriptor: [Follow-Up Studies] explode all trees 67398

#34 ("follow up studies"):ti,ab,kw 68110

#35 ("multivariate analyses" or "multivariate analysis"):ti,ab,kw 20280

#36 (EARLY NEXT DETECTION):ti,ab,kw 4941

#37 MeSH descriptor: [Guideline Adherence] explode all trees 1383

#38 MeSH descriptor: [Observational Study] explode all trees 1538

#39 (observational NEXT stud*):ti,ab,kw 14566

#40 #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 433051

#41 #31 AND #40 with Cochrane Library publication date Between Jan 2012 and Jul 2023, in Cochrane Reviews, Cochrane Protocols, Trials 1573

#42 #41 with Cochrane Library publication date Between Jan 2012 and Jul 2023, in Cochrane Reviews, Cochrane Protocols 3

Appendix 2. Articles selected for full-text screening and reasons for exclusion.

Inadequate study design (1–3)

Insufficient reported data (4–50)

Non-FIT screening (51–80)

Opportunistic screening (81–92)

Overlapped populations (93–95)

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5. Green BB, Wang CY, Anderson ML, Chubak J, Meenan RT, Vernon SW, et al. Automated intervention with stepped increases in support to increase uptake of colorectal cancer screening: A randomized trial. Ann Intern Med. 2013 Mar 5;158(5 PART 1):301–11.

6. de Jonge L, Riggi E, van Duuren LA, Toes-Zoutendijk E, Campari C, Sassatelli R, et al. Risk prediction for advanced neoplasia using longitudinal adherence measures to fecal immunochemical test-based colorectal cancer screening programs. Prev Med (Baltim). 2023 May 1;170.

7. Randel KR, Schult AL, Botteri E, Hoff G, Bretthauer M, Ursin G, et al. Colorectal Cancer Screening With Repeated Fecal Immunochemical Test Versus Sigmoidoscopy: Baseline Results From a Randomized Trial. Gastroenterology. 2021 Mar 1;160(4):1085-1096.e5.

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Supplementary table 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2,3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	5,6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	7, 8
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	7,8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	7, 8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	8, 9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	8, 9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	10, 11
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	10, 11
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	10, 11
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	10, 11
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	10, 11

Section and Topic	Item #	Checklist item	Location where item is reported
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	10
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	10
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	10
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	10
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	12, figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1, appendix 2
Study characteristics	17	Cite each included study and present its characteristics.	Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	S. Fig 1, S. Tables 3
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Suppl Figures 3-17
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	S. Fig 1, S table 3
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Figures 2-6, suppl figures 3-17
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Suppl figures 4,5,7, figure 5
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Suppl figures 4,5,7, figure 5
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	S Fig 1, S Table 3
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Fig 3-7
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	17-20
	23b	Discuss any limitations of the evidence included in the review.	21-22
	23c	Discuss any limitations of the review processes used.	21-22
	23d	Discuss implications of the results for practice, policy, and future research.	20-22

Section and Topic	Item #	Checklist item	Location where item is reported
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	7
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	23
Competing interests	26	Declare any competing interests of review authors.	23
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	23

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

Supplementary table 2. PICO (patient, intervention, comparison, and outcome) questions developed by the authors to identify the issues of greatest importance for the literature revision.

QUESTION	POPULATION	INTERVENTION	COMPARISON	OUTCOME
What is the uptake of participants over time in organized FIT-based CRC screening programs?	Persons eligible for an organized FIT program	Offer of FIT over multiple rounds	Not applicable	- % of participants among eligible for each screening round
What are the participation patterns of FIT adherence over time in organized CRC screening programs?	Persons eligible for an organized FIT program	Offer of FIT over multiple rounds	Not applicable	- % of population that are consistent participants (defined as every cycle) - % of population that are never attenders (defined as 0 cycle) - % of population that are intermittent participants (defined as >0 and <maximum)
What is the yield for adenoma, AA, and CRC over time in organized FIT-based CRC screening programs?	Persons eligible for an organized FIT program	Offer of FIT over multiple rounds	Not applicable	- Positivity rate - CRC detection rate - AA detection rate - Adenoma detection rate

FIT: faecal immunochemical test; CRC: colorectal cancer; AA: advanced adenoma.

Supplementary Table 3. Items for consideration in the Appraisal tool for Cross-Sectional Studies (AXIS)

	Question	Yes	No	Don't know/ Comment
Introduction				
1	Were the aims/objectives of the study clear?			
Methods				
2	Was the study design appropriate for the stated aim(s)?			
3	Was the sample size justified?			
4	Was the target/reference population clearly defined? (Is it clear who the research was about?)			
5	Was the sample frame taken from an appropriate population base so that it closely represented the target/reference population under investigation?			
6	Was the selection process likely to select subjects/participants that were representative of the target/reference population under investigation?			
7	Were measures undertaken to address and categorise non-responders?			
8	Were the risk factor and outcome variables measured appropriate to the aims of the study?			
9	Were the risk factor and outcome variables measured correctly using instruments/measurements that had been trialled, piloted or published previously?			
10	Is it clear what was used to determined statistical significance and/or precision estimates? (e.g. p-values, confidence intervals)			
11	Were the methods (including statistical methods) sufficiently described to enable them to be repeated?			
Results				
12	Were the basic data adequately described?			
13	Does the response rate raise concerns about non-response bias?			
14	If appropriate, was information about non-responders described?			
15	Were the results internally consistent?			
16	Were the results presented for all the analyses described in the methods?			
Discussion				
17	Were the authors' discussions and conclusions justified by the results?			
18	Were the limitations of the study discussed?			
Other				
19	Were there any funding sources or conflicts of interest that may affect the authors' interpretation of the results?			
20	Was ethical approval or consent of participants attained?			

Supplementary Table 4. CRC stage per round over time

Population	CRC stage	1st round n (%)	2nd round n (%)	3rd round n (%)
Overall population	I-II	2555 (72.7)	1364 (73.6)	807 (71.3)
	III-IV	959 (27.3)	489 (26.4)	325 (28.7)
	Total	3514 (100)	1853 (100)	1132
Consistent screeners	I-II	1228 (76.5)	268 (70.3)	193 (73.9)
	III-IV	376 (23.5)	113 (29.7)	68 (26.1)
	Total	1604 (100)	381 (100)	261 (100)

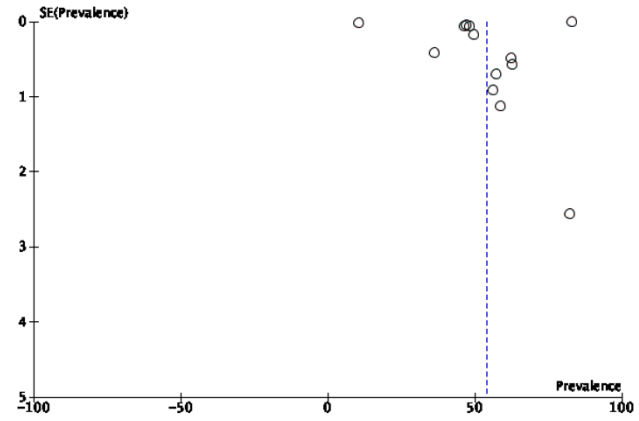
CRC: colorectal cancer

Supplementary figure 1. Risk of bias analysis. Authors' judgement about each risk of bias item presented as percentages across all included studies (A) and presented individually for each included study (B).

	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17	18	19	20
Duncan. 2014 [14]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Schreuders. 2019 [15]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	N/A
Breekveldt. 2023 [16]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Gordon. 2012 [17]	YES	YES	YES	YES	NO	NO	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	N/A
Osborne. 2017 [18]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES
Van Der Vlugt. 2017 [19]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO	YES
Stegeman. 2015 [20]	YES	YES	YES	YES	YES	YES	YES	NO	YES	NO	NO	NO	YES	YES	YES	YES	NO	N/A
Jensen. 2016 [21]	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	NO	NO	YES	YES	YES	YES	NO	N/A
Tepeš. 2007 [22]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	NO	NO	YES
Parente. 2013 [23]	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	NO	YES	YES	YES	NO	NO	N/A
Kapidzic. 2014 [24]	YES	YES	YES	YES	YES	YES	NO	YES	YES	NO	NO	NO	YES	YES	YES	YES	NO	YES
Portillo. 2018 [25]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Choi. 2012 [26]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	N/A
Heyman. 2024 [27]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Crotta. 2012 [28]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Baldacchini. 2021 [29]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Denters. 2013 [30]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO	YES
Benito. 2019 [31]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO	YES
Baldacchini. 2022 [32]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO	YES
Baker. 2015 [33]	YES	YES	YES	YES	YES	NO	YES	NO	YES	YES	NO	YES	YES	YES	YES	YES	NO	N/A
Cole. 2012 [34]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO	YES
Denters. 2012 [35]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Zorzi. 2018 [36]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	N/A
Van Roon. 2013 [37]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	NO	YES	YES	YES	YES	NO	YES

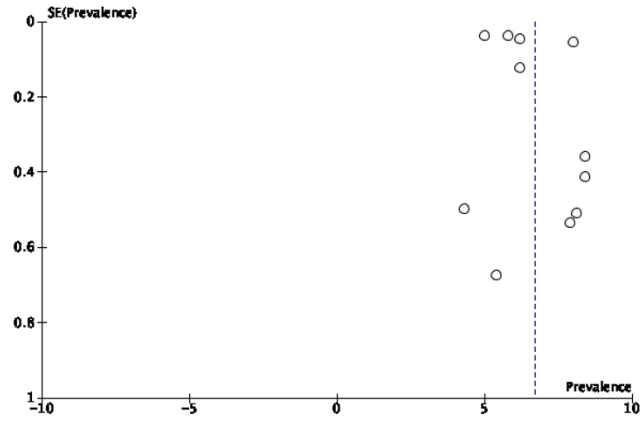
The Appraisal of cross-sectional studies (AXIS) of included manuscripts

Supplementary figure 2. Funnel plots assessing risk of publication bias.



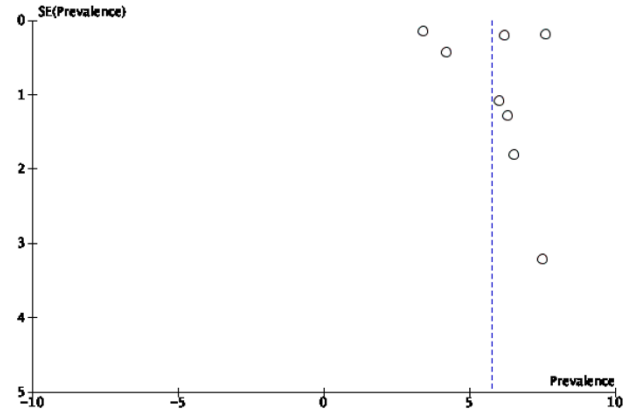
Caption

Funnel plot of comparison: Participation per round.



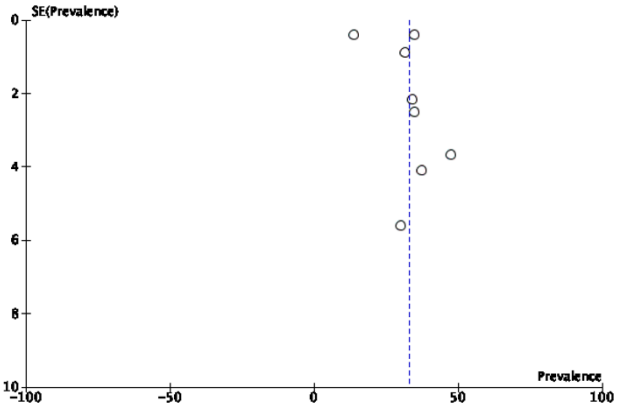
Caption

Funnel plot of comparison: Positivity rate per round – overall population.



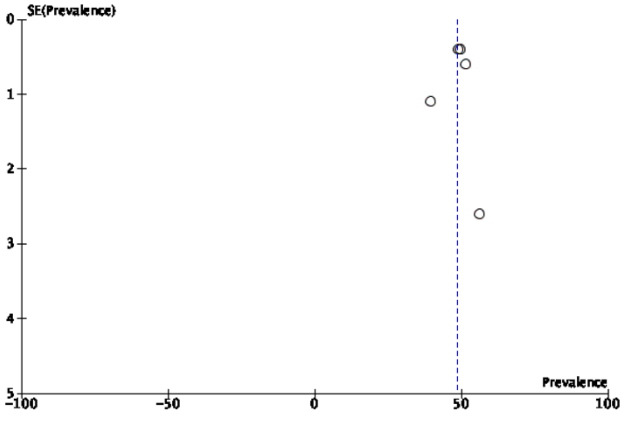
Caption

Funnel plot of comparison: CRC PPV per round – overall population.



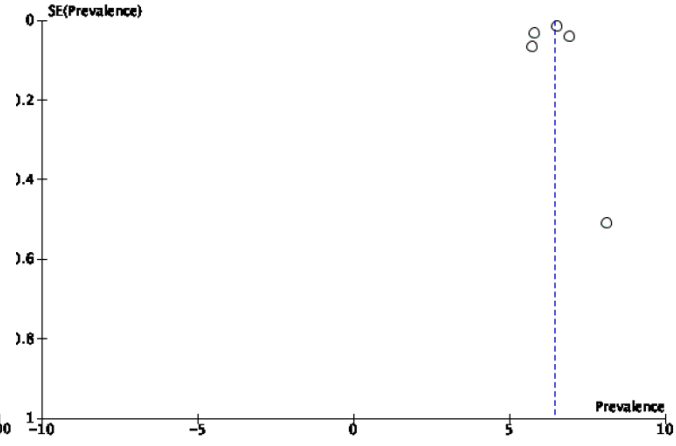
Caption

Funnel plot of comparison: AA PPV per round – overall population.



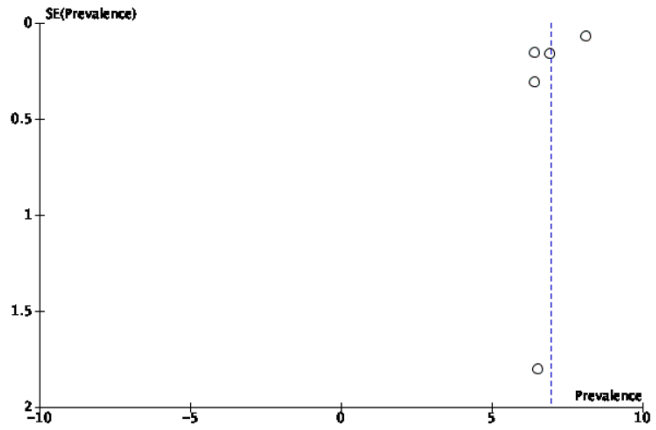
Caption

Funnel plot of comparison: Adenoma PPV per round – overall population.



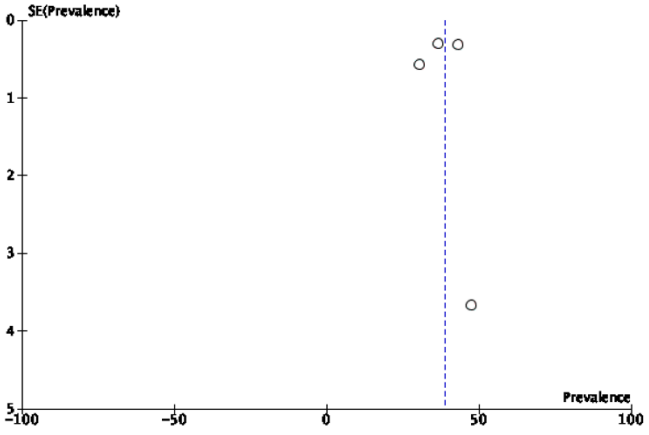
aption

unnel plot of comparison: Positivity rate per round – consistents.



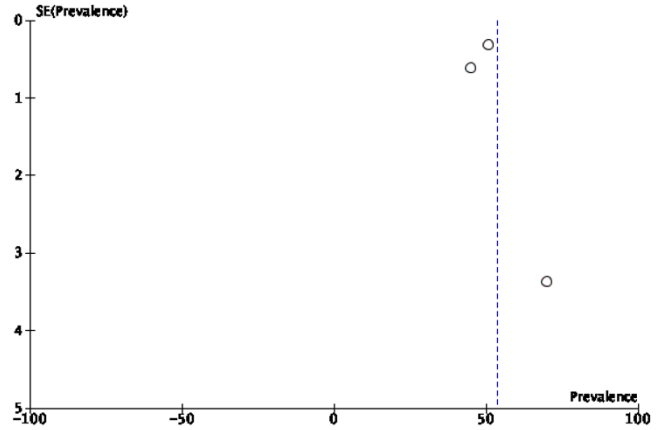
Caption

Funnel plot of comparison: CRC PPV per round – consistents.



Caption

Funnel plot of comparison: AA PPV per round – consistents.



Caption

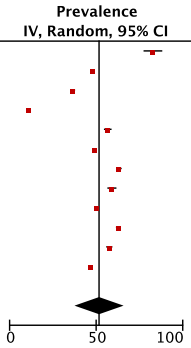
Funnel plot of comparison: Adenoma PPV per round – consistents.

Supplementary figure 3. Participation rates in FIT-based organized CRC screening programs in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

A

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baker 2015 (2 rounds)	225	185	8.3%	82.20 [77.20, 87.20]
Baldacchini 2020 (5 rounds)	845105	398444	8.3%	47.10 [46.99, 47.21]
Benito 2019 (4 rounds)	12727	4625	8.3%	36.00 [35.18, 36.82]
Choi 2012 (5 rounds)	2266656	238376	8.3%	10.50 [10.46, 10.54]
Crotta 2012 (4 rounds)	2959	1660	8.3%	56.10 [54.31, 57.89]
Jensen 2016 (4 rounds)	670841	323349	8.3%	48.20 [48.08, 48.32]
Kapidzic 2014 (3 rounds)	7229	4525	8.3%	62.60 [61.48, 63.72]
Osborne 2017 (4 rounds)	1928	1128	8.3%	58.50 [56.30, 60.70]
Parente 2013 (2 rounds)	78083	38807	8.3%	49.70 [49.35, 50.05]
Schreuders 2019 (4 rounds)	9623	5986	8.3%	62.20 [61.23, 63.17]
Stegeman 2015 (3 rounds)	5038	2871	8.3%	57.00 [55.63, 58.37]
Tepes 2020 (3 rounds)	533040	246916	8.3%	46.30 [46.17, 46.43]
Total (95% CI)	**4433454**	**1266872**	**100.0%**	**51.34 [37.34, 65.33]**

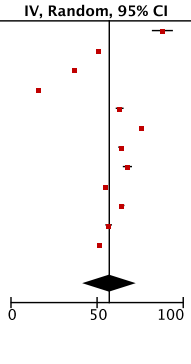
Heterogeneity: $Tau^2 = 611.00$; $Chi^2 = 888799.50$, df = 11 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 7.19 ($P < 0.00001$)



B

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baker 2015 (2 rounds)	133	116	8.3%	87.20 [81.52, 92.88]
Baldacchini 2020 (5 rounds)	990317	496033	8.3%	50.10 [50.00, 50.20]
Benito 2019 (4 rounds)	63824	22981	8.3%	36.00 [35.63, 36.37]
Choi 2012 (5 rounds)	3474440	534470	8.3%	15.40 [15.36, 15.44]
Crotta 2012 (4 rounds)	2566	1600	8.3%	62.40 [60.53, 64.27]
Jensen 2016 (4 rounds)	244476	183988	8.3%	75.30 [75.13, 75.47]
Kapidzic 2014 (3 rounds)	6111	3862	8.3%	63.20 [61.99, 64.41]
Osborne 2017 (4 rounds)	1677	1122	8.3%	66.90 [64.65, 69.15]
Parente 2013 (2 rounds)	81619	44400	8.3%	54.40 [54.06, 54.74]
Schreuders 2019 (4 rounds)	8158	5200	8.3%	63.70 [62.66, 64.74]
Stegeman 2015 (3 rounds)	4211	2359	8.3%	56.00 [54.50, 57.50]
Tepes 2020 (3 rounds)	500516	252653	8.3%	50.50 [50.36, 50.64]
Total (95% CI)	**5378048**	**1548784**	**100.0%**	**56.73 [41.29, 72.17]**

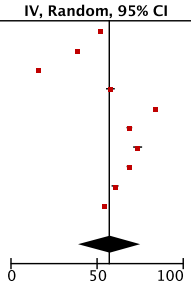
Heterogeneity: $Tau^2 = 743.66$; $Chi^2 = 1004966.94$, df = 11 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 7.20 ($P < 0.00001$)



C

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1018839	527847	10.0%	51.80 [51.70, 51.90]
Benito 2019 (4 rounds)	60906	23325	10.0%	38.30 [38.07, 38.53]
Choi 2012 (5 rounds)	4406826	693230	10.0%	15.70 [15.67, 15.73]
Crotta 2012 (4 rounds)	2056	1179	10.0%	57.30 [55.16, 59.44]
Jensen 2016 (4 rounds)	195693	163281	10.0%	83.40 [83.24, 83.56]
Kapidzic 2014 (3 rounds)	5423	3704	10.0%	68.30 [67.06, 69.54]
Osborne 2017 (4 rounds)	1550	1133	10.0%	73.10 [70.89, 75.31]
Schreuders 2019 (4 rounds)	7326	4998	10.0%	68.20 [67.13, 69.27]
Stegeman 2015 (3 rounds)	10050	3566	10.0%	60.00 [58.39, 61.61]
Tepes 2020 (3 rounds)	499279	268183	10.0%	53.70 [53.56, 53.84]
Total (95% CI)	**6207948**	**1690446**	**100.0%**	**56.98 [38.69, 75.26]**

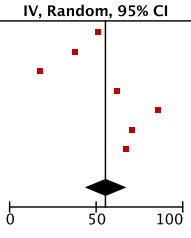
Heterogeneity: $Tau^2 = 870.28$; $Chi^2 = 1264185.24$, df = 9 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 6.11 ($P < 0.00001$)



D

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1084128	557021	14.3%	51.40 [51.31, 51.49]
Benito 2019 (4 rounds)	61428	23366	14.3%	38.30 [37.92, 38.68]
Choi 2012 (5 rounds)	4344969	780651	14.3%	18.00 [17.96, 18.04]
Crotta 2012 (4 rounds)	1862	1165	14.3%	62.60 [62.58, 62.62]
Jensen 2016 (4 rounds)	168853	145452	14.3%	86.10 [86.08, 86.12]
Osborne 2017 (4 rounds)	1487	1061	14.3%	71.40 [71.38, 71.42]
Schreuders 2019 (4 rounds)	6452	4385	14.3%	67.90 [67.89, 67.91]
Total (95% CI)	**5669179**	**1513101**	**100.0%**	**56.53 [44.43, 68.63]**

Heterogeneity: $Tau^2 = 266.81$; $Chi^2 = 13182387.74$, df = 6 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 9.16 ($P < 0.00001$)

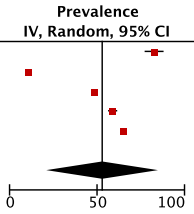


Supplementary figure 4. Participation rates in annual FIT-based organized CRC screening programs in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

A

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baker 2015 (2 rounds)	225	185	19.9%	82.20 [77.20, 87.20]
Choi 2012 (5 rounds)	2266656	238376	20.0%	10.50 [10.46, 10.54]
Jensen 2016 (4 rounds)	670841	323349	20.0%	48.20 [48.08, 48.32]
Osborne 2017 (4 rounds)	1928	1128	20.0%	58.50 [56.30, 60.70]
Van Roon 2013 (2 rounds)	2385	1543	20.0%	64.70 [64.68, 64.72]
Total (95% CI)	**2942035**	**564581**	**100.0%**	**52.79 [20.84, 84.74]**

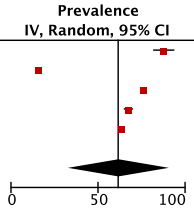
Heterogeneity: $Tau^2 = 1326.90$; $Chi^2 = 6121825.72$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 3.24 (P = 0.001)



B

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baker 2015 (2 rounds)	133	116	19.9%	87.20 [81.52, 92.88]
Choi 2012 (5 rounds)	3474440	534470	20.0%	15.40 [15.36, 15.44]
Jensen 2016 (4 rounds)	244476	183988	20.0%	75.30 [75.13, 75.47]
Osborne 2017 (4 rounds)	1677	1122	20.0%	66.90 [64.65, 69.15]
Van Roon 2013 (2 rounds)	2034	1286	20.0%	63.20 [63.18, 63.22]
Total (95% CI)	**3722760**	**720982**	**100.0%**	**61.56 [32.43, 90.69]**

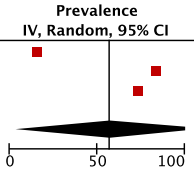
Heterogeneity: $Tau^2 = 1102.51$; $Chi^2 = 4704355.38$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 4.14 (P < 0.0001)



C

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Choi 2012 (5 rounds)	4406826	693230	33.3%	15.70 [15.67, 15.73]
Jensen 2016 (4 rounds)	195693	163281	33.3%	83.40 [83.24, 83.56]
Osborne 2017 (4 rounds)	1550	1133	33.3%	73.10 [70.89, 75.31]
Total (95% CI)	**4604069**	**857644**	**100.0%**	**57.40 [3.27, 111.52]**

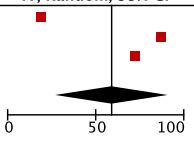
Heterogeneity: $Tau^2 = 2287.56$; $Chi^2 = 624939.57$, df = 2 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 2.08 (P = 0.04)



D

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Choi 2012 (5 rounds)	4344969	780651	33.3%	18.00 [17.96, 18.04]
Jensen 2016 (4 rounds)	168853	145452	33.3%	86.10 [86.08, 86.12]
Osborne 2017 (4 rounds)	1487	1061	33.3%	71.40 [71.38, 71.42]
Total (95% CI)	**4515309**	**927164**	**100.0%**	**58.50 [26.51, 90.49]**

Heterogeneity: $Tau^2 = 799.28$; $Chi^2 = 12021508.92$, df = 2 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 3.58 (P = 0.0003)

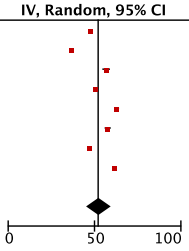


Supplementary figure 5. Participation rates in biennial FIT-based organized CRC screening programs in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

A

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	845105	398444	12.5%	47.10 [46.99, 47.21]
Benito 2019 (4 rounds)	12727	4625	12.5%	36.00 [35.18, 36.82]
Crotta 2012 (4 rounds)	2959	1660	12.4%	56.10 [54.31, 57.89]
Parente 2013 (2 rounds)	78083	38807	12.5%	49.70 [49.35, 50.05]
Schreuders 2019 (4 rounds)	9623	5986	12.5%	62.20 [61.23, 63.17]
Stegeman 2015 (3 rounds)	5038	2871	12.5%	57.00 [55.63, 58.37]
Tepes 2020 (3 rounds)	533040	246916	12.5%	46.30 [46.17, 46.43]
Van Roon 2013 (2 rounds)	2428	1481	12.5%	61.00 [60.98, 61.02]
Total (95% CI)	**1489003**	**700790**	**100.0%**	**51.92 [45.03, 58.81]**

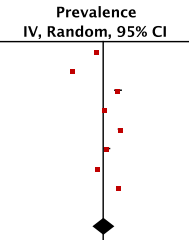
Heterogeneity: $Tau^2 = 98.69$; $Chi^2 = 115073.44$, df = 7 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 14.77$ ($P < 0.00001$)



B

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	990317	496033	12.5%	50.10 [50.00, 50.20]
Benito 2019 (4 rounds)	63824	22981	12.5%	36.00 [35.63, 36.37]
Crotta 2012 (4 rounds)	2566	1600	12.4%	62.40 [60.53, 64.27]
Parente 2013 (2 rounds)	81619	44400	12.5%	54.40 [54.06, 54.74]
Schreuders 2019 (4 rounds)	8158	5200	12.5%	63.70 [62.66, 64.74]
Stegeman 2015 (3 rounds)	4211	2359	12.4%	56.00 [54.50, 57.50]
Tepes 2020 (3 rounds)	500516	252653	12.5%	50.50 [50.36, 50.64]
Van Roon 2013 (2 rounds)	2048	1280	12.5%	62.50 [62.48, 62.52]
Total (95% CI)	**1653259**	**826506**	**100.0%**	**54.43 [48.12, 60.75]**

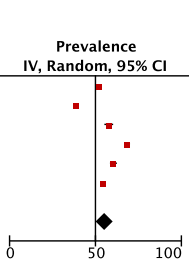
Heterogeneity: $Tau^2 = 82.74$; $Chi^2 = 103734.89$, df = 7 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 16.90$ ($P < 0.00001$)



C

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1018839	527847	16.8%	51.80 [51.70, 51.90]
Benito 2019 (4 rounds)	60906	23325	16.8%	38.30 [38.07, 38.53]
Crotta 2012 (4 rounds)	2056	1179	16.3%	57.30 [55.16, 59.44]
Schreuders 2019 (4 rounds)	7326	4998	16.7%	68.20 [67.13, 69.27]
Stegeman 2015 (3 rounds)	10050	3566	16.5%	60.00 [58.39, 61.61]
Tepes 2020 (3 rounds)	499279	268183	16.8%	53.70 [53.56, 53.84]
Total (95% CI)	**1598456**	**829098**	**100.0%**	**54.84 [49.91, 59.77]**

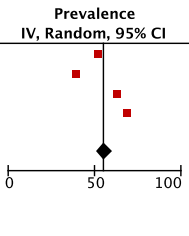
Heterogeneity: $Tau^2 = 37.67$; $Chi^2 = 14137.09$, df = 5 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 21.78$ ($P < 0.00001$)



D

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1084128	557021	25.0%	51.40 [51.31, 51.49]
Benito 2019 (4 rounds)	61428	23366	25.0%	38.30 [37.92, 38.68]
Crotta 2012 (4 rounds)	1862	1165	25.0%	62.60 [62.58, 62.62]
Schreuders 2019 (4 rounds)	6452	4385	25.0%	67.90 [67.89, 67.91]
Total (95% CI)	**1153870**	**585937**	**100.0%**	**55.06 [50.46, 59.65]**

Heterogeneity: $Tau^2 = 21.97$; $Chi^2 = 318191.43$, df = 3 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 23.49$ ($P < 0.00001$)

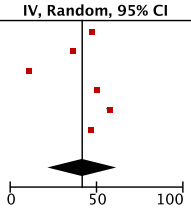


Supplementary figure 6. Participation rates in FIT-based organized CRC screening programs within open cohorts in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

A

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	845105	398444	16.7%	47.10 [46.99, 47.21]
Benito 2019 (4 rounds)	12727	4625	16.7%	36.00 [35.18, 36.82]
Choi 2012 (5 rounds)	2266656	238376	16.7%	10.50 [10.46, 10.54]
Parente 2013 (2 rounds)	78083	38807	16.7%	49.70 [49.35, 50.05]
Stegeman 2015 (3 rounds)	5038	2871	16.7%	57.00 [55.63, 58.37]
Tepes 2020 (3 rounds)	533040	246916	16.7%	46.30 [46.17, 46.43]
Total (95% CI)	**3740649**	**930039**	**100.0%**	**41.10 [21.13, 61.07]**

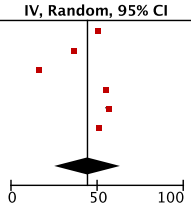
Heterogeneity: $Tau^2 = 622.78$; $Chi^2 = 639853.59$, df = 5 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 4.03$ ($P < 0.0001$)



B

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	990317	496033	16.7%	50.10 [50.00, 50.20]
Benito 2019 (4 rounds)	63824	22981	16.7%	36.00 [35.63, 36.37]
Choi 2012 (5 rounds)	3474440	534470	16.7%	15.40 [15.36, 15.44]
Parente 2013 (2 rounds)	81619	44400	16.7%	54.40 [54.06, 54.74]
Stegeman 2015 (3 rounds)	4211	2359	16.7%	56.00 [54.50, 57.50]
Tepes 2020 (3 rounds)	500516	252653	16.7%	50.50 [50.36, 50.64]
Total (95% CI)	**5114927**	**1352896**	**100.0%**	**43.73 [24.76, 62.70]**

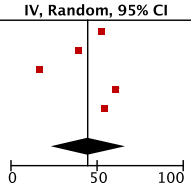
Heterogeneity: $Tau^2 = 562.09$; $Chi^2 = 633429.29$, df = 5 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 4.52$ ($P < 0.00001$)



C

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1018839	527847	20.0%	51.80 [51.70, 51.90]
Benito 2019 (4 rounds)	60906	23325	20.0%	38.30 [38.07, 38.53]
Choi 2012 (5 rounds)	4406826	693230	20.0%	15.70 [15.67, 15.73]
Stegeman 2015 (3 rounds)	10050	3566	20.0%	60.00 [58.39, 61.61]
Tepes 2020 (3 rounds)	499279	268183	20.0%	53.70 [53.56, 53.84]
Total (95% CI)	**5995900**	**1516151**	**100.0%**	**43.90 [22.39, 65.41]**

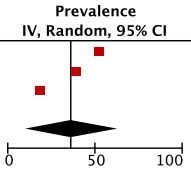
Heterogeneity: $Tau^2 = 602.09$; $Chi^2 = 726389.75$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 4.00$ ($P < 0.0001$)



D

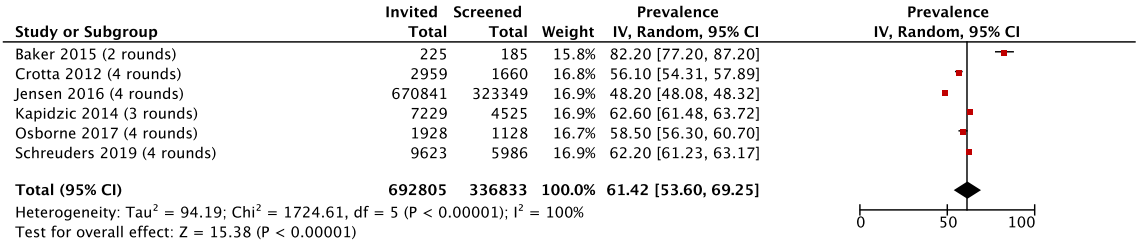
Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1084128	557021	33.3%	51.40 [51.31, 51.49]
Benito 2019 (4 rounds)	61428	23366	33.3%	38.30 [37.92, 38.68]
Choi 2012 (5 rounds)	4344969	780651	33.3%	18.00 [17.96, 18.04]
Total (95% CI)	**5490525**	**1361038**	**100.0%**	**35.90 [9.74, 62.06]**

Heterogeneity: $Tau^2 = 534.31$; $Chi^2 = 431256.59$, df = 2 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 2.69$ ($P = 0.007$)

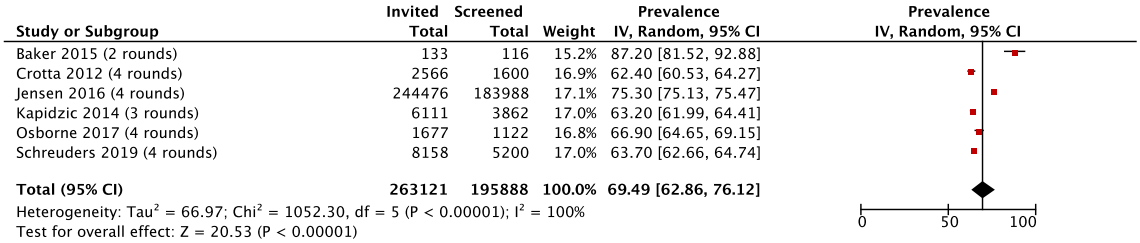


Supplementary figure 7. Participation rates in FIT-based organized CRC screening programs within closed cohorts in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

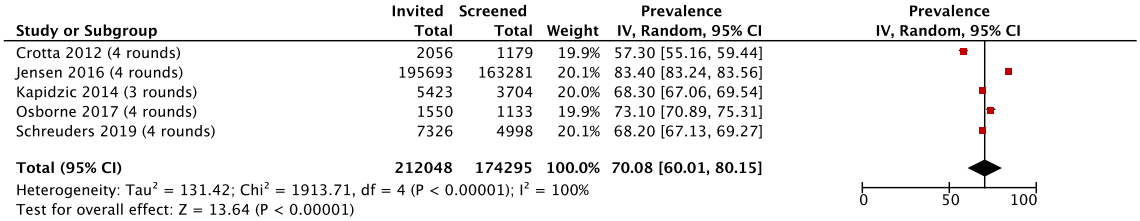
A



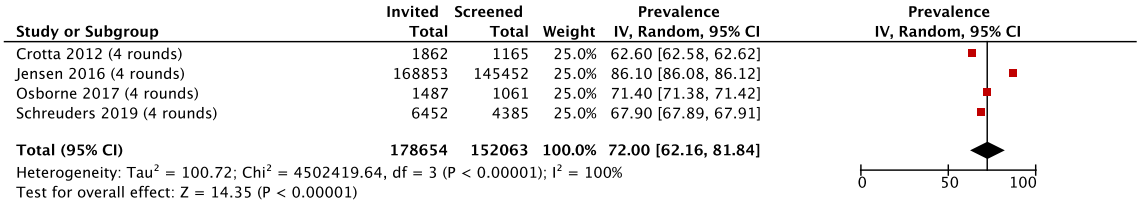
B



C

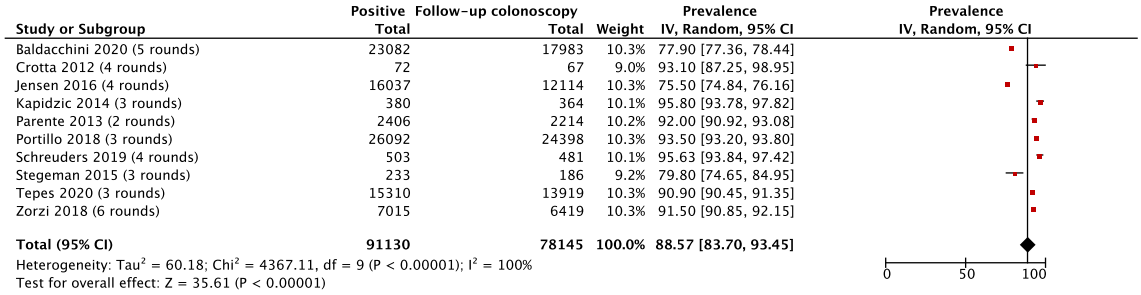


D

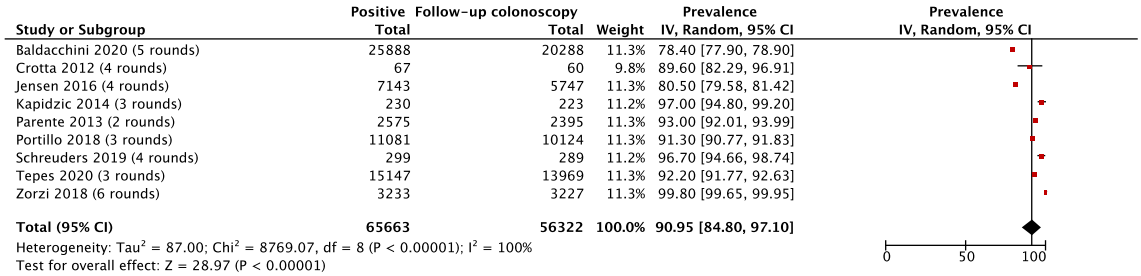


Supplementary figure 8. Follow-up colonoscopy completion rates in FIT-based organized CRC screening programs in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

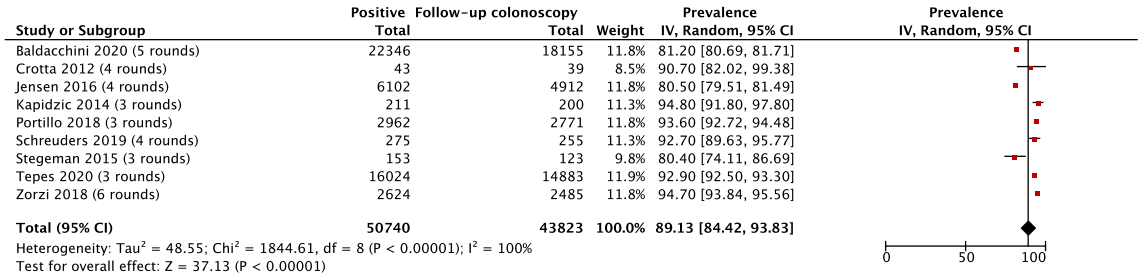
A



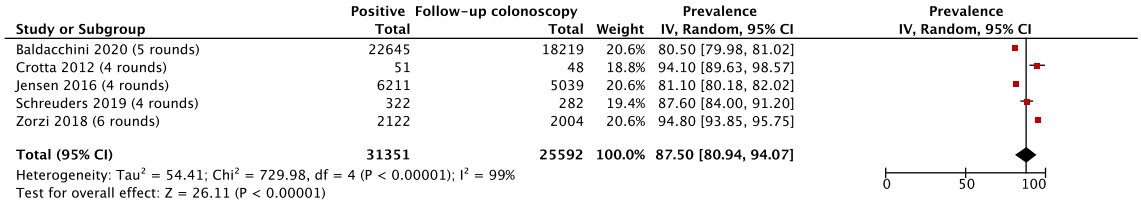
B



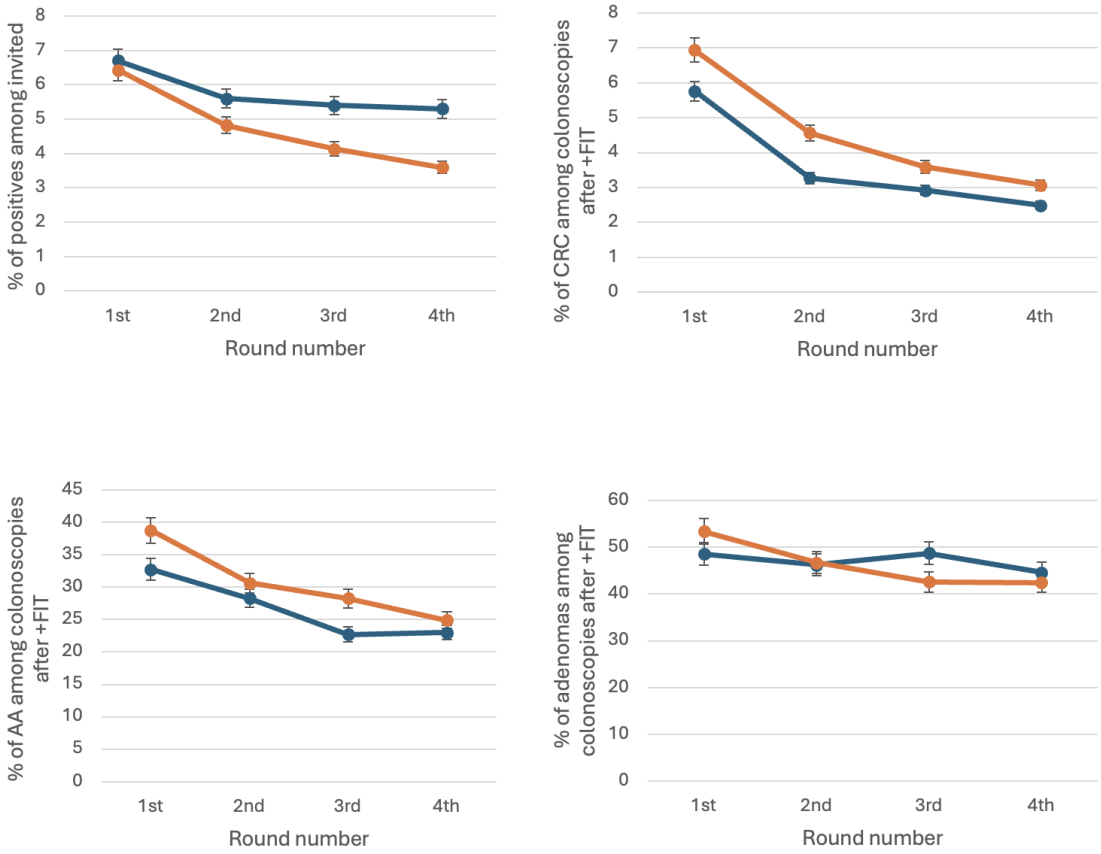
C



D



Supplementary figure 9. Comparison of the behaviors of positivity rate (A) and PPV for CRC (B), AA (C) and adenoma (D) of FIT across rounds in overall (blue) vs consistent (orange) screeners participating in FIT-based organized screening programs.



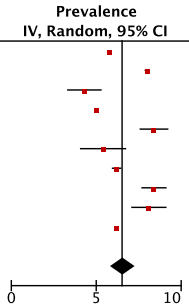
PPV: positive predictive value; CRC: colorectal cancer; AA: advanced adenoma; FIT: fecal immunochemical test.

Supplementary figure 10. Positivity rates for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	398444	23082	10.9%	5.79 [5.72, 5.86]
Choi 2012 (5 rounds)	238376	19070	10.9%	8.00 [7.89, 8.11]
Crotta 2012 (4 rounds)	1660	72	9.1%	4.30 [3.32, 5.28]
Jensen 2016 (4 rounds)	323349	16037	10.9%	5.00 [4.93, 5.07]
Kapidzic 2014 (3 rounds)	4525	380	9.6%	8.40 [7.59, 9.21]
Osborne 2017 (4 rounds)	1128	61	8.0%	5.40 [4.08, 6.72]
Parente 2013 (2 rounds)	38807	2406	10.8%	6.20 [5.96, 6.44]
Schreuders 2019 (4 rounds)	5986	503	9.9%	8.40 [7.70, 9.10]
Stegeman 2015 (3 rounds)	2871	233	9.0%	8.10 [7.10, 9.10]
Tepes 2020 (3 rounds)	246916	15310	10.9%	6.20 [6.11, 6.29]
Total (95% CI)	**1262062**	**77154**	**100.0%**	**6.58 [5.86, 7.31]**

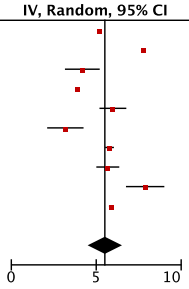
Heterogeneity: $Tau^2 = 1.25$; $Chi^2 = 2126.15$, df = 9 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 17.81$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	496033	25888	10.4%	5.20 [5.14, 5.26]
Choi 2012 (5 rounds)	534470	41688	10.4%	7.80 [7.73, 7.87]
Crotta 2012 (4 rounds)	1600	67	9.5%	4.20 [3.22, 5.18]
Jensen 2016 (4 rounds)	183988	7143	10.4%	3.90 [3.81, 3.99]
Kapidzic 2014 (3 rounds)	3864	230	9.8%	6.00 [5.25, 6.75]
Osborne 2017 (4 rounds)	1122	37	9.4%	3.20 [2.17, 4.23]
Parente 2013 (2 rounds)	44400	2575	10.4%	5.80 [5.58, 6.02]
Schreuders 2019 (4 rounds)	5200	299	10.0%	5.70 [5.07, 6.33]
Stegeman 2015 (3 rounds)	2359	186	9.3%	7.90 [6.81, 8.99]
Tepes 2020 (3 rounds)	252653	15147	10.4%	5.90 [5.81, 5.99]
Total (95% CI)	**1525689**	**93260**	**100.0%**	**5.57 [4.57, 6.56]**

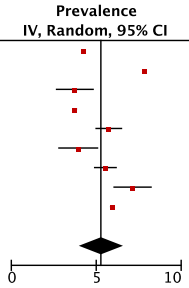
Heterogeneity: $Tau^2 = 2.48$; $Chi^2 = 5109.80$, df = 9 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 10.96$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	527847	22346	11.5%	4.20 [4.15, 4.25]
Choi 2012 (5 rounds)	693239	54072	11.5%	7.80 [7.74, 7.86]
Crotta 2012 (4 rounds)	1179	43	10.7%	3.70 [2.62, 4.78]
Jensen 2016 (4 rounds)	163281	6102	11.5%	3.70 [3.61, 3.79]
Kapidzic 2014 (3 rounds)	3704	211	11.1%	5.70 [4.95, 6.45]
Osborne 2017 (4 rounds)	1133	45	10.6%	3.90 [2.76, 5.04]
Schreuders 2019 (4 rounds)	4998	275	11.2%	5.50 [4.87, 6.13]
Stegeman 2015 (3 rounds)	2142	153	10.6%	7.10 [6.01, 8.19]
Tepes 2020 (3 rounds)	268183	16024	11.5%	5.90 [5.81, 5.99]
Total (95% CI)	**1665706**	**99271**	**100.0%**	**5.29 [3.97, 6.60]**

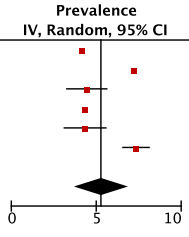
Heterogeneity: $Tau^2 = 3.92$; $Chi^2 = 8953.50$, df = 8 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 7.88$ ($P < 0.00001$)



D

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	557021	22645	17.3%	4.10 [4.05, 4.15]
Choi 2012 (5 rounds)	780651	56207	17.3%	7.20 [7.14, 7.26]
Crotta 2012 (4 rounds)	1165	51	15.8%	4.40 [3.22, 5.58]
Jensen 2016 (4 rounds)	145452	6211	17.3%	4.30 [4.20, 4.40]
Osborne 2017 (4 rounds)	1061	46	15.7%	4.30 [3.08, 5.52]
Schreuders 2019 (4 rounds)	4385	322	16.6%	7.30 [6.53, 8.07]
Total (95% CI)	**1489735**	**85482**	**100.0%**	**5.28 [3.70, 6.86]**

Heterogeneity: $Tau^2 = 3.76$; $Chi^2 = 6630.53$, df = 5 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 6.54$ ($P < 0.00001$)

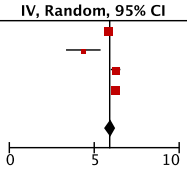


Supplementary figure 11. Positivity rates for the overall population participating in biennial FIT-based organized CRC screening programs with a cut-off of 20 µg Hb/g feces from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	398444	23082	32.0%	5.79 [5.72, 5.86]
Crotta 2012 (4 rounds)	1660	72	8.6%	4.30 [3.32, 5.28]
Parente 2013 (2 rounds)	38807	2406	27.8%	6.20 [5.96, 6.44]
Tepes 2020 (3 rounds)	246916	15310	31.6%	6.20 [6.11, 6.29]
Total (95% CI)	**685827**	**40870**	**100.0%**	**5.91 [5.57, 6.24]**

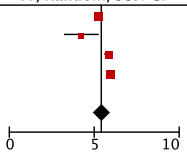
Heterogeneity: $Tau^2 = 0.09$; $Chi^2 = 61.00$, df = 3 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 34.65$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	496033	25888	29.3%	5.20 [5.14, 5.26]
Crotta 2012 (4 rounds)	1600	67	13.8%	4.20 [3.22, 5.18]
Parente 2013 (2 rounds)	44400	2575	27.8%	5.80 [5.58, 6.02]
Tepes 2020 (3 rounds)	252653	15147	29.1%	5.90 [5.81, 5.99]
Total (95% CI)	**794686**	**43677**	**100.0%**	**5.43 [4.93, 5.93]**

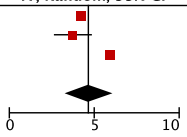
Heterogeneity: $Tau^2 = 0.22$; $Chi^2 = 168.68$, df = 3 ($P < 0.00001$); $I^2 = 98\%$
Test for overall effect: $Z = 21.25$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	527847	22346	35.4%	4.20 [4.15, 4.25]
Crotta 2012 (4 rounds)	1179	43	29.3%	3.70 [2.62, 4.78]
Tepes 2020 (3 rounds)	268183	16024	35.4%	5.90 [5.81, 5.99]
Total (95% CI)	**797209**	**38413**	**100.0%**	**4.65 [3.26, 6.05]**

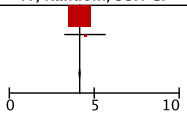
Heterogeneity: $Tau^2 = 1.44$; $Chi^2 = 999.60$, df = 2 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 6.53$ ($P < 0.00001$)



D

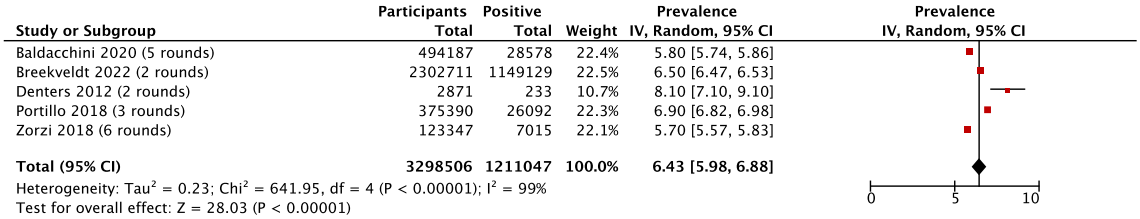
Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	557021	22645	99.8%	4.10 [4.05, 4.15]
Crotta 2012 (4 rounds)	1165	51	0.2%	4.40 [3.22, 5.58]
Total (95% CI)	**558186**	**22696**	**100.0%**	**4.10 [4.05, 4.15]**

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.25$, df = 1 ($P = 0.62$); $I^2 = 0\%$
Test for overall effect: $Z = 152.03$ ($P < 0.00001$)

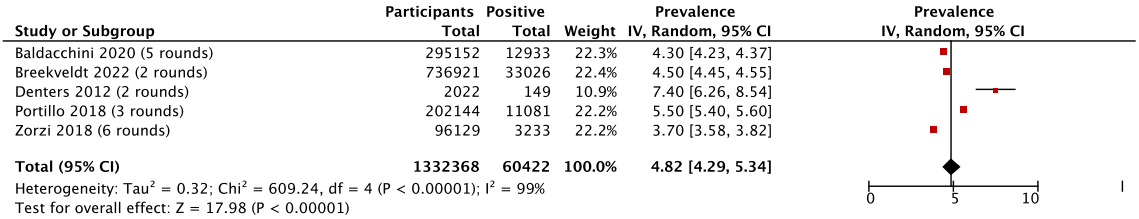


Supplementary figure 12. Positivity rates for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

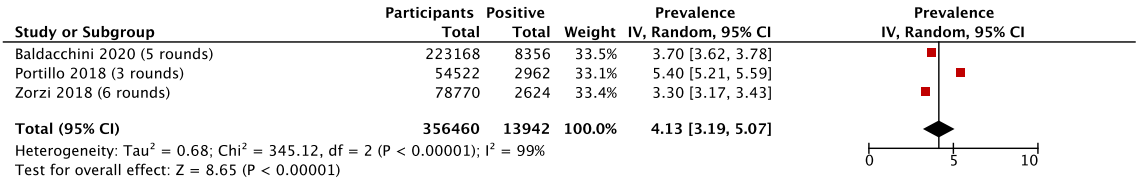
A



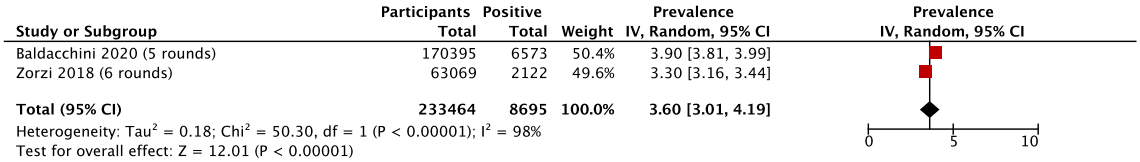
B



C



D

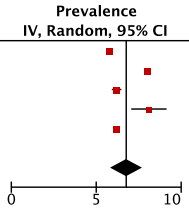


Supplementary figure 13. Positivity rates for the overall population participating in FIT-based organized CRC screening programs within open cohorts from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	398444	23082	20.9%	5.79 [5.72, 5.86]
Choi 2012 (5 rounds)	238376	19070	20.8%	8.00 [7.89, 8.11]
Parente 2013 (2 rounds)	38807	2406	20.6%	6.20 [5.96, 6.44]
Stegeman 2015 (3 rounds)	2871	233	16.9%	8.10 [7.10, 9.10]
Tepes 2020 (3 rounds)	246916	15310	20.8%	6.20 [6.11, 6.29]
Total (95% CI)	**925414**	**60101**	**100.0%**	**6.81 [5.87, 7.75]**

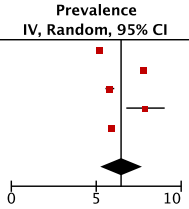
Heterogeneity: $Tau^2 = 1.09$; $Chi^2 = 1118.79$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 14.27$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	496033	25888	20.6%	5.20 [5.14, 5.26]
Choi 2012 (5 rounds)	534470	41688	20.6%	7.80 [7.73, 7.87]
Parente 2013 (2 rounds)	44400	2575	20.5%	5.80 [5.58, 6.02]
Stegeman 2015 (3 rounds)	2359	186	17.8%	7.90 [6.81, 8.99]
Tepes 2020 (3 rounds)	252653	15147	20.6%	5.90 [5.81, 5.99]
Total (95% CI)	**1329915**	**85484**	**100.0%**	**6.48 [5.24, 7.73]**

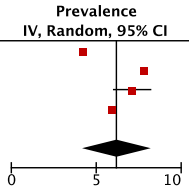
Heterogeneity: $Tau^2 = 1.95$; $Chi^2 = 2909.27$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 10.22$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	527847	22346	25.4%	4.20 [4.15, 4.25]
Choi 2012 (5 rounds)	693239	54072	25.4%	7.80 [7.74, 7.86]
Stegeman 2015 (3 rounds)	2142	153	23.7%	7.10 [6.01, 8.19]
Tepes 2020 (3 rounds)	268183	16024	25.4%	5.90 [5.81, 5.99]
Total (95% CI)	**1491411**	**92595**	**100.0%**	**6.24 [4.20, 8.27]**

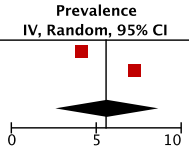
Heterogeneity: $Tau^2 = 4.22$; $Chi^2 = 7181.29$, df = 3 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 6.02$ ($P < 0.00001$)



D

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	557021	22645	50.0%	4.10 [4.05, 4.15]
Choi 2012 (5 rounds)	780651	56207	50.0%	7.20 [7.14, 7.26]
Total (95% CI)	**1337672**	**78852**	**100.0%**	**5.65 [2.61, 8.69]**

Heterogeneity: $Tau^2 = 4.80$; $Chi^2 = 6121.02$, df = 1 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 3.65$ ($P = 0.0003$)

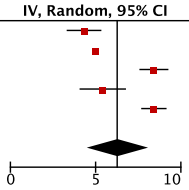


Supplementary figure 14. Positivity rates for the overall population participating in FIT-based organized CRC screening programs within closed cohorts from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Crotta 2012 (4 rounds)	1660	72	19.7%	4.30 [3.32, 5.28]
Jensen 2016 (4 rounds)	323349	16037	21.0%	5.00 [4.93, 5.07]
Kapidzic 2014 (3 rounds)	4525	380	20.1%	8.40 [7.59, 9.21]
Osborne 2017 (4 rounds)	1128	61	18.8%	5.40 [4.08, 6.72]
Schreuders 2019 (4 rounds)	5986	503	20.3%	8.40 [7.70, 9.10]
Total (95% CI)	**336648**	**17053**	**100.0%**	**6.31 [4.53, 8.09]**

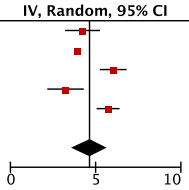
Heterogeneity: $Tau^2 = 3.93$; $Chi^2 = 157.83$, df = 4 ($P < 0.00001$); $I^2 = 97\%$
Test for overall effect: $Z = 6.95$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Crotta 2012 (4 rounds)	1600	67	18.6%	4.20 [3.22, 5.18]
Jensen 2016 (4 rounds)	183988	7143	22.4%	3.90 [3.81, 3.99]
Kapidzic 2014 (3 rounds)	3864	230	20.0%	6.00 [5.25, 6.75]
Osborne 2017 (4 rounds)	1122	37	18.3%	3.20 [2.17, 4.23]
Schreuders 2019 (4 rounds)	5200	299	20.7%	5.70 [5.07, 6.33]
Total (95% CI)	**195774**	**7776**	**100.0%**	**4.62 [3.59, 5.65]**

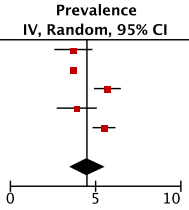
Heterogeneity: $Tau^2 = 1.24$; $Chi^2 = 61.64$, df = 4 ($P < 0.00001$); $I^2 = 94\%$
Test for overall effect: $Z = 8.78$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Crotta 2012 (4 rounds)	1179	43	18.2%	3.70 [2.62, 4.78]
Jensen 2016 (4 rounds)	163281	6102	22.7%	3.70 [3.61, 3.79]
Kapidzic 2014 (3 rounds)	3704	211	20.3%	5.70 [4.95, 6.45]
Osborne 2017 (4 rounds)	1133	45	17.8%	3.90 [2.76, 5.04]
Schreuders 2019 (4 rounds)	4998	275	21.0%	5.50 [4.87, 6.13]
Total (95% CI)	**174295**	**6676**	**100.0%**	**4.52 [3.50, 5.54]**

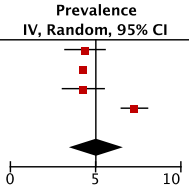
Heterogeneity: $Tau^2 = 1.19$; $Chi^2 = 56.89$, df = 4 ($P < 0.00001$); $I^2 = 93\%$
Test for overall effect: $Z = 8.66$ ($P < 0.00001$)



D

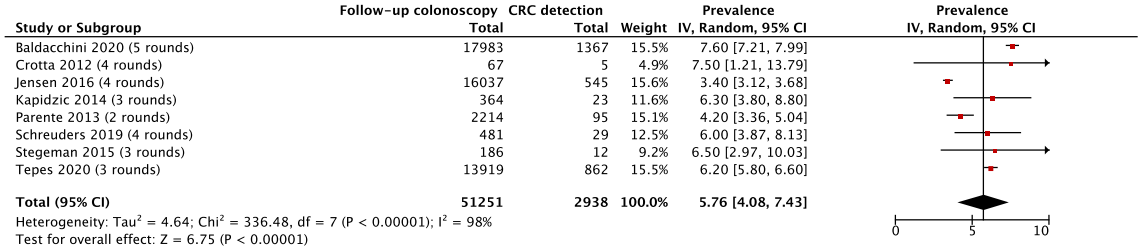
Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Crotta 2012 (4 rounds)	1165	51	23.7%	4.40 [3.22, 5.58]
Jensen 2016 (4 rounds)	145452	6211	27.3%	4.30 [4.20, 4.40]
Osborne 2017 (4 rounds)	1061	46	23.4%	4.30 [3.08, 5.52]
Schreuders 2019 (4 rounds)	4385	322	25.6%	7.30 [6.53, 8.07]
Total (95% CI)	**152063**	**6630**	**100.0%**	**5.09 [3.52, 6.66]**

Heterogeneity: $Tau^2 = 2.35$; $Chi^2 = 57.24$, df = 3 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 6.36$ ($P < 0.00001$)

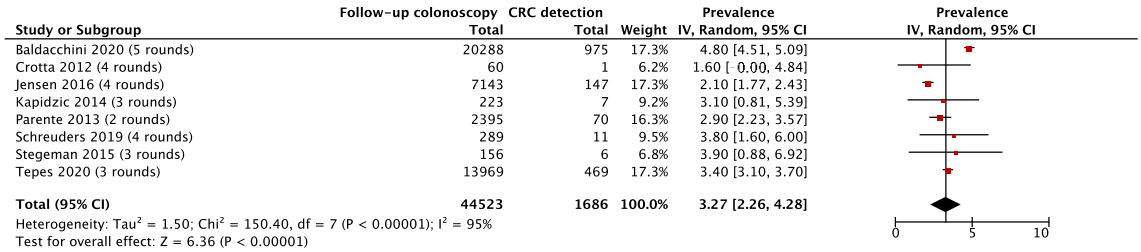


Supplementary figure 15. CRC PPV for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

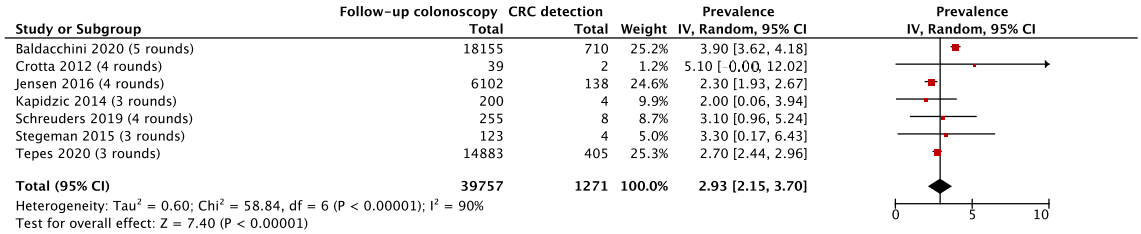
A



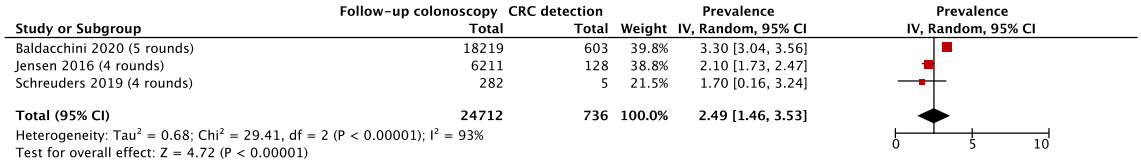
B



C

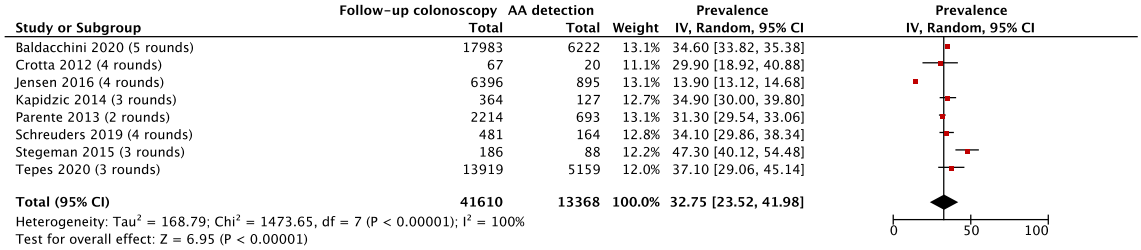


D

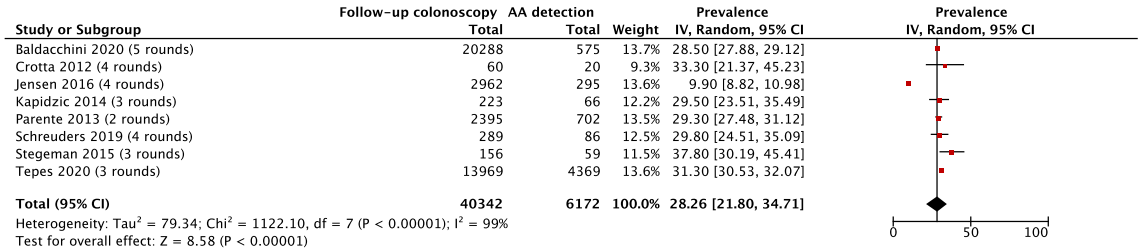


Supplementary figure 16. AA PPV for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

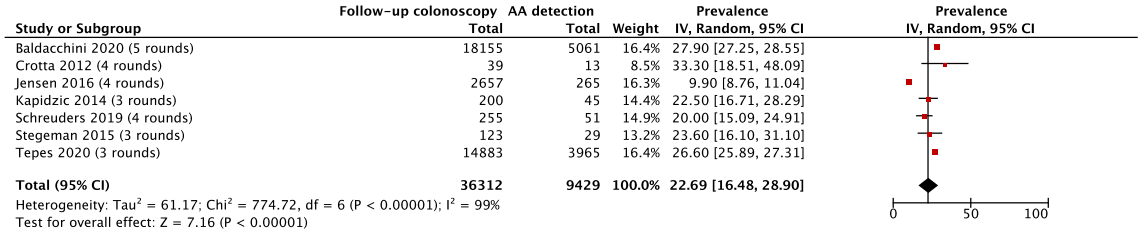
A



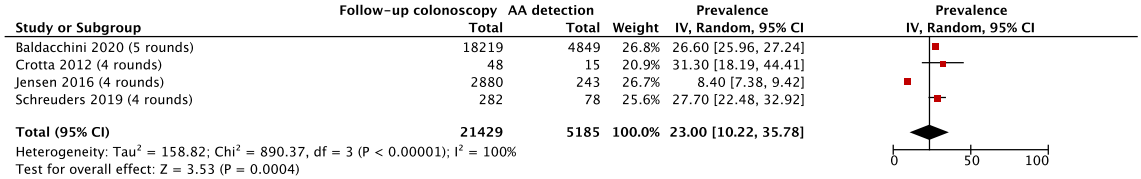
B



C

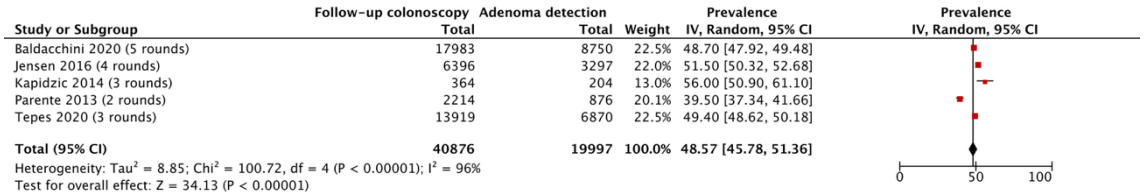


D

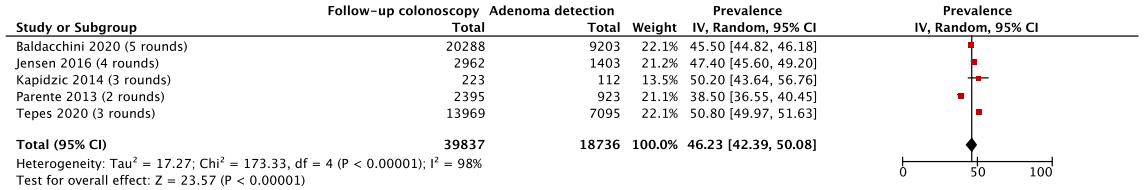


Supplementary figure 17. Adenoma PPV for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

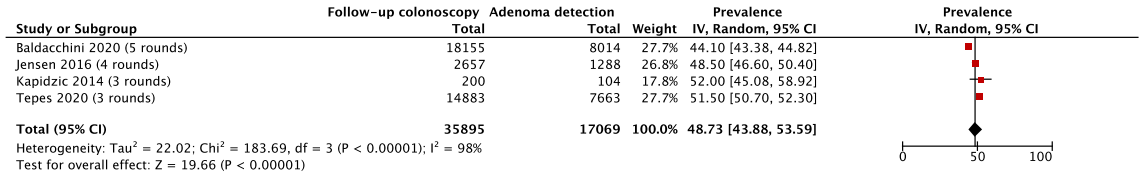
A



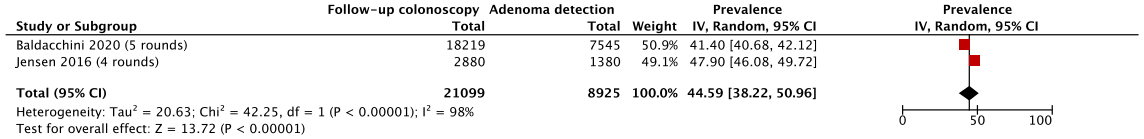
B



C



D

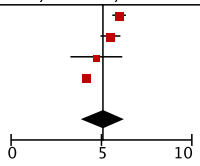


Supplementary figure 18. Positivity rate (A) and PPV for CRC (B), AA (C) and adenoma (D) of FIT per round in the overall screening population participating in biennial FIT-based organized screening programs with a cut-off of 20 µg Hb/g feces.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
1st round (4 studies)	685827	40870	26.8%	5.91 [5.58, 6.24]
2nd round (4 studies)	794686	43677	26.0%	5.43 [4.92, 5.94]
3rd round (3 studies)	797209	38413	19.8%	4.65 [3.26, 6.04]
4rd round (2 studies)	558186	22696	27.4%	4.10 [4.04, 4.16]
Total (95% CI)	**2835908**	**145656**	**100.0%**	**5.04 [3.86, 6.22]**

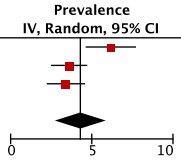
Heterogeneity: $Tau^2 = 1.32$; $Chi^2 = 134.14$, df = 3 ($P < 0.00001$); $I^2 = 98\%$
Test for overall effect: $Z = 8.38$ ($P < 0.00001$)



B

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (4 studies)	34183	2329	30.2%	6.12 [4.61, 7.63]
2nd round (4 studies)	36712	1515	35.2%	3.55 [2.47, 4.63]
3rd round (3 studies)	33077	1117	34.5%	3.35 [2.21, 4.49]
Total (95% CI)	**103972**	**4961**	**100.0%**	**4.26 [2.71, 5.80]**

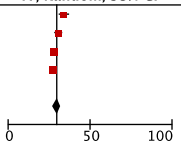
Heterogeneity: $Tau^2 = 1.46$; $Chi^2 = 9.50$, df = 2 ($P = 0.009$); $I^2 = 79\%$
Test for overall effect: $Z = 5.40$ ($P < 0.00001$)



C

Study or Subgroup	Follow-up colonoscopy Total	AA detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (4 studies)	34183	12094	20.1%	33.27 [30.51, 36.03]
2nd round (4 studies)	36712	5666	23.6%	29.81 [27.77, 31.85]
3rd round (3 studies)	33077	9039	27.2%	27.30 [26.07, 28.53]
4rd round (2 studies)	18267	4864	29.1%	26.61 [25.96, 27.26]
Total (95% CI)	**122239**	**31663**	**100.0%**	**28.89 [26.73, 31.05]**

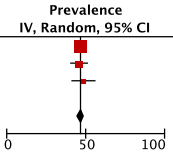
Heterogeneity: $Tau^2 = 4.08$; $Chi^2 = 28.05$, df = 3 ($P < 0.00001$); $I^2 = 89\%$
Test for overall effect: $Z = 26.17$ ($P < 0.00001$)



D

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (3 studies)	34116	15803	58.7%	46.08 [42.51, 49.65]
2nd round (3 studies)	36652	17221	27.1%	45.01 [39.76, 50.26]
3rd round (2 studies)	33038	15677	14.2%	47.80 [40.55, 55.05]
Total (95% CI)	**103806**	**48701**	**100.0%**	**46.03 [43.30, 48.77]**

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.37$, df = 2 ($P = 0.83$); $I^2 = 0\%$
Test for overall effect: $Z = 33.01$ ($P < 0.00001$)

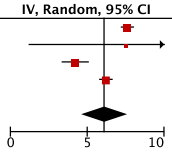


Supplementary figure 19. CRC PPV for the overall population participating in biennial FIT-based organized CRC screening programs with a cut-off of 20 µg Hb/g feces from A) 1st round, B) 2nd round, to C) 3rd round.

A

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	17983	1367	32.5%	7.60 [7.21, 7.99]
Crotta 2012 (4 rounds)	67	5	4.9%	7.50 [1.21, 13.79]
Parente 2013 (2 rounds)	2214	95	30.1%	4.20 [3.36, 5.04]
Tepes 2020 (3 rounds)	13919	862	32.5%	6.20 [5.80, 6.60]
Total (95% CI)	**34183**	**2329**	**100.0%**	**6.12 [4.61, 7.63]**

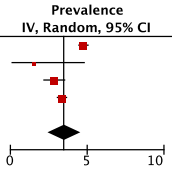
Heterogeneity: $Tau^2 = 1.79$; $Chi^2 = 60.22$, df = 3 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 7.94$ ($P < 0.00001$)



B

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	20288	975	31.5%	4.80 [4.51, 5.09]
Crotta 2012 (4 rounds)	60	1	8.3%	1.60 [0.00, 4.84]
Parente 2013 (2 rounds)	2395	70	28.7%	2.90 [2.23, 3.57]
Tepes 2020 (3 rounds)	13969	469	31.5%	3.40 [3.10, 3.70]
Total (95% CI)	**36712**	**1515**	**100.0%**	**3.55 [2.46, 4.63]**

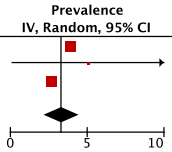
Heterogeneity: $Tau^2 = 0.95$; $Chi^2 = 56.36$, df = 3 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 6.40$ ($P < 0.00001$)



C

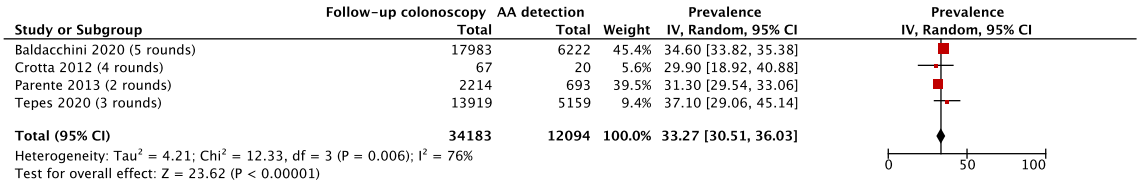
Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	18155	710	48.6%	3.90 [3.62, 4.18]
Crotta 2012 (4 rounds)	39	2	2.6%	5.10 [0.00, 12.02]
Tepes 2020 (3 rounds)	14883	405	48.8%	2.70 [2.44, 2.96]
Total (95% CI)	**33077**	**1117**	**100.0%**	**3.35 [2.20, 4.49]**

Heterogeneity: $Tau^2 = 0.69$; $Chi^2 = 37.75$, df = 2 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 5.71$ ($P < 0.00001$)

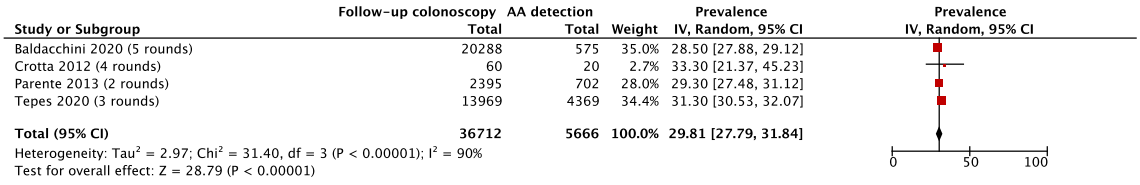


Supplementary figure 20. AA PPV for the overall population participating in biennial FIT-based organized CRC screening programs with a cut-off of 20 µg Hb/g feces from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

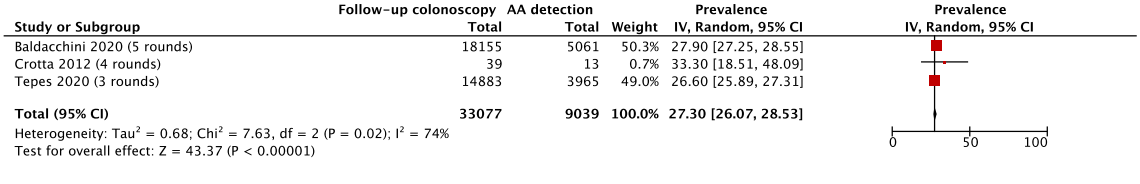
A



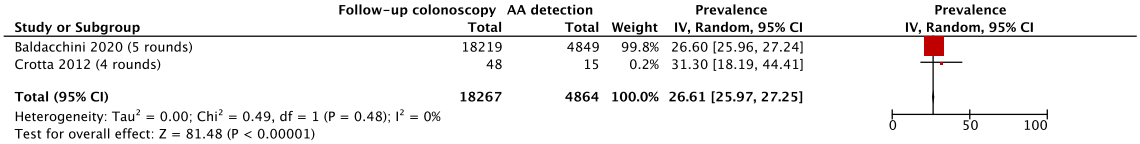
B



C

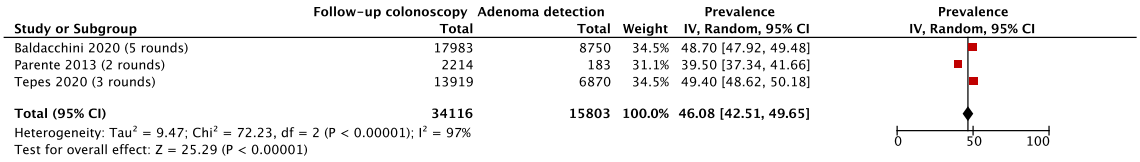


D

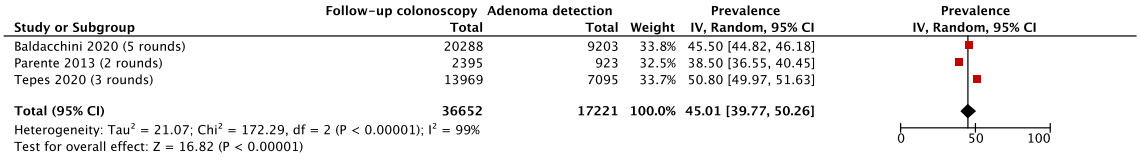


Supplementary figure 21. Adenoma PPV for the overall population participating in biennial FIT-based organized CRC screening programs with a cut-off of 20 µg Hb/g feces from A) 1st round, B) 2nd round, to C) 3rd round.

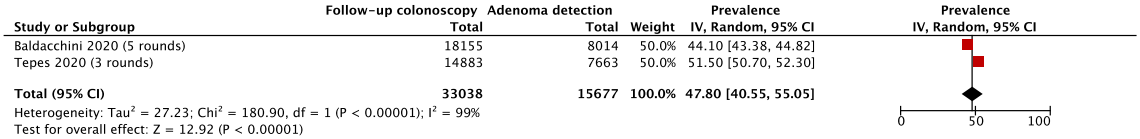
A



B



C

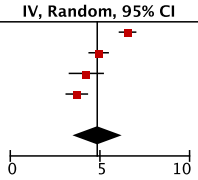


Supplementary figure 22. Positivity rate (A) and PPV for CRC (B), AA (C) and adenoma (D) of FIT per round for the consistent screeners participating in FIT-based organized screening programs.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
1st round (5 studies)	3298506	1211047	25.8%	6.43 [5.98, 6.88]
2nd round (5 studies)	1332368	60422	25.5%	4.82 [4.29, 5.35]
3rd round (3 studies)	356460	13942	23.5%	4.13 [3.19, 5.07]
4rd round (3 studies)	233464	8695	25.3%	3.60 [3.01, 4.19]
Total (95% CI)	**5220798**	**1294106**	**100.0%**	**4.76 [3.41, 6.12]**

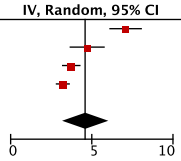
Heterogeneity: Tau² = 1.79; Chi² = 63.59, df = 3 (P < 0.00001); I² = 95%
Test for overall effect: Z = 6.91 (P < 0.00001)



B

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (5 studies)	204346	15856	24.0%	6.94 [5.98, 7.90]
2nd round (5 studies)	57628	2897	23.5%	4.57 [3.53, 5.61]
3rd round (3 studies)	12615	463	26.0%	3.59 [3.06, 4.12]
4rd round (2 studies)	7863	243	26.5%	3.07 [2.68, 3.46]
Total (95% CI)	**282452**	**19459**	**100.0%**	**4.49 [3.09, 5.89]**

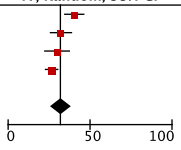
Heterogeneity: Tau² = 1.89; Chi² = 56.50, df = 3 (P < 0.00001); I² = 95%
Test for overall effect: Z = 6.28 (P < 0.00001)



C

Study or Subgroup	Follow-up colonoscopy Total	AA detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (4 studies)	55217	21421	25.2%	38.74 [32.94, 44.54]
2nd round (4 studies)	24602	7304	24.1%	30.62 [24.21, 37.03]
3rd round (3 studies)	12615	3328	22.2%	28.24 [20.73, 35.75]
4rd round (2 studies)	7863	1897	28.6%	24.91 [21.19, 28.63]
Total (95% CI)	**100297**	**33950**	**100.0%**	**30.51 [24.02, 36.99]**

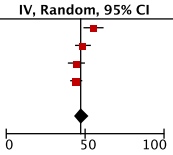
Heterogeneity: Tau² = 34.77; Chi² = 15.74, df = 3 (P = 0.001); I² = 81%
Test for overall effect: Z = 9.21 (P < 0.00001)



D

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (3 studies)	30819	15259	21.6%	53.41 [47.47, 59.35]
2nd round (3 studies)	14478	6211	25.5%	46.72 [42.15, 51.29]
3rd round (2 studies)	9844	4075	24.2%	42.58 [37.58, 47.58]
4rd round (2 studies)	7863	3280	28.8%	42.43 [39.00, 45.86]
Total (95% CI)	**63004**	**28825**	**100.0%**	**45.93 [41.43, 50.42]**

Heterogeneity: Tau² = 15.22; Chi² = 11.31, df = 3 (P = 0.01); I² = 73%
Test for overall effect: Z = 20.03 (P < 0.00001)

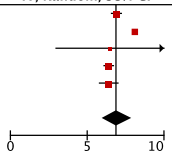


Supplementary figure 23. CRC PPV for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	24214	1682	23.8%	6.90 [6.58, 7.22]
Breekveldt 2022 (2 rounds)	149129	12183	24.3%	8.10 [7.96, 8.24]
Denters 2012 (2 rounds)	186	12	5.7%	6.50 [2.97, 10.03]
Portillo 2018 (3 rounds)	24398	1566	23.9%	6.40 [6.09, 6.71]
Zorzi 2018 (6 rounds)	6419	413	22.3%	6.40 [5.80, 7.00]
Total (95% CI)	**204346**	**15856**	**100.0%**	**6.94 [5.98, 7.90]**

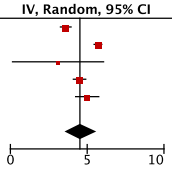
Heterogeneity: $Tau^2 = 0.99$; $Chi^2 = 142.59$, df = 4 ($P < 0.00001$); $I^2 = 97\%$
Test for overall effect: $Z = 14.12$ ($P < 0.00001$)



B

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	11121	402	23.5%	3.60 [3.25, 3.95]
Breekveldt 2022 (2 rounds)	33026	1874	23.8%	5.70 [5.45, 5.95]
Denters 2012 (2 rounds)	130	4	8.2%	3.10 [0.13, 6.07]
Portillo 2018 (3 rounds)	10124	455	23.2%	4.50 [4.10, 4.90]
Zorzi 2018 (6 rounds)	3227	162	21.4%	5.00 [4.25, 5.75]
Total (95% CI)	**57628**	**2897**	**100.0%**	**4.57 [3.53, 5.61]**

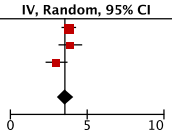
Heterogeneity: $Tau^2 = 1.17$; $Chi^2 = 98.86$, df = 4 ($P < 0.00001$); $I^2 = 96\%$
Test for overall effect: $Z = 8.59$ ($P < 0.00001$)



C

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	7359	278	42.3%	3.80 [3.36, 4.24]
Portillo 2018 (3 rounds)	2771	110	27.7%	3.90 [3.17, 4.63]
Zorzi 2018 (6 rounds)	2485	75	30.0%	3.00 [2.33, 3.67]
Total (95% CI)	**12615**	**463**	**100.0%**	**3.59 [3.07, 4.11]**

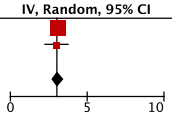
Heterogeneity: $Tau^2 = 0.12$; $Chi^2 = 4.49$, df = 2 ($P = 0.11$); $I^2 = 55\%$
Test for overall effect: $Z = 13.50$ ($P < 0.00001$)



D

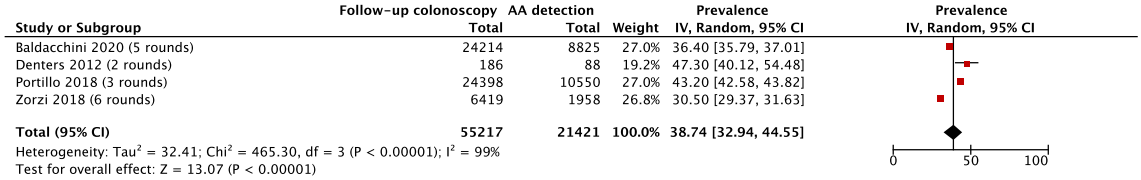
Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	5859	182	74.3%	3.10 [2.66, 3.54]
Zorzi 2018 (6 rounds)	2004	61	25.7%	3.00 [2.25, 3.75]
Total (95% CI)	**7863**	**243**	**100.0%**	**3.07 [2.69, 3.46]**

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.05$, df = 1 ($P = 0.82$); $I^2 = 0\%$
Test for overall effect: $Z = 15.78$ ($P < 0.00001$)

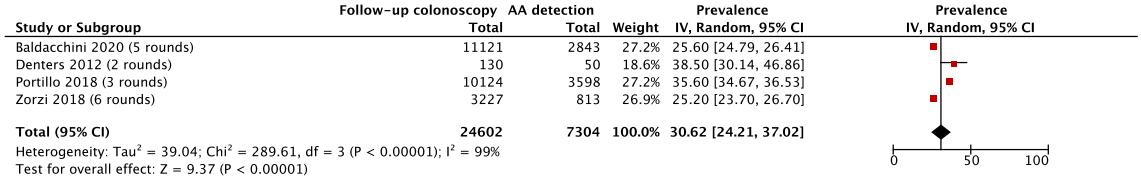


Supplementary figure 24. AA PPV for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

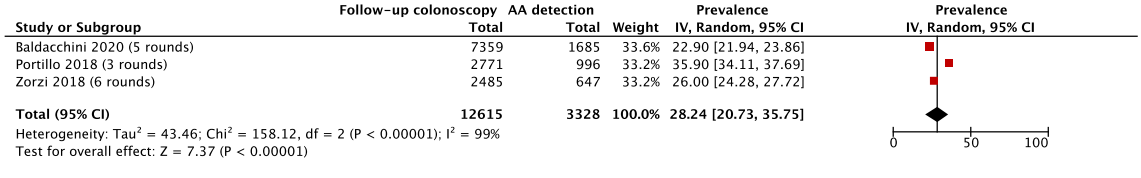
A



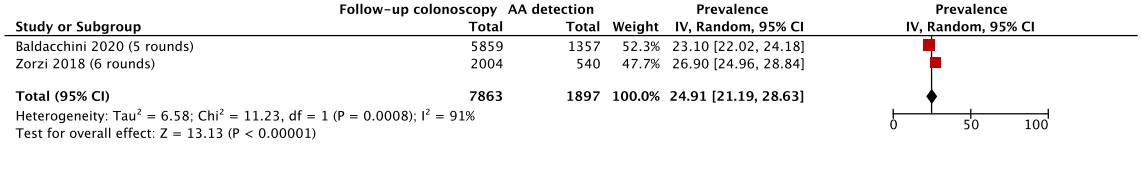
B



C



D

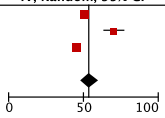


Supplementary figure 25. Adenoma PPV for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	24214	12262	37.4%	50.60 [49.97, 51.23]
Denters 2012 (2 rounds)	186	120	25.7%	69.90 [63.31, 76.49]
Zorzi 2018 (6 rounds)	6419	2877	37.0%	44.80 [43.58, 46.02]
Total (95% CI)	**30819**	**15259**	**100.0%**	**53.41 [47.47, 59.35]**

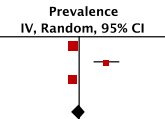
Heterogeneity: $Tau^2 = 24.49$; $Chi^2 = 105.81$, df = 2 ($P < 0.00001$); $I^2 = 98\%$
Test for overall effect: $Z = 17.62$ ($P < 0.00001$)



B

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	11121	4753	41.8%	42.70 [41.78, 43.62]
Denters 2012 (2 rounds)	130	85	18.0%	65.40 [57.22, 73.58]
Zorzi 2018 (6 rounds)	3227	1373	40.2%	42.50 [40.79, 44.21]
Total (95% CI)	**14478**	**6211**	**100.0%**	**46.72 [42.14, 51.29]**

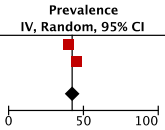
Heterogeneity: $Tau^2 = 12.83$; $Chi^2 = 29.48$, df = 2 ($P < 0.00001$); $I^2 = 93\%$
Test for overall effect: $Z = 20.00$ ($P < 0.00001$)



C

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	7359	2953	51.3%	40.10 [38.98, 41.22]
Zorzi 2018 (6 rounds)	2485	1122	48.7%	45.20 [43.24, 47.16]
Total (95% CI)	**9844**	**4075**	**100.0%**	**42.58 [37.59, 47.58]**

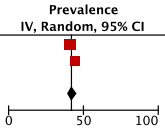
Heterogeneity: $Tau^2 = 12.34$; $Chi^2 = 19.67$, df = 1 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 16.71$ ($P < 0.00001$)



D

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	5859	2392	53.3%	40.80 [39.54, 42.06]
Zorzi 2018 (6 rounds)	2004	888	46.7%	44.30 [42.13, 46.47]
Total (95% CI)	**7863**	**3280**	**100.0%**	**42.43 [39.01, 45.86]**

Heterogeneity: $Tau^2 = 5.30$; $Chi^2 = 7.46$, df = 1 ($P = 0.006$); $I^2 = 87\%$
Test for overall effect: $Z = 24.30$ ($P < 0.00001$)

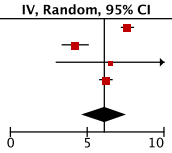


Supplementary figure 26. CRC PPV for the overall population participating in FIT-based organized CRC screening programs within open cohorts from A) 1st round, B) 2nd round, to C) 3rd round.

A

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	17983	1367	30.5%	7.60 [7.21, 7.99]
Parente 2013 (2 rounds)	2214	95	28.2%	4.20 [3.36, 5.04]
Stegeman 2015 (3 rounds)	186	12	11.0%	6.50 [2.97, 10.03]
Tepes 2020 (3 rounds)	13919	862	30.4%	6.20 [5.80, 6.60]
Total (95% CI)	**34302**	**2336**	**100.0%**	**6.10 [4.64, 7.55]**

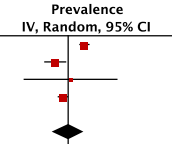
Heterogeneity: $Tau^2 = 1.77$; $Chi^2 = 60.16$, df = 3 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 8.22$ ($P < 0.00001$)



B

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	20288	975	31.3%	4.80 [4.51, 5.09]
Parente 2013 (2 rounds)	2395	70	28.4%	2.90 [2.23, 3.57]
Stegeman 2015 (3 rounds)	156	6	8.9%	3.90 [0.88, 6.92]
Tepes 2020 (3 rounds)	13969	469	31.3%	3.40 [3.10, 3.70]
Total (95% CI)	**36808**	**1520**	**100.0%**	**3.74 [2.68, 4.80]**

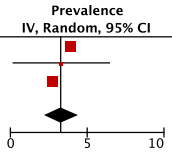
Heterogeneity: $Tau^2 = 0.91$; $Chi^2 = 54.25$, df = 3 ($P < 0.00001$); $I^2 = 94\%$
Test for overall effect: $Z = 6.90$ ($P < 0.00001$)



C

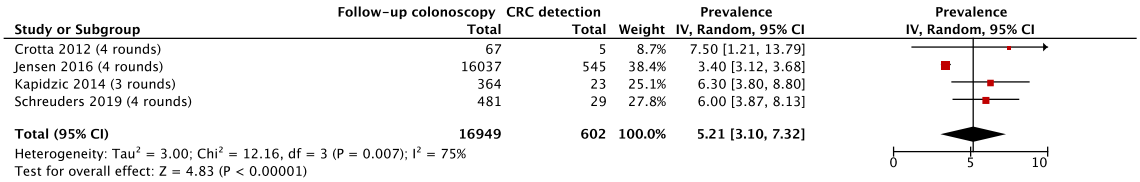
Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	18155	710	45.1%	3.90 [3.62, 4.18]
Stegeman 2015 (3 rounds)	123	4	9.7%	3.30 [0.17, 6.43]
Tepes 2020 (3 rounds)	14883	405	45.3%	2.70 [2.44, 2.96]
Total (95% CI)	**33161**	**1119**	**100.0%**	**3.30 [2.20, 4.40]**

Heterogeneity: $Tau^2 = 0.67$; $Chi^2 = 37.48$, df = 2 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 5.90$ ($P < 0.00001$)

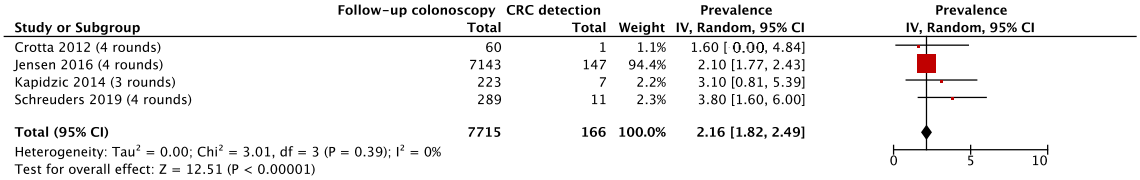


Supplementary figure 27. CRC PPV for the overall population participating in FIT-based organized CRC screening programs within closed cohorts from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

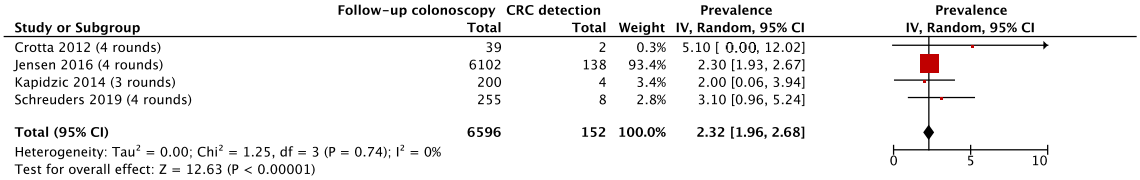
A



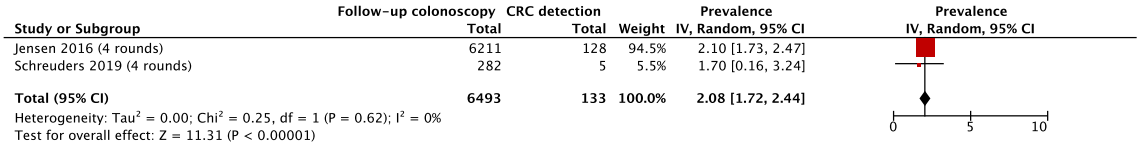
B



C

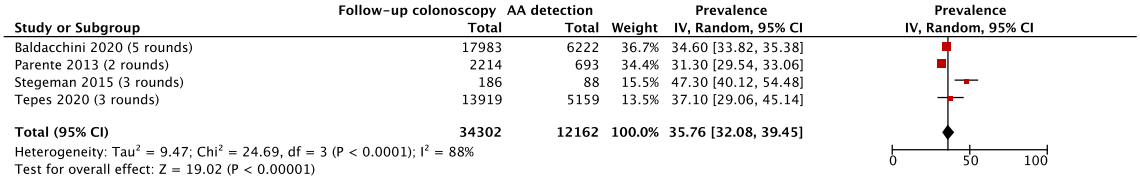


D

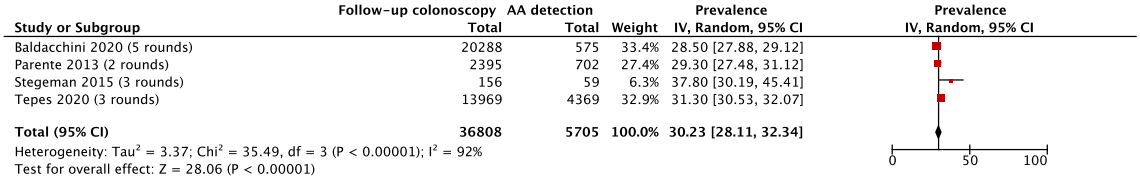


Supplementary figure 28. AA PPV for the overall population participating in FIT-based organized CRC screening programs within open cohorts from A) 1st round, B) 2nd round, to C) 3rd round.

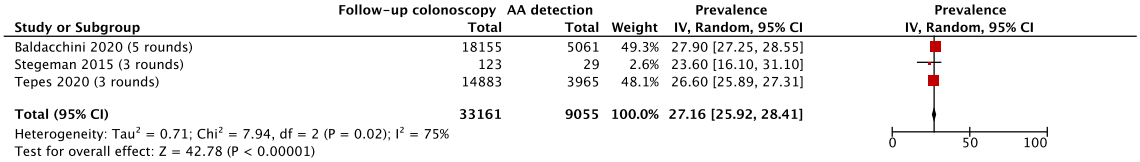
A



B

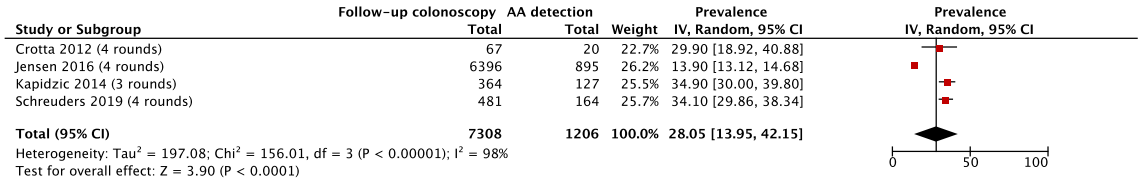


C

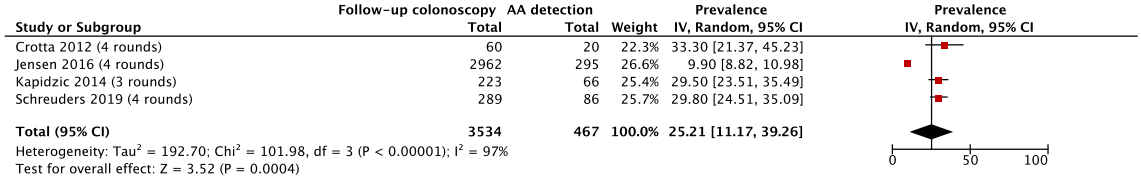


Supplementary figure 29. AA PPV for the overall population participating in FIT-based organized CRC screening programs within closed cohorts from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

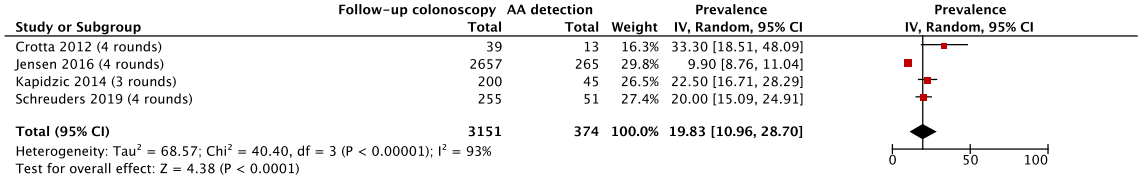
A



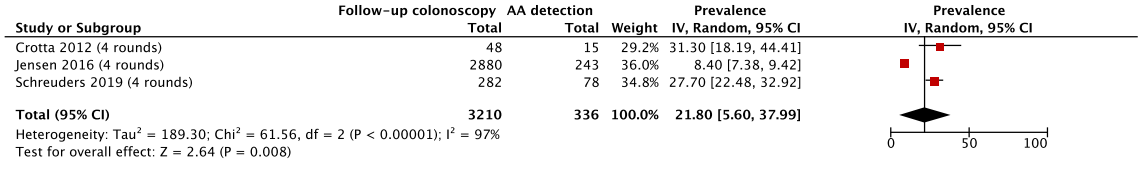
B



C

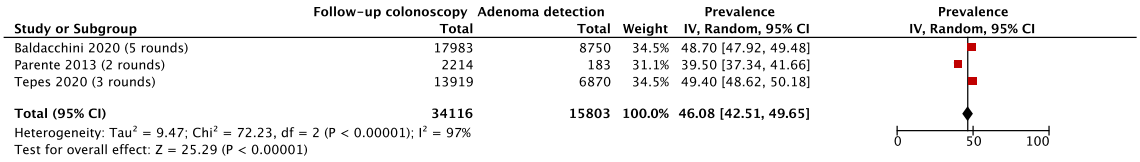


D

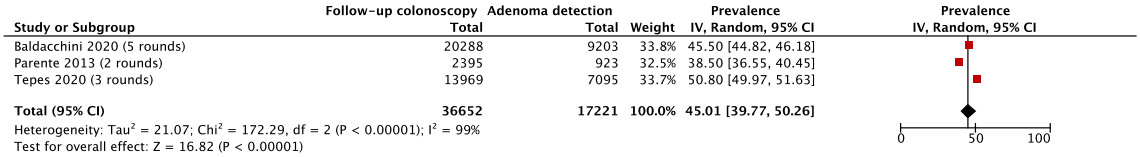


Supplementary figure 30. Adenoma PPV for the overall population participating in FIT-based organized CRC screening programs within open cohorts from A) 1st round, B) 2nd round, to C) 3rd round.

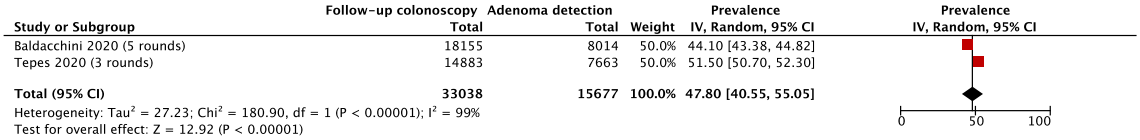
A



B

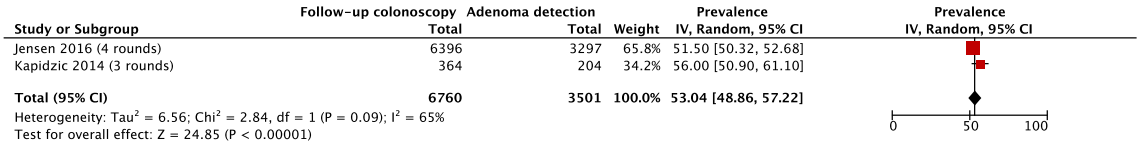


C

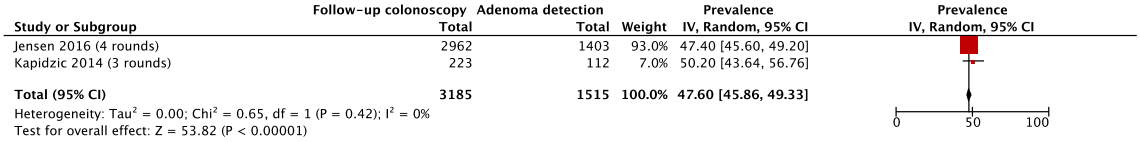


Supplementary figure 31. Adenoma PPV for the overall population participating in FIT-based organized CRC screening programs within closed cohorts from A) 1st round, B) 2nd round, to C) 3rd round.

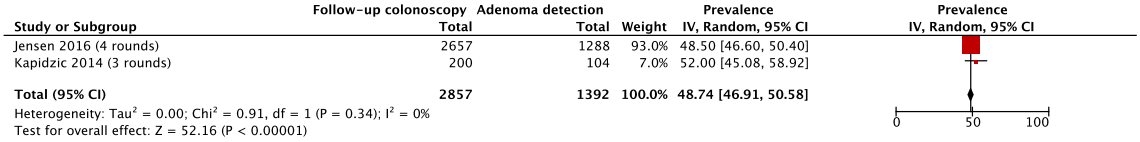
A



B

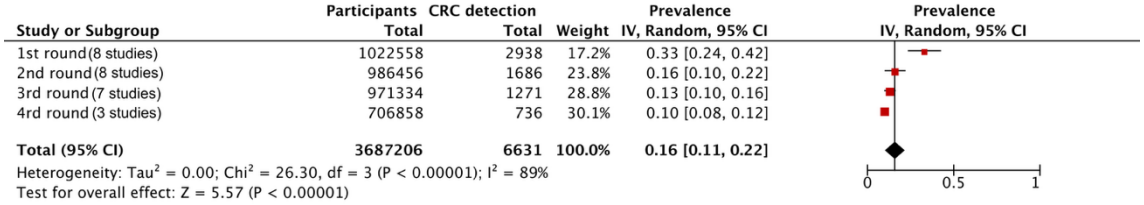


C

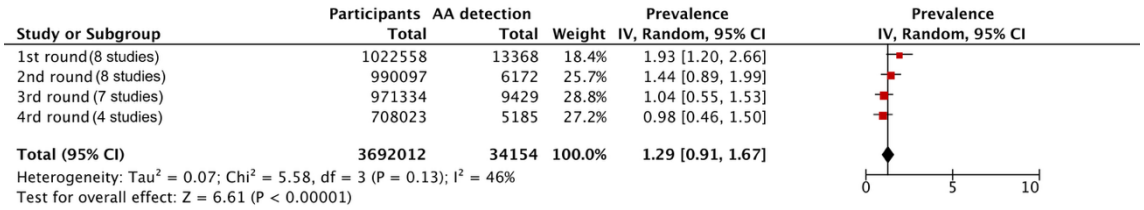


Supplementary figure 32. DR for CRC (A), AA (B) and adenoma (C) of FIT per round in the overall screening population participating in FIT-based organized screening programs.

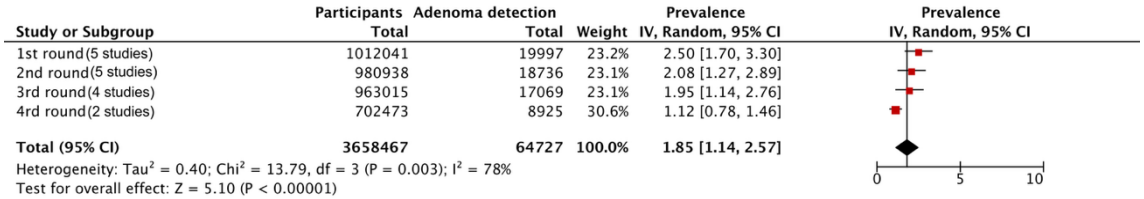
A



B

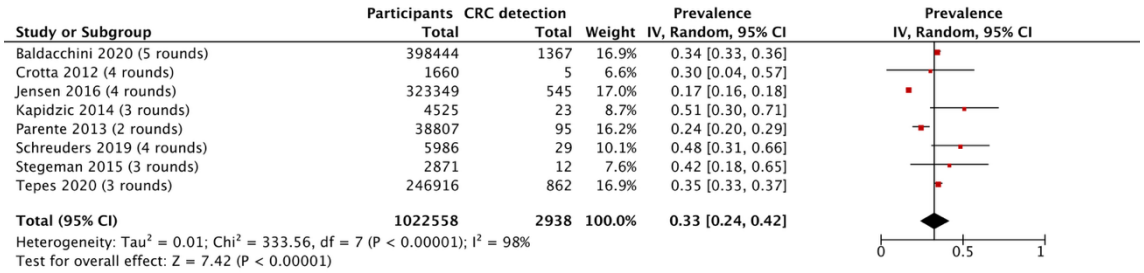


C

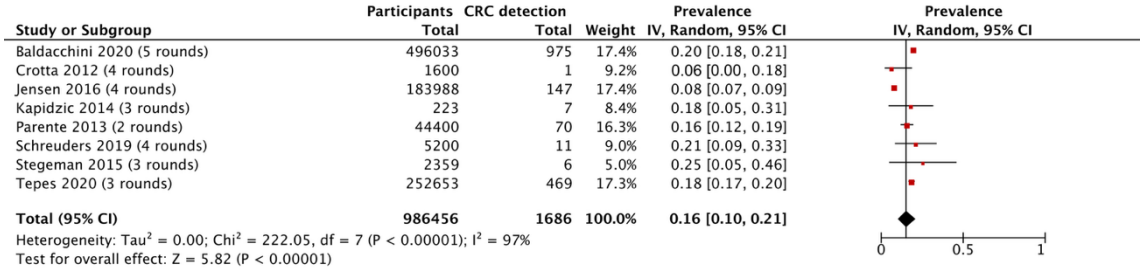


Supplementary figure 33. CRC DR for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

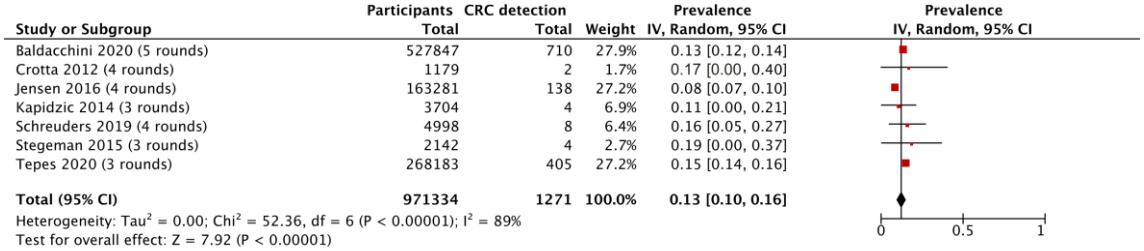
A



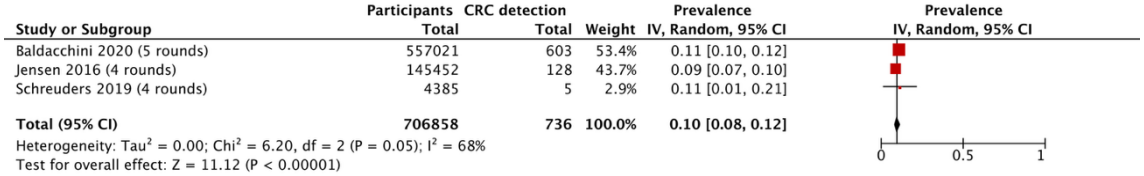
B



C

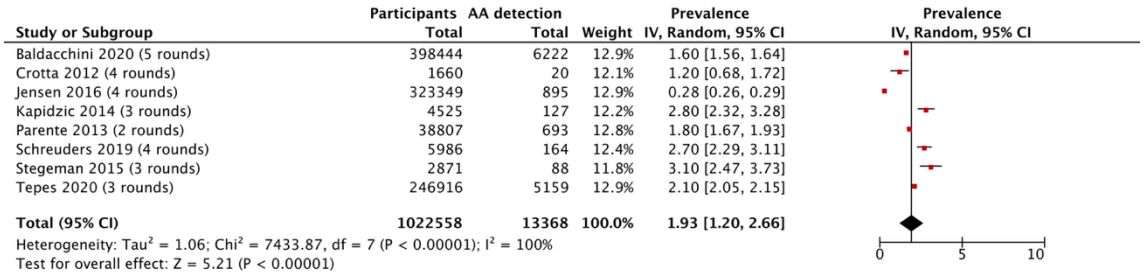


D

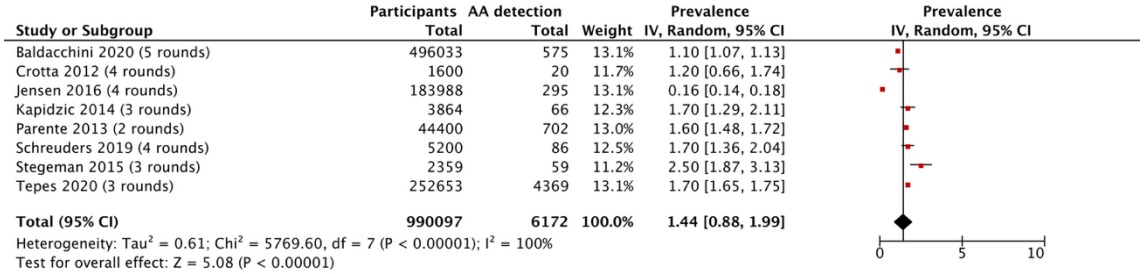


Supplementary figure 34. AA DR for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

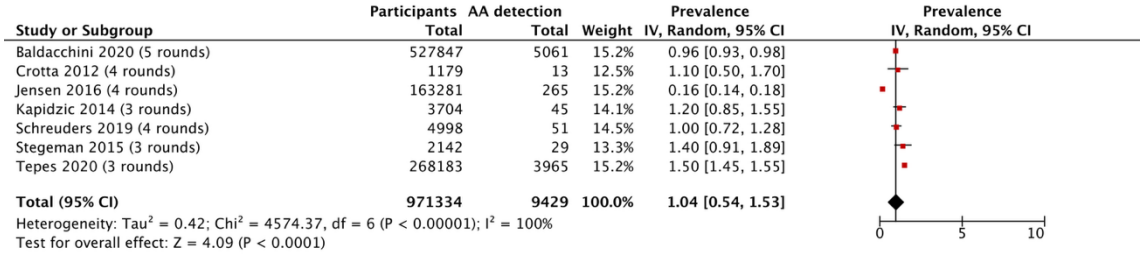
A



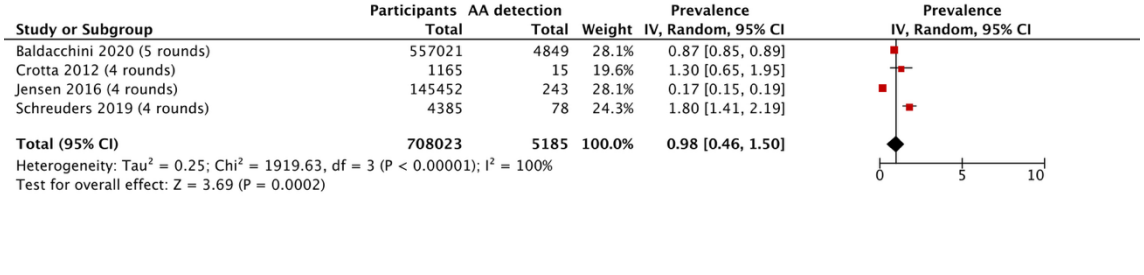
B



C



D

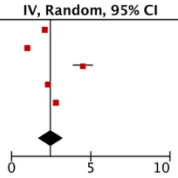


Supplementary figure 35. Adenoma DR for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	398444	8750	20.4%	2.10 [2.05, 2.15]
Jensen 2016 (4 rounds)	323349	3297	20.5%	1.00 [0.97, 1.03]
Kapidzic 2014 (3 rounds)	4525	204	18.3%	4.50 [3.90, 5.10]
Parente 2013 (2 rounds)	38807	876	20.3%	2.30 [2.15, 2.45]
Tepes 2020 (3 rounds)	246916	6870	20.4%	2.80 [2.74, 2.86]
Total (95% CI)	**1012041**	**19997**	**100.0%**	**2.50 [1.70, 3.30]**

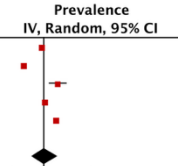
Heterogeneity: $Tau^2 = 0.82$; $Chi^2 = 3213.88$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 6.10$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	496033	9203	20.3%	1.90 [1.86, 1.94]
Jensen 2016 (4 rounds)	183988	1403	20.3%	0.76 [0.72, 0.80]
Kapidzic 2014 (3 rounds)	3864	112	18.7%	2.90 [2.37, 3.43]
Parente 2013 (2 rounds)	44400	923	20.2%	2.10 [1.97, 2.23]
Tepes 2020 (3 rounds)	252653	7095	20.3%	2.80 [2.74, 2.86]
Total (95% CI)	**980938**	**18736**	**100.0%**	**2.08 [1.27, 2.89]**

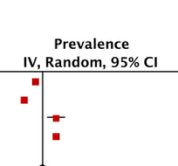
Heterogeneity: $Tau^2 = 0.83$; $Chi^2 = 3485.78$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 5.05$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	527847	8014	25.6%	1.50 [1.47, 1.53]
Jensen 2016 (4 rounds)	163281	1288	25.6%	0.79 [0.75, 0.83]
Kapidzic 2014 (3 rounds)	3704	104	23.1%	2.80 [2.27, 3.33]
Tepes 2020 (3 rounds)	268183	7663	25.6%	2.80 [2.74, 2.86]
Total (95% CI)	**963015**	**17069**	**100.0%**	**1.95 [1.14, 2.76]**

Heterogeneity: $Tau^2 = 0.67$; $Chi^2 = 2807.98$, df = 3 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 4.70$ ($P < 0.00001$)



D

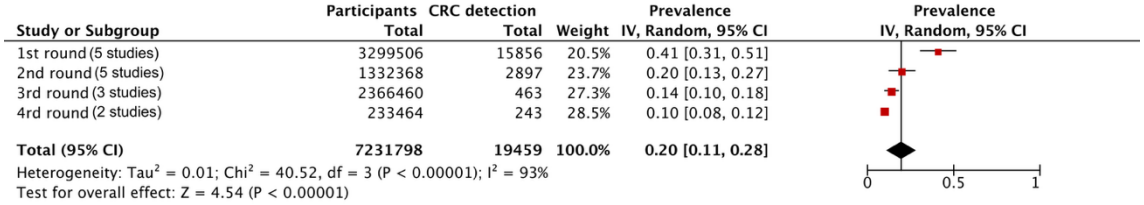
Study or Subgroup	Participants Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	557021	7545	50.2%	1.30 [1.27, 1.33]
Jensen 2016 (4 rounds)	145452	1380	49.8%	0.95 [0.90, 1.00]
Total (95% CI)	**702473**	**8925**	**100.0%**	**1.12 [0.78, 1.47]**

Heterogeneity: $Tau^2 = 0.06$; $Chi^2 = 145.77$, df = 1 ($P < 0.00001$); $I^2 = 99\%$
Test for overall effect: $Z = 6.39$ ($P < 0.00001$)

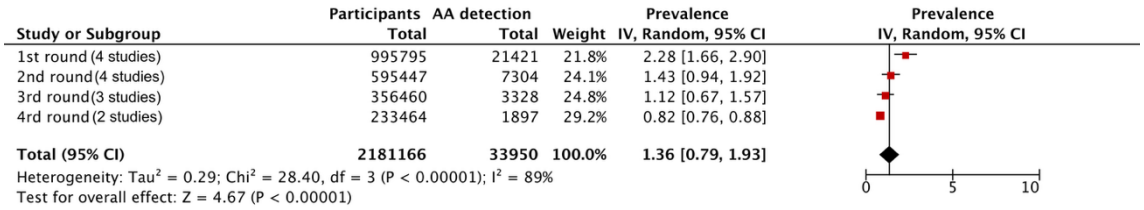


Supplementary figure 36. DR for CRC (A), AA (B) and adenoma (C) of FIT per round in the consistent screeners participating in FIT-based organized screening programs.

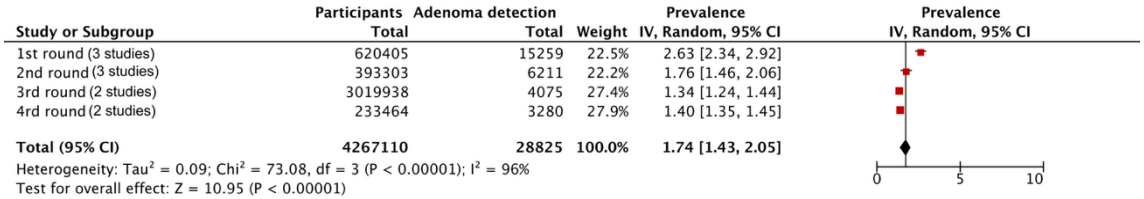
A



B



C

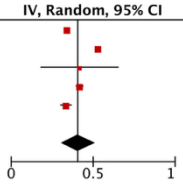


Supplementary figure 37. CRC DR for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	494187	1682	22.5%	0.34 [0.32, 0.36]
Breekveldt 2022 (2 rounds)	2302711	12183	22.5%	0.53 [0.52, 0.54]
Denters 2012 (2 rounds)	2871	12	10.5%	0.42 [0.18, 0.65]
Portillo 2018 (3 rounds)	376390	1566	22.4%	0.42 [0.40, 0.44]
Zorzi 2018 (6 rounds)	123347	413	22.1%	0.33 [0.30, 0.37]
Total (95% CI)	**3299506**	**15856**	**100.0%**	**0.41 [0.30, 0.51]**

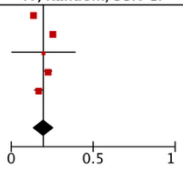
Heterogeneity: $Tau^2 = 0.01$; $Chi^2 = 578.36$, df = 4 ($P < 0.00001$); $I^2 = 99\%$
Test for overall effect: $Z = 7.64$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	295152	402	23.4%	0.14 [0.12, 0.15]
Breekveldt 2022 (2 rounds)	736921	1874	23.4%	0.25 [0.24, 0.26]
Denters 2012 (2 rounds)	2022	4	7.6%	0.20 [0.00, 0.39]
Portillo 2018 (3 rounds)	202144	455	22.9%	0.23 [0.20, 0.25]
Zorzi 2018 (6 rounds)	96129	162	22.7%	0.17 [0.14, 0.19]
Total (95% CI)	**1332368**	**2897**	**100.0%**	**0.20 [0.13, 0.26]**

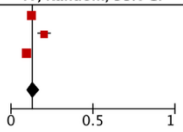
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 239.70$, df = 4 ($P < 0.00001$); $I^2 = 98\%$
Test for overall effect: $Z = 5.99$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	2233168	278	35.4%	0.13 [0.11, 0.14]
Portillo 2018 (3 rounds)	54522	110	27.5%	0.20 [0.16, 0.24]
Zorzi 2018 (6 rounds)	78770	75	37.0%	0.10 [0.09, 0.10]
Total (95% CI)	**2366460**	**463**	**100.0%**	**0.14 [0.10, 0.17]**

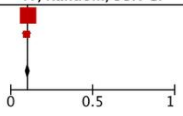
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 49.28$, df = 2 ($P < 0.00001$); $I^2 = 96\%$
Test for overall effect: $Z = 6.88$ ($P < 0.00001$)



D

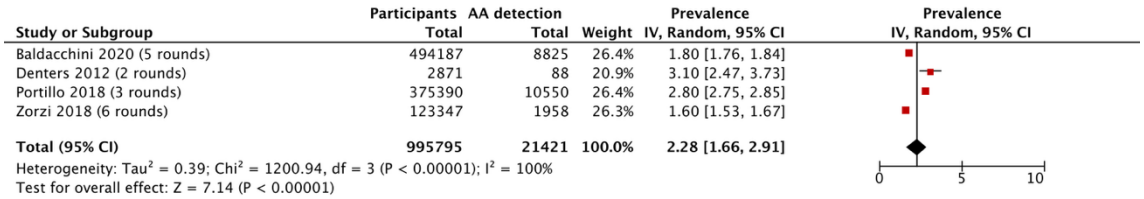
Study or Subgroup	Participants Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	170395	182	74.6%	0.11 [0.09, 0.12]
Zorzi 2018 (6 rounds)	63069	61	25.4%	0.10 [0.07, 0.12]
Total (95% CI)	**233464**	**243**	**100.0%**	**0.10 [0.09, 0.12]**

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.52$, df = 1 ($P = 0.47$); $I^2 = 0\%$
Test for overall effect: $Z = 17.11$ ($P < 0.00001$)

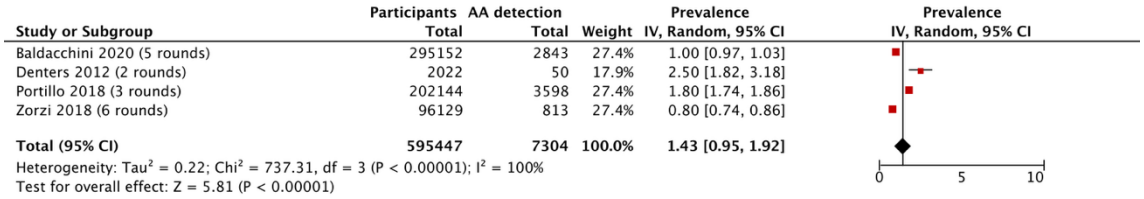


Supplementary figure 38. AA DR for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

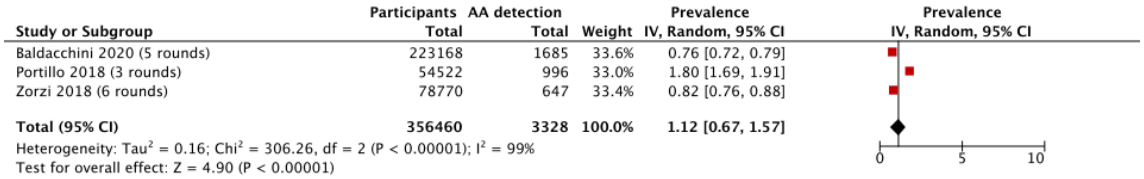
A



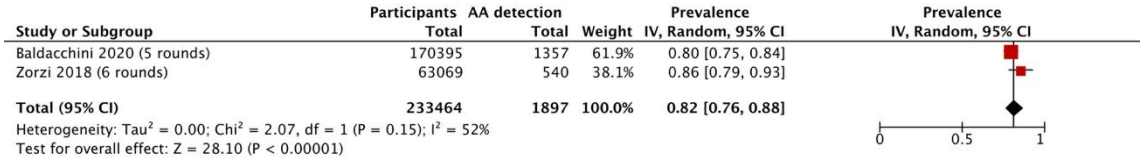
B



C

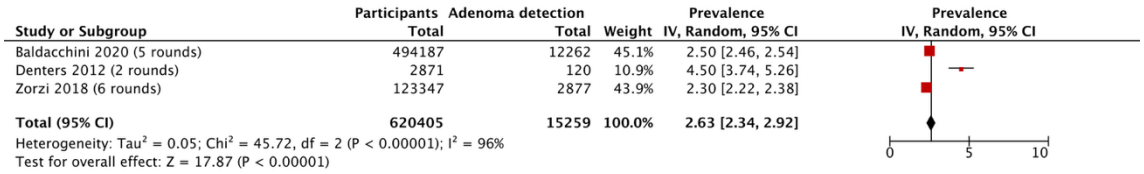


D

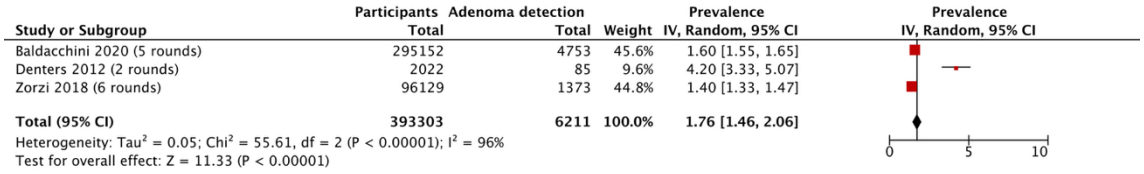


Supplementary figure 39. Adenoma DR for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

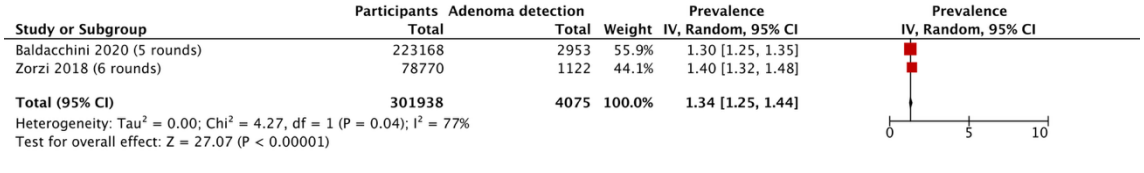
A



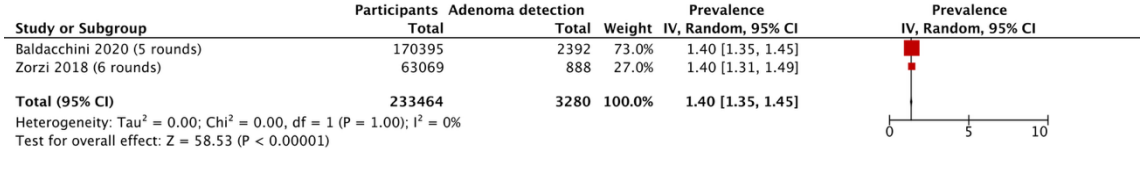
B



C



D



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Appendix 1. Search strategies in Pubmed, EMBASE and Cochrane databases.

PUBMED (NLM)

Search: ((((("Early Detection of Cancer"[Mesh]) OR "Prospective Studies"[Mesh]) OR (((((("Clinical Study" [Publication Type] OR "Observational Study" [Publication Type] OR "Clinical Trial" [Publication Type]) OR (("Follow-Up Studies"[Mesh]) OR "Multivariate Analysis"[Mesh])) OR (("multivariate analysis"[Title]) OR ("multivariate analysis"[Other Term]))) OR (("observational study"[Title]) OR ("observational study"[Other Term]))) OR (("clinical trial"[Title]) OR (clinical trial[Other Term])))) AND (((((("Mass Screening"[Mesh]) OR ("Patient Compliance"[Mesh] OR "Treatment Adherence and Compliance"[Mesh])) OR ("mass screening"[Title] OR "screening"[Title] OR "round?"[Title] OR "MULTIPLE ROUND?"[Title] OR "PARTICIPATION"[Title] OR "ADHERENCE" [Title] OR "ATTENDANCE"[Title] OR "YIELD"[Title])) OR ("mass screening" [Other Term] OR "screening" [Other Term] OR "round?"[Other Term] OR "PARTICIPATION"[Other Term] OR "ADHERENCE" [Other Term] OR "ATTENDANCE"[Other Term] OR "YIELD"[Other Term])) OR (((((("patient compliance" [Title])) OR (patient compliance [Other Term])) OR (compliance [Other Term])) OR (compliance [Title]))))) AND (((((((((("Colonoscopy"[Mesh]) OR ("Occult Blood"[Mesh])) OR ("Endoscopy"[Mesh])) OR ("lmmunochemistry"[Mesh])) OR (("FAECAL IMMUNOCHEMICAL TEST"[Title]) OR ("FECAL IMMUNOCHEMICAL TEST"[Title]))) OR ((((((("faecal haemoglobin" [Title]) OR ("fecal hemoglobin"[Title])) OR ("faecal occult blood" [Title])) OR ("fecal occult blood"[Title])) OR ("haemoccult"[Title])) OR ("hemoccult"[Title])) OR (guaiac[Title])) OR ("immunochemical testing"[Title]))) OR ((((((("faecal haemoglobin"[Other Term]) OR ("fecal hemoglobin"[Other Term])) OR ("faecal occult blood"[Other Term])) OR ("fecal occult blood"[Other Term])) OR ("hemoccult" [Other Term])) OR (guaiac[Other Term])) OR ("immunochemical testing"[Other Term]))) OR ((((colonoscopy[Title]) OR (endoscopy[Title])) OR (occult blood[Title])) OR (immunochemistry[Title]))) OR ((((colonoscopy[Other Term]) OR (endoscopy[Other Term)) OR (occult blood[Other Term))) OR (immunochemistry[Other Term)))) AND ((((((((((((((((((((((("Colonic Neoplasms"[Mesh]) OR ("Colorectal Neoplasms"[Mesh))) OR ("Rectal Neoplasms"[Mesh])) OR ("Intestinal Neoplasms"[Mesh])) OR ("COLON TUMOR"[Other Term])) OR ("COLORECTAL TUMOR"[Other Term])) OR ("RECTUM TUMOR"[Other Term])) OR ("COLORECTAL CANCER"[Other Term])) OR ("INTESTINE TUMOR"[Other Term])) OR ("COLORECTAL ADENOMA"[Other Term))) OR ("BOWEL CANCER" [Other Term])) OR ("COLORECTAL CANCER"[Title])) OR ("BOWEL CANCER"[Title])) OR ("COLORECTAL ADENOMA"[Text Word])) OR ("COLON TUMOR"[Title])) OR ("COLORECTAL TUMOR"[Title])) OR ("RECTUM TUMOR"[Title])) OR ("INTESTINE TUMOR"[Title])) OR ("INTESTINAL TUMOR"[Title])) OR ("RECTAL TUMOR"[Title])) OR ("RECTAL TUMOR"[Title])) OR ("RECTAL TUMOR"[Other Term])) OR ("INTESTINAL TUMOR"[Other Term])))) Filters: Humans, from 2012/1/1 - 2023/7/31 Sort by: Publication Date

EMBASE (Elsevier)

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#3 'colorectal tumor'/exp

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#6 'intestine tumor'/exp

#7 'intestine cancer'/exp

#8 'colorectal cancer cell line'/exp

#9 'colorectal cancer':ti OR 'colorectal cancer':kw

#10 'bowel cancer':ti OR 'bowel cancer':kw

#11 'colorectal tumor':ti OR 'colorectal tumor':kw

#12 'colorectal adenoma':ti OR 'colorectal adenoma':kw

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#14 'faecal haemoglobin':ti,kw OR 'fecal hemoglobin':ti,kw

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#16 'occult blood test'/exp

#17 'colonoscopy'/exp

#18 'endoscopy'/exp

#19 'immunochemistry'

#20 'occult blood'

#21 #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20

#22 #13AND #21

#23 'mass screening':ti,kw OR screening:ti,kw OR round:ti,kw OR 'population-based screening':ti,kw OR 'multiple round*':ti,kw OR participation*:ti,kw OR adherence:ti,kw OR attendance:ti,kw OR yield:ti,kw

#24 'protocol compliance'/exp

#25 'patient compliance'/exp OR 'patient compliance'

#26 #23 OR #24 OR #25

#27 'major clinical study'/exp OR 'major clinical stud*'

#28 'follow up'/exp OR 'follow up'

#29 'multivariate analysis'/exp OR 'multivariate analysis'

#30 #27 OR #28 OR #29

#31 #22 AND #26 ANO #30

#32 #31 AND (2012:py OR 2013:py OR 2014:py OR 2015:py OR 2016:py OR 2017:py OR 2018:py OR 2019:py OR 2020:py OR 2021:py OR 2022:py OR 2023:py) AND ('clinical trial'/de OR 'human'/de OR 'major clinical study'/de OR 'observational study'/de OR 'randomized controlled trial'/de) AND ([aged]/lim OR [middle aged]/lim) AND 'Article'/it

#33 #31 AND (2012:py OR 2013:py OR 2014:py OR 2015:py OR 2016:py OR 2017:py OR 2018:py OR 2019:py OR 2020:py OR 2021:py OR 2022:py OR 2023:py) AND ('clinical trial'/de OR 'human'/de OR 'major clinical study'/de OR 'observational study'/de OR 'randomized controlled trial'/de) AND ([aged]/lim OR [middle aged]/lim) AND 'article'/it AND (english:la OR spanish:la)

Cochrane Library

ID	Search	Hits
#1	MeSH descriptor: [Colonic Neoplasms] explode all trees	2270
#2	(colorectal NEXT tumor*):ti,ab,kw	772

#3 (COLON* NEXT TUMOR*):ti,ab,kw 216

#4 (RECTUM NEXT TUMOR*):ti,ab,kw 145

#5 (RECTAL NEXT TUMOR*):ti,ab,kw 107

#6 (COLORECTAL NEXT CANCER):ti,ab,kw 15319

#7 (BOWEL NEXT CANCER):ti,ab,kw 332

#8 (COLORECTAL NEXT ADENOMA):ti,ab,kw 592

#9 #1 RO #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 16132

#10 (IMMUNOCHEMICAL NEXT TEST*):ti,ab,kw 481

#11 (FECAL OR FAECAL OR FECES):ti,ab,kw 17298

#12 (OCCULT NEXT BLOOD):ti,ab,kw 1565

#13 ("haemoccult" OR HEMOCCULT):ti,ab,kw 127

#14 (HAEMOGLOBIN OR HEMOGLOBIN):ti,ab,kw 44578

#15 (COLONOSCOP*):ti,ab,kw 8700

#16 (ENDOSCOP*):ti,ab,kw 33252

#17 (IMMUNOCHEM*):ti,ab,kw 1359

#18 #10 OR #11 OR #12 OR #13 OR #14 OR |5 OR #16 OR #17 967344

#19 #9 AND #18 10656

#20 (MASS NEXT SCREENING):ti,ab,kw 4978

#21 (SCREENING):ti,ab,kw 72823

#22 ("round"):ti,ab,kw 2978

#23 ("population"):ti,ab,kw 145441

#24 (ROUNDS):ti,ab,kw 1937

#25 (PARTICIPATION):ti,ab,kw 40757

#26 (ADHERENCE):ti,ab,kw 44242

#27 ("attendance"):ti,ab,kw 8352

#28 (YIELD):ti,ab,kw 8646

#29 ("patient compliance"):ti,ab,kw 22411

#30 #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 295976

#31 #19 AND #30 3528

#32 ("clinical studies" OR "clinical study"):ti,ab,kw 353504

#33 MeSH descriptor: [Follow-Up Studies] explode all trees 67398

#34 ("follow up studies"):ti,ab,kw 68110

#35 ("multivariate analyses" or "multivariate analysis"):ti,ab,kw 20280

#36 (EARLY NEXT DETECTION):ti,ab,kw 4941

#37 MeSH descriptor: [Guideline Adherence] explode all trees 1383

#38 MeSH descriptor: [Observational Study] explode all trees 1538

#39 (observational NEXT stud*):ti,ab,kw 14566

#40 #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 433051

#41 #31 AND #40 with Cochrane Library publication date Between Jan 2012 and Jul 2023, in Cochrane Reviews, Cochrane Protocols, Trials 1573

#42 #41 with Cochrane Library publication date Between Jan 2012 and Jul 2023, in Cochrane Reviews, Cochrane Protocols 3

Appendix 2. Articles selected for full-text screening and reasons for exclusion.

Inadequate study design (1–3)

Insufficient reported data (4–50)

Non-FIT screening (51–80)

Opportunistic screening (81–92)

Overlapped populations (93–95)

1. Sali L, Ventura L, Mascalchi M, Falchini M, Mallardi B, Carozzi F, et al. Single CT colonography versus three rounds of faecal immunochemical test for population-based screening of colorectal cancer (SAVE): a randomised controlled trial. Lancet Gastroenterol Hepatol. 2022 Nov 1;7(11):1016–23.

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Supplementary table 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2,3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	5,6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	7, 8
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	7,8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	7, 8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	8, 9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	8, 9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	10, 11
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	10, 11
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	10, 11
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	10, 11
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	10, 11

Section and Topic	Item #	Checklist item	Location where item is reported
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	10
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	10
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	10
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	10
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	12, figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1, appendix 2
Study characteristics	17	Cite each included study and present its characteristics.	Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	S. Fig 1, S. Tables 3
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Suppl Figures 3-17
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	S. Fig 1, S table 3
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Figures 2-6, suppl figures 3-17
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Suppl figures 4,5,7, figure 5
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Suppl figures 4,5,7, figure 5
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	S Fig 1, S Table 3
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Fig 3-7
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	17-20
	23b	Discuss any limitations of the evidence included in the review.	21-22
	23c	Discuss any limitations of the review processes used.	21-22
	23d	Discuss implications of the results for practice, policy, and future research.	20-22

Section and Topic	Item #	Checklist item	Location where item is reported
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	7
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	23
Competing interests	26	Declare any competing interests of review authors.	23
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	23

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

Supplementary table 2. PICO (patient, intervention, comparison, and outcome) questions developed by the authors to identify the issues of greatest importance for the literature revision.

QUESTION	POPULATION	INTERVENTION	COMPARISON	OUTCOME
What is the uptake of participants over time in organized FIT-based CRC screening programs?	Persons eligible for an organized FIT program	Offer of FIT over multiple rounds	Not applicable	- % of participants among eligible for each screening round
What are the participation patterns of FIT adherence over time in organized CRC screening programs?	Persons eligible for an organized FIT program	Offer of FIT over multiple rounds	Not applicable	- % of population that are consistent participants (defined as every cycle) - % of population that are never attenders (defined as 0 cycle) - % of population that are intermittent participants (defined as >0 and <maximum)
What is the yield for adenoma, AA, and CRC over time in organized FIT-based CRC screening programs?	Persons eligible for an organized FIT program	Offer of FIT over multiple rounds	Not applicable	- Positivity rate - CRC detection rate - AA detection rate - Adenoma detection rate

FIT: faecal immunochemical test; CRC: colorectal cancer; AA: advanced adenoma.

Supplementary Table 3. Items for consideration in the Appraisal tool for Cross-Sectional Studies (AXIS)

	Question	Yes	No	Don't know/ Comment
Introduction				
1	Were the aims/objectives of the study clear?			
Methods				
2	Was the study design appropriate for the stated aim(s)?			
3	Was the sample size justified?			
4	Was the target/reference population clearly defined? (Is it clear who the research was about?)			
5	Was the sample frame taken from an appropriate population base so that it closely represented the target/reference population under investigation?			
6	Was the selection process likely to select subjects/participants that were representative of the target/reference population under investigation?			
7	Were measures undertaken to address and categorise non-responders?			
8	Were the risk factor and outcome variables measured appropriate to the aims of the study?			
9	Were the risk factor and outcome variables measured correctly using instruments/measurements that had been trialled, piloted or published previously?			
10	Is it clear what was used to determined statistical significance and/or precision estimates? (e.g. p-values, confidence intervals)			
11	Were the methods (including statistical methods) sufficiently described to enable them to be repeated?			
Results				
12	Were the basic data adequately described?			
13	Does the response rate raise concerns about non-response bias?			
14	If appropriate, was information about non-responders described?			
15	Were the results internally consistent?			
16	Were the results presented for all the analyses described in the methods?			
Discussion				
17	Were the authors' discussions and conclusions justified by the results?			
18	Were the limitations of the study discussed?			
Other				
19	Were there any funding sources or conflicts of interest that may affect the authors' interpretation of the results?			
20	Was ethical approval or consent of participants attained?			

Supplementary Table 4. CRC stage per round over time

Population	CRC stage	1st round n (%)	2nd round n (%)	3rd round n (%)
Overall population	I-II	2555 (72.7)	1364 (73.6)	807 (71.3)
	III-IV	959 (27.3)	489 (26.4)	325 (28.7)
	Total	3514 (100)	1853 (100)	1132
Consistent screeners	I-II	1228 (76.5)	268 (70.3)	193 (73.9)
	III-IV	376 (23.5)	113 (29.7)	68 (26.1)
	Total	1604 (100)	381 (100)	261 (100)

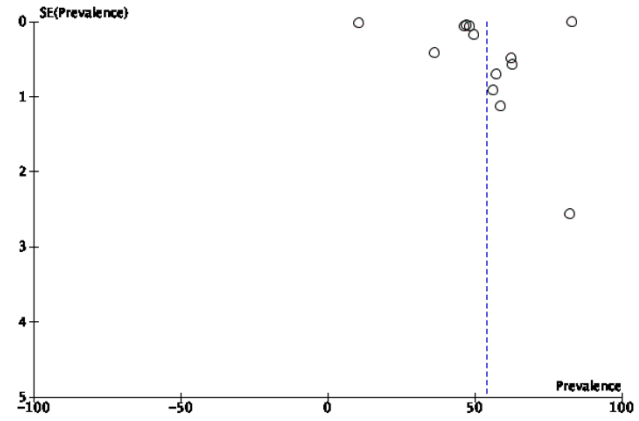
CRC: colorectal cancer

Supplementary figure 1. Risk of bias analysis. Authors' judgement about each risk of bias item presented as percentages across all included studies (A) and presented individually for each included study (B).

	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17	18	19	20
Duncan. 2014 [14]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Schreuders. 2019 [15]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	N/A
Breekveldt. 2023 [16]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Gordon. 2012 [17]	YES	YES	YES	YES	NO	NO	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	N/A
Osborne. 2017 [18]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES
Van Der Vlugt. 2017 [19]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO	YES
Stegeman. 2015 [20]	YES	YES	YES	YES	YES	YES	YES	NO	YES	NO	NO	NO	YES	YES	YES	YES	NO	N/A
Jensen. 2016 [21]	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	NO	NO	YES	YES	YES	YES	NO	N/A
Tepeš. 2007 [22]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	NO	NO	YES
Parente. 2013 [23]	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	NO	YES	YES	YES	NO	NO	N/A
Kapidzic. 2014 [24]	YES	YES	YES	YES	YES	YES	NO	YES	YES	NO	NO	NO	YES	YES	YES	YES	NO	YES
Portillo. 2018 [25]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Choi. 2012 [26]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	N/A
Heyman. 2024 [27]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Crotta. 2012 [28]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Baldacchini. 2021 [29]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Denters. 2013 [30]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO	YES
Benito. 2019 [31]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO	YES
Baldacchini. 2022 [32]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO	YES
Baker. 2015 [33]	YES	YES	YES	YES	YES	NO	YES	NO	YES	YES	NO	YES	YES	YES	YES	YES	NO	N/A
Cole. 2012 [34]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	NO	YES
Denters. 2012 [35]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	YES
Zorzi. 2018 [36]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES	YES	YES	YES	NO	N/A
Van Roon. 2013 [37]	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	NO	YES	YES	YES	YES	NO	YES

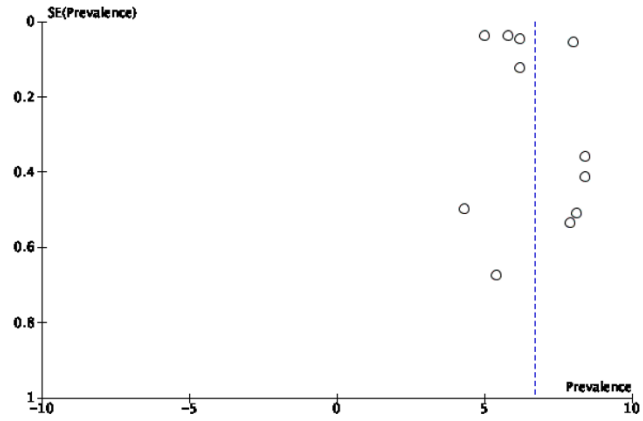
The Appraisal of cross-sectional studies (AXIS) of included manuscripts

Supplementary figure 2. Funnel plots assessing risk of publication bias.



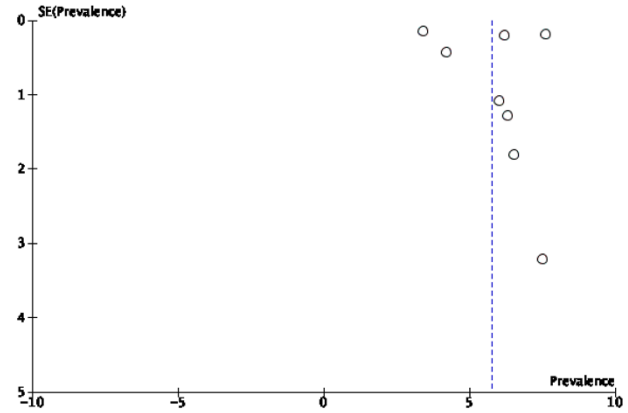
Caption

Funnel plot of comparison: Participation per round.



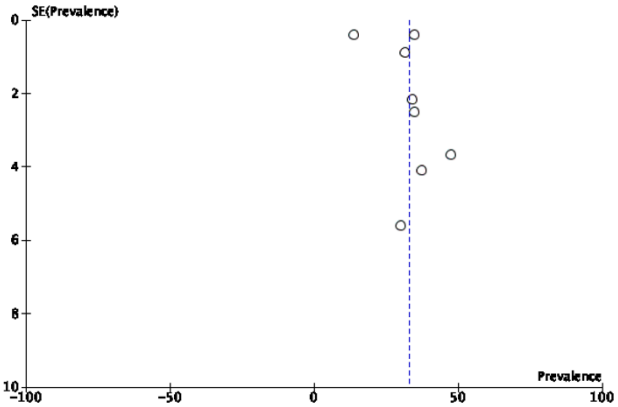
Caption

Funnel plot of comparison: Positivity rate per round – overall population.



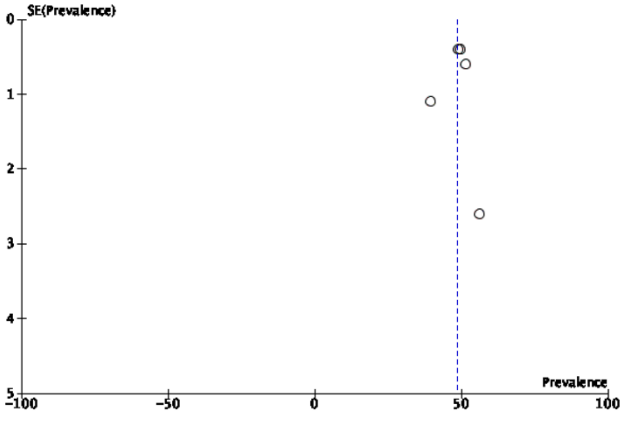
Caption

Funnel plot of comparison: CRC PPV per round – overall population.



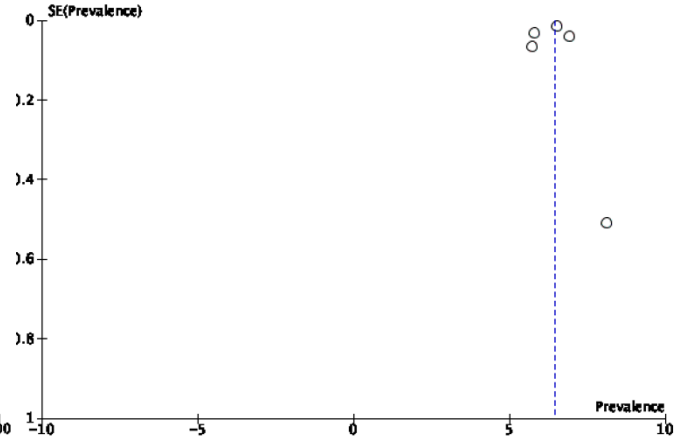
Caption

Funnel plot of comparison: AA PPV per round – overall population.



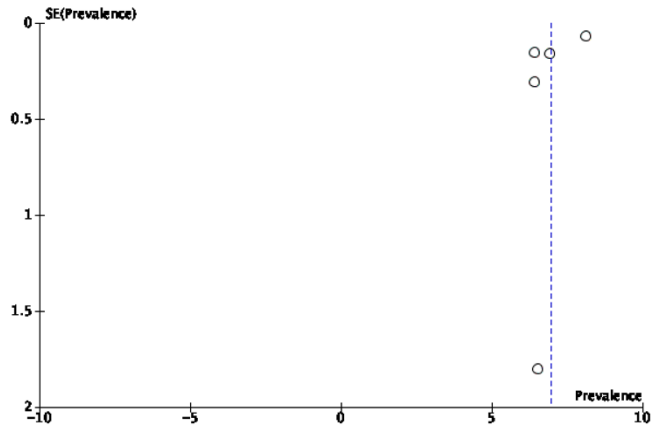
Caption

Funnel plot of comparison: Adenoma PPV per round – overall population.



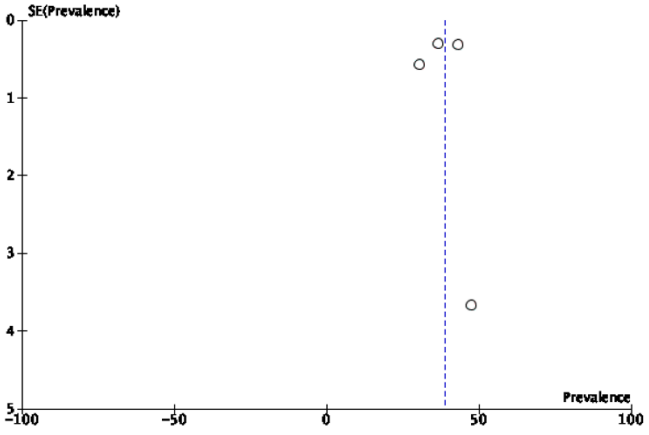
aption

unnel plot of comparison: Positivity rate per round – consistents.



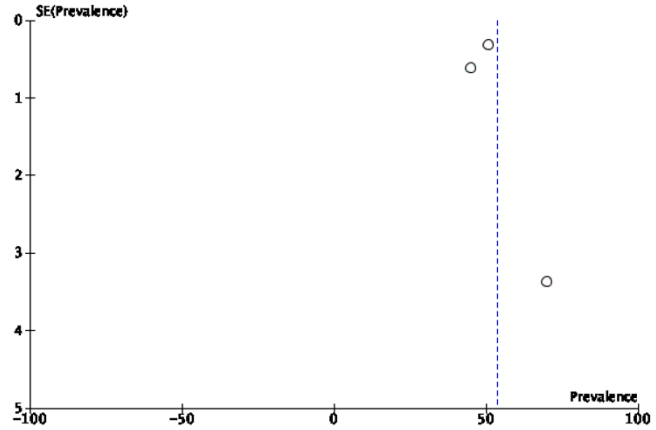
Caption

Funnel plot of comparison: CRC PPV per round – consistents.



Caption

Funnel plot of comparison: AA PPV per round – consistents.



Caption

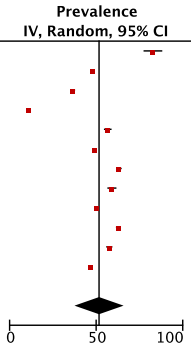
Funnel plot of comparison: Adenoma PPV per round – consistents.

Supplementary figure 3. Participation rates in FIT-based organized CRC screening programs in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

A

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baker 2015 (2 rounds)	225	185	8.3%	82.20 [77.20, 87.20]
Baldacchini 2020 (5 rounds)	845105	398444	8.3%	47.10 [46.99, 47.21]
Benito 2019 (4 rounds)	12727	4625	8.3%	36.00 [35.18, 36.82]
Choi 2012 (5 rounds)	2266656	238376	8.3%	10.50 [10.46, 10.54]
Crotta 2012 (4 rounds)	2959	1660	8.3%	56.10 [54.31, 57.89]
Jensen 2016 (4 rounds)	670841	323349	8.3%	48.20 [48.08, 48.32]
Kapidzic 2014 (3 rounds)	7229	4525	8.3%	62.60 [61.48, 63.72]
Osborne 2017 (4 rounds)	1928	1128	8.3%	58.50 [56.30, 60.70]
Parente 2013 (2 rounds)	78083	38807	8.3%	49.70 [49.35, 50.05]
Schreuders 2019 (4 rounds)	9623	5986	8.3%	62.20 [61.23, 63.17]
Stegeman 2015 (3 rounds)	5038	2871	8.3%	57.00 [55.63, 58.37]
Tepes 2020 (3 rounds)	533040	246916	8.3%	46.30 [46.17, 46.43]
Total (95% CI)	**4433454**	**1266872**	**100.0%**	**51.34 [37.34, 65.33]**

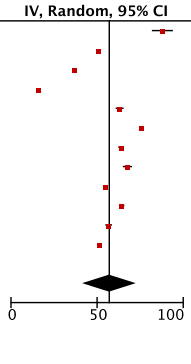
Heterogeneity: $Tau^2 = 611.00$; $Chi^2 = 888799.50$, df = 11 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 7.19 ($P < 0.00001$)



B

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baker 2015 (2 rounds)	133	116	8.3%	87.20 [81.52, 92.88]
Baldacchini 2020 (5 rounds)	990317	496033	8.3%	50.10 [50.00, 50.20]
Benito 2019 (4 rounds)	63824	22981	8.3%	36.00 [35.63, 36.37]
Choi 2012 (5 rounds)	3474440	534470	8.3%	15.40 [15.36, 15.44]
Crotta 2012 (4 rounds)	2566	1600	8.3%	62.40 [60.53, 64.27]
Jensen 2016 (4 rounds)	244476	183988	8.3%	75.30 [75.13, 75.47]
Kapidzic 2014 (3 rounds)	6111	3862	8.3%	63.20 [61.99, 64.41]
Osborne 2017 (4 rounds)	1677	1122	8.3%	66.90 [64.65, 69.15]
Parente 2013 (2 rounds)	81619	44400	8.3%	54.40 [54.06, 54.74]
Schreuders 2019 (4 rounds)	8158	5200	8.3%	63.70 [62.66, 64.74]
Stegeman 2015 (3 rounds)	4211	2359	8.3%	56.00 [54.50, 57.50]
Tepes 2020 (3 rounds)	500516	252653	8.3%	50.50 [50.36, 50.64]
Total (95% CI)	**5378048**	**1548784**	**100.0%**	**56.73 [41.29, 72.17]**

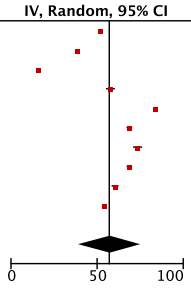
Heterogeneity: $Tau^2 = 743.66$; $Chi^2 = 1004966.94$, df = 11 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 7.20 ($P < 0.00001$)



C

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1018839	527847	10.0%	51.80 [51.70, 51.90]
Benito 2019 (4 rounds)	60906	23325	10.0%	38.30 [38.07, 38.53]
Choi 2012 (5 rounds)	4406826	693230	10.0%	15.70 [15.67, 15.73]
Crotta 2012 (4 rounds)	2056	1179	10.0%	57.30 [55.16, 59.44]
Jensen 2016 (4 rounds)	195693	163281	10.0%	83.40 [83.24, 83.56]
Kapidzic 2014 (3 rounds)	5423	3704	10.0%	68.30 [67.06, 69.54]
Osborne 2017 (4 rounds)	1550	1133	10.0%	73.10 [70.89, 75.31]
Schreuders 2019 (4 rounds)	7326	4998	10.0%	68.20 [67.13, 69.27]
Stegeman 2015 (3 rounds)	10050	3566	10.0%	60.00 [58.39, 61.61]
Tepes 2020 (3 rounds)	499279	268183	10.0%	53.70 [53.56, 53.84]
Total (95% CI)	**6207948**	**1690446**	**100.0%**	**56.98 [38.69, 75.26]**

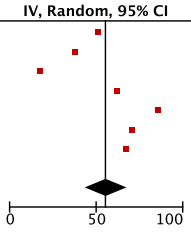
Heterogeneity: $Tau^2 = 870.28$; $Chi^2 = 1264185.24$, df = 9 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 6.11 ($P < 0.00001$)



D

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1084128	557021	14.3%	51.40 [51.31, 51.49]
Benito 2019 (4 rounds)	61428	23366	14.3%	38.30 [37.92, 38.68]
Choi 2012 (5 rounds)	4344969	780651	14.3%	18.00 [17.96, 18.04]
Crotta 2012 (4 rounds)	1862	1165	14.3%	62.60 [62.58, 62.62]
Jensen 2016 (4 rounds)	168853	145452	14.3%	86.10 [86.08, 86.12]
Osborne 2017 (4 rounds)	1487	1061	14.3%	71.40 [71.38, 71.42]
Schreuders 2019 (4 rounds)	6452	4385	14.3%	67.90 [67.89, 67.91]
Total (95% CI)	**5669179**	**1513101**	**100.0%**	**56.53 [44.43, 68.63]**

Heterogeneity: $Tau^2 = 266.81$; $Chi^2 = 13182387.74$, df = 6 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 9.16 ($P < 0.00001$)

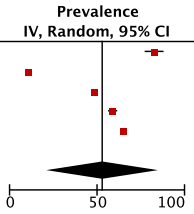


Supplementary figure 4. Participation rates in annual FIT-based organized CRC screening programs in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

A

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baker 2015 (2 rounds)	225	185	19.9%	82.20 [77.20, 87.20]
Choi 2012 (5 rounds)	2266656	238376	20.0%	10.50 [10.46, 10.54]
Jensen 2016 (4 rounds)	670841	323349	20.0%	48.20 [48.08, 48.32]
Osborne 2017 (4 rounds)	1928	1128	20.0%	58.50 [56.30, 60.70]
Van Roon 2013 (2 rounds)	2385	1543	20.0%	64.70 [64.68, 64.72]
Total (95% CI)	**2942035**	**564581**	**100.0%**	**52.79 [20.84, 84.74]**

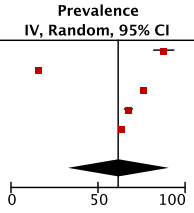
Heterogeneity: $Tau^2 = 1326.90$; $Chi^2 = 6121825.72$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 3.24 (P = 0.001)



B

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baker 2015 (2 rounds)	133	116	19.9%	87.20 [81.52, 92.88]
Choi 2012 (5 rounds)	3474440	534470	20.0%	15.40 [15.36, 15.44]
Jensen 2016 (4 rounds)	244476	183988	20.0%	75.30 [75.13, 75.47]
Osborne 2017 (4 rounds)	1677	1122	20.0%	66.90 [64.65, 69.15]
Van Roon 2013 (2 rounds)	2034	1286	20.0%	63.20 [63.18, 63.22]
Total (95% CI)	**3722760**	**720982**	**100.0%**	**61.56 [32.43, 90.69]**

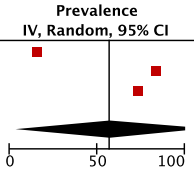
Heterogeneity: $Tau^2 = 1102.51$; $Chi^2 = 4704355.38$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 4.14 (P < 0.0001)



C

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Choi 2012 (5 rounds)	4406826	693230	33.3%	15.70 [15.67, 15.73]
Jensen 2016 (4 rounds)	195693	163281	33.3%	83.40 [83.24, 83.56]
Osborne 2017 (4 rounds)	1550	1133	33.3%	73.10 [70.89, 75.31]
Total (95% CI)	**4604069**	**857644**	**100.0%**	**57.40 [3.27, 111.52]**

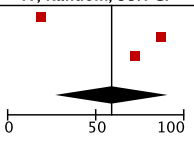
Heterogeneity: $Tau^2 = 2287.56$; $Chi^2 = 624939.57$, df = 2 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 2.08 (P = 0.04)



D

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Choi 2012 (5 rounds)	4344969	780651	33.3%	18.00 [17.96, 18.04]
Jensen 2016 (4 rounds)	168853	145452	33.3%	86.10 [86.08, 86.12]
Osborne 2017 (4 rounds)	1487	1061	33.3%	71.40 [71.38, 71.42]
Total (95% CI)	**4515309**	**927164**	**100.0%**	**58.50 [26.51, 90.49]**

Heterogeneity: $Tau^2 = 799.28$; $Chi^2 = 12021508.92$, df = 2 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: Z = 3.58 (P = 0.0003)

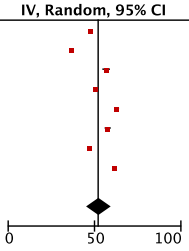


Supplementary figure 5. Participation rates in biennial FIT-based organized CRC screening programs in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

A

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	845105	398444	12.5%	47.10 [46.99, 47.21]
Benito 2019 (4 rounds)	12727	4625	12.5%	36.00 [35.18, 36.82]
Crotta 2012 (4 rounds)	2959	1660	12.4%	56.10 [54.31, 57.89]
Parente 2013 (2 rounds)	78083	38807	12.5%	49.70 [49.35, 50.05]
Schreuders 2019 (4 rounds)	9623	5986	12.5%	62.20 [61.23, 63.17]
Stegeman 2015 (3 rounds)	5038	2871	12.5%	57.00 [55.63, 58.37]
Tepes 2020 (3 rounds)	533040	246916	12.5%	46.30 [46.17, 46.43]
Van Roon 2013 (2 rounds)	2428	1481	12.5%	61.00 [60.98, 61.02]
Total (95% CI)	**1489003**	**700790**	**100.0%**	**51.92 [45.03, 58.81]**

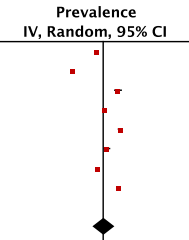
Heterogeneity: $Tau^2 = 98.69$; $Chi^2 = 115073.44$, df = 7 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 14.77$ ($P < 0.00001$)



B

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	990317	496033	12.5%	50.10 [50.00, 50.20]
Benito 2019 (4 rounds)	63824	22981	12.5%	36.00 [35.63, 36.37]
Crotta 2012 (4 rounds)	2566	1600	12.4%	62.40 [60.53, 64.27]
Parente 2013 (2 rounds)	81619	44400	12.5%	54.40 [54.06, 54.74]
Schreuders 2019 (4 rounds)	8158	5200	12.5%	63.70 [62.66, 64.74]
Stegeman 2015 (3 rounds)	4211	2359	12.4%	56.00 [54.50, 57.50]
Tepes 2020 (3 rounds)	500516	252653	12.5%	50.50 [50.36, 50.64]
Van Roon 2013 (2 rounds)	2048	1280	12.5%	62.50 [62.48, 62.52]
Total (95% CI)	**1653259**	**826506**	**100.0%**	**54.43 [48.12, 60.75]**

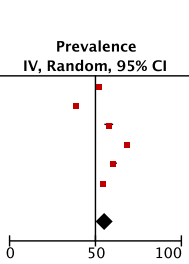
Heterogeneity: $Tau^2 = 82.74$; $Chi^2 = 103734.89$, df = 7 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 16.90$ ($P < 0.00001$)



C

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1018839	527847	16.8%	51.80 [51.70, 51.90]
Benito 2019 (4 rounds)	60906	23325	16.8%	38.30 [38.07, 38.53]
Crotta 2012 (4 rounds)	2056	1179	16.3%	57.30 [55.16, 59.44]
Schreuders 2019 (4 rounds)	7326	4998	16.7%	68.20 [67.13, 69.27]
Stegeman 2015 (3 rounds)	10050	3566	16.5%	60.00 [58.39, 61.61]
Tepes 2020 (3 rounds)	499279	268183	16.8%	53.70 [53.56, 53.84]
Total (95% CI)	**1598456**	**829098**	**100.0%**	**54.84 [49.91, 59.77]**

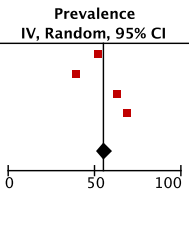
Heterogeneity: $Tau^2 = 37.67$; $Chi^2 = 14137.09$, df = 5 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 21.78$ ($P < 0.00001$)



D

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1084128	557021	25.0%	51.40 [51.31, 51.49]
Benito 2019 (4 rounds)	61428	23366	25.0%	38.30 [37.92, 38.68]
Crotta 2012 (4 rounds)	1862	1165	25.0%	62.60 [62.58, 62.62]
Schreuders 2019 (4 rounds)	6452	4385	25.0%	67.90 [67.89, 67.91]
Total (95% CI)	**1153870**	**585937**	**100.0%**	**55.06 [50.46, 59.65]**

Heterogeneity: $Tau^2 = 21.97$; $Chi^2 = 318191.43$, df = 3 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 23.49$ ($P < 0.00001$)

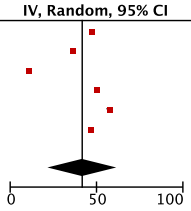


Supplementary figure 6. Participation rates in FIT-based organized CRC screening programs within open cohorts in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

A

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	845105	398444	16.7%	47.10 [46.99, 47.21]
Benito 2019 (4 rounds)	12727	4625	16.7%	36.00 [35.18, 36.82]
Choi 2012 (5 rounds)	2266656	238376	16.7%	10.50 [10.46, 10.54]
Parente 2013 (2 rounds)	78083	38807	16.7%	49.70 [49.35, 50.05]
Stegeman 2015 (3 rounds)	5038	2871	16.7%	57.00 [55.63, 58.37]
Tepes 2020 (3 rounds)	533040	246916	16.7%	46.30 [46.17, 46.43]
Total (95% CI)	**3740649**	**930039**	**100.0%**	**41.10 [21.13, 61.07]**

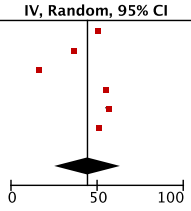
Heterogeneity: $Tau^2 = 622.78$; $Chi^2 = 639853.59$, df = 5 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 4.03$ ($P < 0.0001$)



B

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	990317	496033	16.7%	50.10 [50.00, 50.20]
Benito 2019 (4 rounds)	63824	22981	16.7%	36.00 [35.63, 36.37]
Choi 2012 (5 rounds)	3474440	534470	16.7%	15.40 [15.36, 15.44]
Parente 2013 (2 rounds)	81619	44400	16.7%	54.40 [54.06, 54.74]
Stegeman 2015 (3 rounds)	4211	2359	16.7%	56.00 [54.50, 57.50]
Tepes 2020 (3 rounds)	500516	252653	16.7%	50.50 [50.36, 50.64]
Total (95% CI)	**5114927**	**1352896**	**100.0%**	**43.73 [24.76, 62.70]**

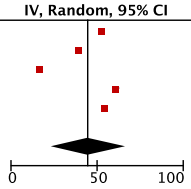
Heterogeneity: $Tau^2 = 562.09$; $Chi^2 = 633429.29$, df = 5 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 4.52$ ($P < 0.00001$)



C

Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1018839	527847	20.0%	51.80 [51.70, 51.90]
Benito 2019 (4 rounds)	60906	23325	20.0%	38.30 [38.07, 38.53]
Choi 2012 (5 rounds)	4406826	693230	20.0%	15.70 [15.67, 15.73]
Stegeman 2015 (3 rounds)	10050	3566	20.0%	60.00 [58.39, 61.61]
Tepes 2020 (3 rounds)	499279	268183	20.0%	53.70 [53.56, 53.84]
Total (95% CI)	**5995900**	**1516151**	**100.0%**	**43.90 [22.39, 65.41]**

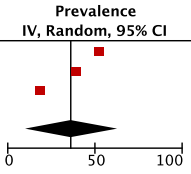
Heterogeneity: $Tau^2 = 602.09$; $Chi^2 = 726389.75$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 4.00$ ($P < 0.0001$)



D

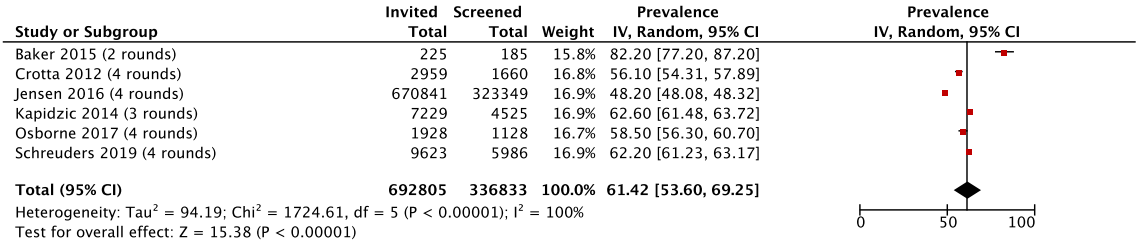
Study or Subgroup	Invited Total	Screened Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	1084128	557021	33.3%	51.40 [51.31, 51.49]
Benito 2019 (4 rounds)	61428	23366	33.3%	38.30 [37.92, 38.68]
Choi 2012 (5 rounds)	4344969	780651	33.3%	18.00 [17.96, 18.04]
Total (95% CI)	**5490525**	**1361038**	**100.0%**	**35.90 [9.74, 62.06]**

Heterogeneity: $Tau^2 = 534.31$; $Chi^2 = 431256.59$, df = 2 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 2.69$ ($P = 0.007$)

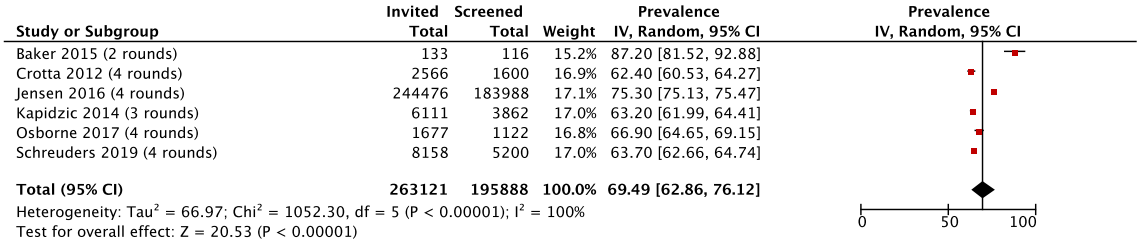


Supplementary figure 7. Participation rates in FIT-based organized CRC screening programs within closed cohorts in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

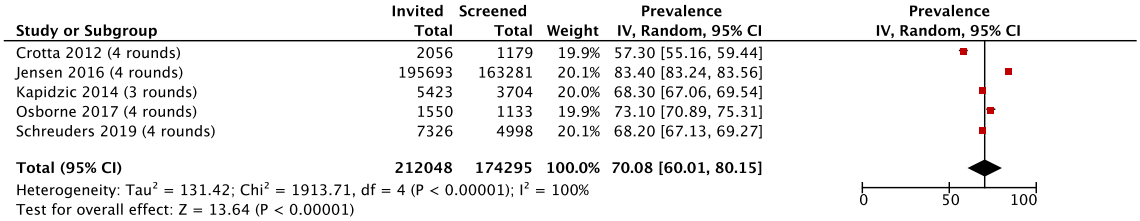
A



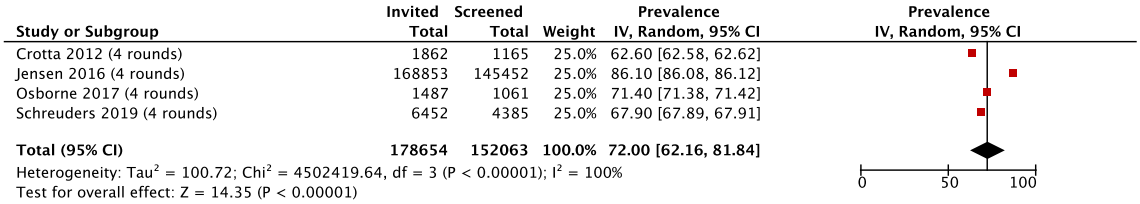
B



C

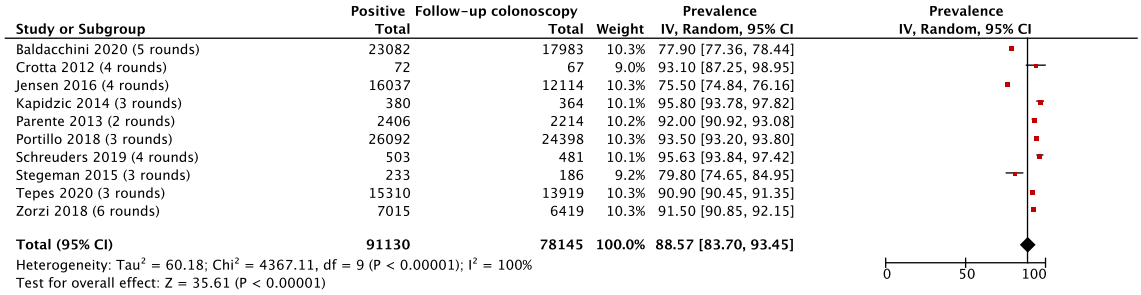


D

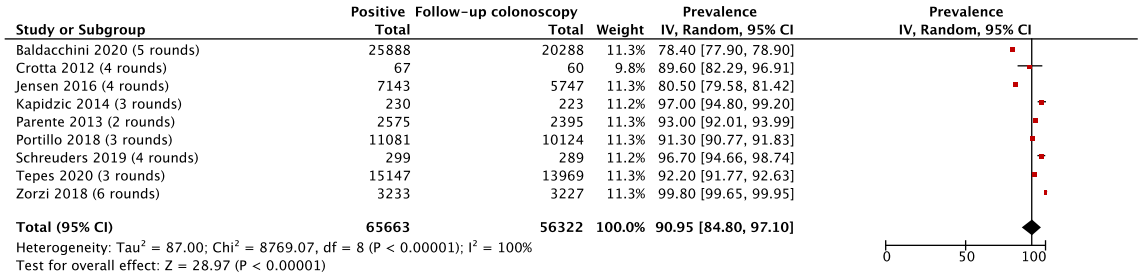


Supplementary figure 8. Follow-up colonoscopy completion rates in FIT-based organized CRC screening programs in A) 1st round, B) 2nd round, C) 3rd round, D) 4th round.

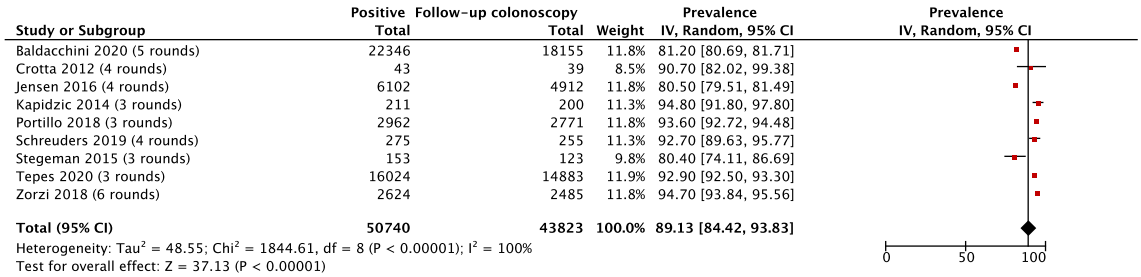
A



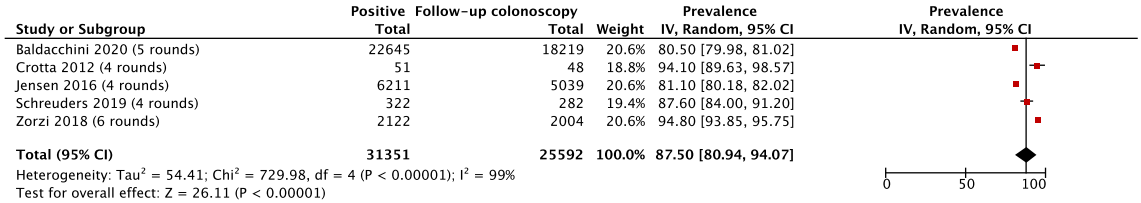
B



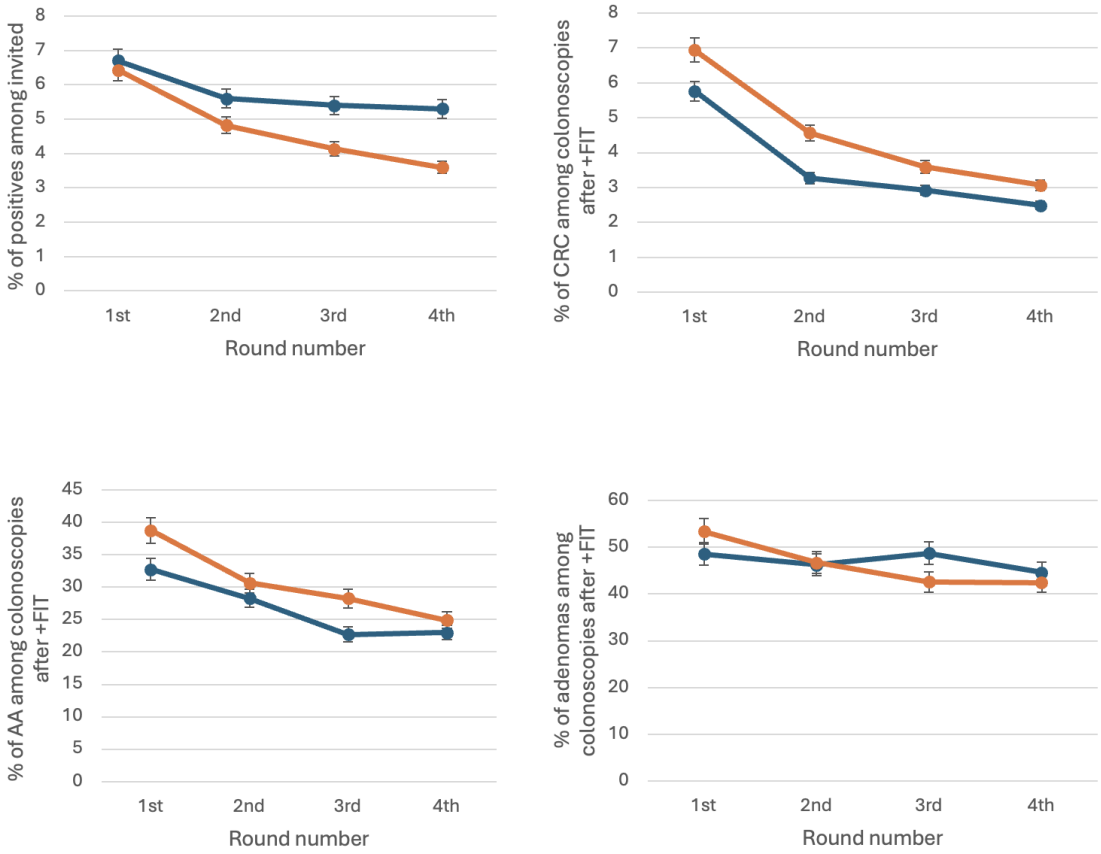
C



D



Supplementary figure 9. Comparison of the behaviors of positivity rate (A) and PPV for CRC (B), AA (C) and adenoma (D) of FIT across rounds in overall (blue) vs consistent (orange) screeners participating in FIT-based organized screening programs.



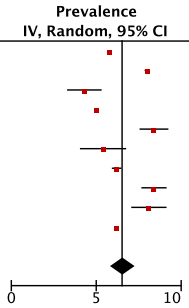
PPV: positive predictive value; CRC: colorectal cancer; AA: advanced adenoma; FIT: fecal immunochemical test.

Supplementary figure 10. Positivity rates for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	398444	23082	10.9%	5.79 [5.72, 5.86]
Choi 2012 (5 rounds)	238376	19070	10.9%	8.00 [7.89, 8.11]
Crotta 2012 (4 rounds)	1660	72	9.1%	4.30 [3.32, 5.28]
Jensen 2016 (4 rounds)	323349	16037	10.9%	5.00 [4.93, 5.07]
Kapidzic 2014 (3 rounds)	4525	380	9.6%	8.40 [7.59, 9.21]
Osborne 2017 (4 rounds)	1128	61	8.0%	5.40 [4.08, 6.72]
Parente 2013 (2 rounds)	38807	2406	10.8%	6.20 [5.96, 6.44]
Schreuders 2019 (4 rounds)	5986	503	9.9%	8.40 [7.70, 9.10]
Stegeman 2015 (3 rounds)	2871	233	9.0%	8.10 [7.10, 9.10]
Tepes 2020 (3 rounds)	246916	15310	10.9%	6.20 [6.11, 6.29]
Total (95% CI)	**1262062**	**77154**	**100.0%**	**6.58 [5.86, 7.31]**

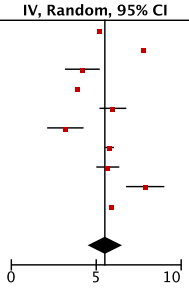
Heterogeneity: $Tau^2 = 1.25$; $Chi^2 = 2126.15$, df = 9 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 17.81$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	496033	25888	10.4%	5.20 [5.14, 5.26]
Choi 2012 (5 rounds)	534470	41688	10.4%	7.80 [7.73, 7.87]
Crotta 2012 (4 rounds)	1600	67	9.5%	4.20 [3.22, 5.18]
Jensen 2016 (4 rounds)	183988	7143	10.4%	3.90 [3.81, 3.99]
Kapidzic 2014 (3 rounds)	3864	230	9.8%	6.00 [5.25, 6.75]
Osborne 2017 (4 rounds)	1122	37	9.4%	3.20 [2.17, 4.23]
Parente 2013 (2 rounds)	44400	2575	10.4%	5.80 [5.58, 6.02]
Schreuders 2019 (4 rounds)	5200	299	10.0%	5.70 [5.07, 6.33]
Stegeman 2015 (3 rounds)	2359	186	9.3%	7.90 [6.81, 8.99]
Tepes 2020 (3 rounds)	252653	15147	10.4%	5.90 [5.81, 5.99]
Total (95% CI)	**1525689**	**93260**	**100.0%**	**5.57 [4.57, 6.56]**

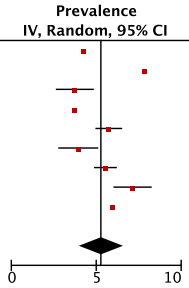
Heterogeneity: $Tau^2 = 2.48$; $Chi^2 = 5109.80$, df = 9 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 10.96$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	527847	22346	11.5%	4.20 [4.15, 4.25]
Choi 2012 (5 rounds)	693239	54072	11.5%	7.80 [7.74, 7.86]
Crotta 2012 (4 rounds)	1179	43	10.7%	3.70 [2.62, 4.78]
Jensen 2016 (4 rounds)	163281	6102	11.5%	3.70 [3.61, 3.79]
Kapidzic 2014 (3 rounds)	3704	211	11.1%	5.70 [4.95, 6.45]
Osborne 2017 (4 rounds)	1133	45	10.6%	3.90 [2.76, 5.04]
Schreuders 2019 (4 rounds)	4998	275	11.2%	5.50 [4.87, 6.13]
Stegeman 2015 (3 rounds)	2142	153	10.6%	7.10 [6.01, 8.19]
Tepes 2020 (3 rounds)	268183	16024	11.5%	5.90 [5.81, 5.99]
Total (95% CI)	**1665706**	**99271**	**100.0%**	**5.29 [3.97, 6.60]**

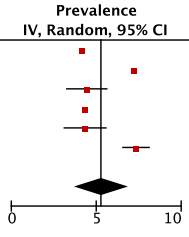
Heterogeneity: $Tau^2 = 3.92$; $Chi^2 = 8953.50$, df = 8 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 7.88$ ($P < 0.00001$)



D

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	557021	22645	17.3%	4.10 [4.05, 4.15]
Choi 2012 (5 rounds)	780651	56207	17.3%	7.20 [7.14, 7.26]
Crotta 2012 (4 rounds)	1165	51	15.8%	4.40 [3.22, 5.58]
Jensen 2016 (4 rounds)	145452	6211	17.3%	4.30 [4.20, 4.40]
Osborne 2017 (4 rounds)	1061	46	15.7%	4.30 [3.08, 5.52]
Schreuders 2019 (4 rounds)	4385	322	16.6%	7.30 [6.53, 8.07]
Total (95% CI)	**1489735**	**85482**	**100.0%**	**5.28 [3.70, 6.86]**

Heterogeneity: $Tau^2 = 3.76$; $Chi^2 = 6630.53$, df = 5 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 6.54$ ($P < 0.00001$)

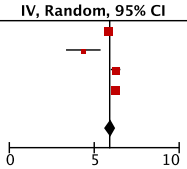


Supplementary figure 11. Positivity rates for the overall population participating in biennial FIT-based organized CRC screening programs with a cut-off of 20 µg Hb/g feces from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	398444	23082	32.0%	5.79 [5.72, 5.86]
Crotta 2012 (4 rounds)	1660	72	8.6%	4.30 [3.32, 5.28]
Parente 2013 (2 rounds)	38807	2406	27.8%	6.20 [5.96, 6.44]
Tepes 2020 (3 rounds)	246916	15310	31.6%	6.20 [6.11, 6.29]
Total (95% CI)	**685827**	**40870**	**100.0%**	**5.91 [5.57, 6.24]**

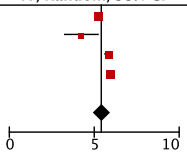
Heterogeneity: $Tau^2 = 0.09$; $Chi^2 = 61.00$, df = 3 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 34.65$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	496033	25888	29.3%	5.20 [5.14, 5.26]
Crotta 2012 (4 rounds)	1600	67	13.8%	4.20 [3.22, 5.18]
Parente 2013 (2 rounds)	44400	2575	27.8%	5.80 [5.58, 6.02]
Tepes 2020 (3 rounds)	252653	15147	29.1%	5.90 [5.81, 5.99]
Total (95% CI)	**794686**	**43677**	**100.0%**	**5.43 [4.93, 5.93]**

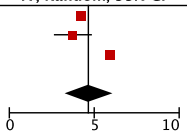
Heterogeneity: $Tau^2 = 0.22$; $Chi^2 = 168.68$, df = 3 ($P < 0.00001$); $I^2 = 98\%$
Test for overall effect: $Z = 21.25$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	527847	22346	35.4%	4.20 [4.15, 4.25]
Crotta 2012 (4 rounds)	1179	43	29.3%	3.70 [2.62, 4.78]
Tepes 2020 (3 rounds)	268183	16024	35.4%	5.90 [5.81, 5.99]
Total (95% CI)	**797209**	**38413**	**100.0%**	**4.65 [3.26, 6.05]**

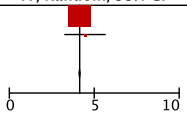
Heterogeneity: $Tau^2 = 1.44$; $Chi^2 = 999.60$, df = 2 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 6.53$ ($P < 0.00001$)



D

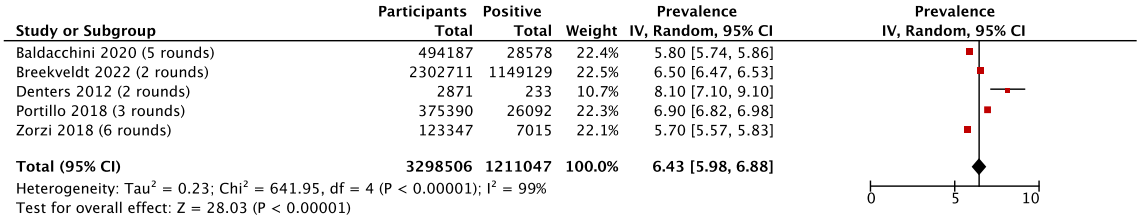
Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	557021	22645	99.8%	4.10 [4.05, 4.15]
Crotta 2012 (4 rounds)	1165	51	0.2%	4.40 [3.22, 5.58]
Total (95% CI)	**558186**	**22696**	**100.0%**	**4.10 [4.05, 4.15]**

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.25$, df = 1 ($P = 0.62$); $I^2 = 0\%$
Test for overall effect: $Z = 152.03$ ($P < 0.00001$)

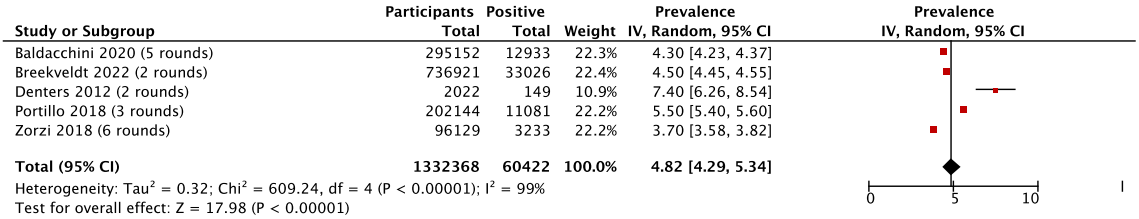


Supplementary figure 12. Positivity rates for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

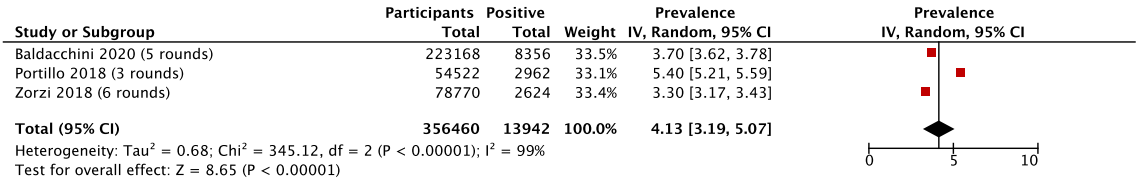
A



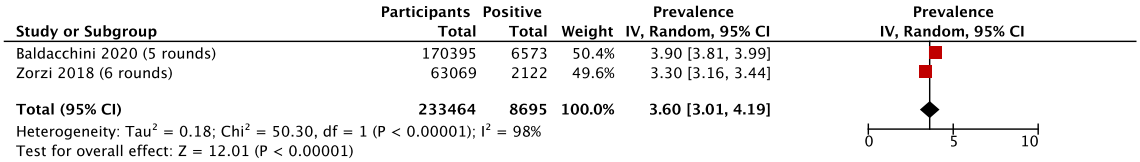
B



C



D

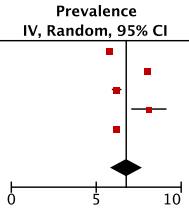


Supplementary figure 13. Positivity rates for the overall population participating in FIT-based organized CRC screening programs within open cohorts from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	398444	23082	20.9%	5.79 [5.72, 5.86]
Choi 2012 (5 rounds)	238376	19070	20.8%	8.00 [7.89, 8.11]
Parente 2013 (2 rounds)	38807	2406	20.6%	6.20 [5.96, 6.44]
Stegeman 2015 (3 rounds)	2871	233	16.9%	8.10 [7.10, 9.10]
Tepes 2020 (3 rounds)	246916	15310	20.8%	6.20 [6.11, 6.29]
Total (95% CI)	**925414**	**60101**	**100.0%**	**6.81 [5.87, 7.75]**

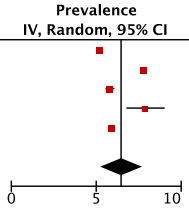
Heterogeneity: $Tau^2 = 1.09$; $Chi^2 = 1118.79$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 14.27$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	496033	25888	20.6%	5.20 [5.14, 5.26]
Choi 2012 (5 rounds)	534470	41688	20.6%	7.80 [7.73, 7.87]
Parente 2013 (2 rounds)	44400	2575	20.5%	5.80 [5.58, 6.02]
Stegeman 2015 (3 rounds)	2359	186	17.8%	7.90 [6.81, 8.99]
Tepes 2020 (3 rounds)	252653	15147	20.6%	5.90 [5.81, 5.99]
Total (95% CI)	**1329915**	**85484**	**100.0%**	**6.48 [5.24, 7.73]**

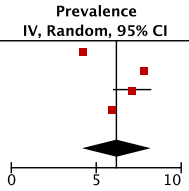
Heterogeneity: $Tau^2 = 1.95$; $Chi^2 = 2909.27$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 10.22$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	527847	22346	25.4%	4.20 [4.15, 4.25]
Choi 2012 (5 rounds)	693239	54072	25.4%	7.80 [7.74, 7.86]
Stegeman 2015 (3 rounds)	2142	153	23.7%	7.10 [6.01, 8.19]
Tepes 2020 (3 rounds)	268183	16024	25.4%	5.90 [5.81, 5.99]
Total (95% CI)	**1491411**	**92595**	**100.0%**	**6.24 [4.20, 8.27]**

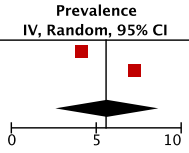
Heterogeneity: $Tau^2 = 4.22$; $Chi^2 = 7181.29$, df = 3 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 6.02$ ($P < 0.00001$)



D

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	557021	22645	50.0%	4.10 [4.05, 4.15]
Choi 2012 (5 rounds)	780651	56207	50.0%	7.20 [7.14, 7.26]
Total (95% CI)	**1337672**	**78852**	**100.0%**	**5.65 [2.61, 8.69]**

Heterogeneity: $Tau^2 = 4.80$; $Chi^2 = 6121.02$, df = 1 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 3.65$ ($P = 0.0003$)

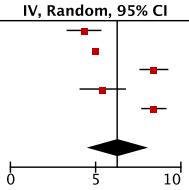


Supplementary figure 14. Positivity rates for the overall population participating in FIT-based organized CRC screening programs within closed cohorts from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Crotta 2012 (4 rounds)	1660	72	19.7%	4.30 [3.32, 5.28]
Jensen 2016 (4 rounds)	323349	16037	21.0%	5.00 [4.93, 5.07]
Kapidzic 2014 (3 rounds)	4525	380	20.1%	8.40 [7.59, 9.21]
Osborne 2017 (4 rounds)	1128	61	18.8%	5.40 [4.08, 6.72]
Schreuders 2019 (4 rounds)	5986	503	20.3%	8.40 [7.70, 9.10]
Total (95% CI)	**336648**	**17053**	**100.0%**	**6.31 [4.53, 8.09]**

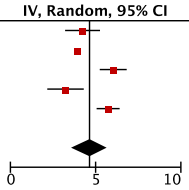
Heterogeneity: $Tau^2 = 3.93$; $Chi^2 = 157.83$, df = 4 ($P < 0.00001$); $I^2 = 97\%$
Test for overall effect: $Z = 6.95$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Crotta 2012 (4 rounds)	1600	67	18.6%	4.20 [3.22, 5.18]
Jensen 2016 (4 rounds)	183988	7143	22.4%	3.90 [3.81, 3.99]
Kapidzic 2014 (3 rounds)	3864	230	20.0%	6.00 [5.25, 6.75]
Osborne 2017 (4 rounds)	1122	37	18.3%	3.20 [2.17, 4.23]
Schreuders 2019 (4 rounds)	5200	299	20.7%	5.70 [5.07, 6.33]
Total (95% CI)	**195774**	**7776**	**100.0%**	**4.62 [3.59, 5.65]**

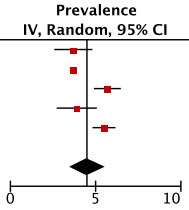
Heterogeneity: $Tau^2 = 1.24$; $Chi^2 = 61.64$, df = 4 ($P < 0.00001$); $I^2 = 94\%$
Test for overall effect: $Z = 8.78$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Crotta 2012 (4 rounds)	1179	43	18.2%	3.70 [2.62, 4.78]
Jensen 2016 (4 rounds)	163281	6102	22.7%	3.70 [3.61, 3.79]
Kapidzic 2014 (3 rounds)	3704	211	20.3%	5.70 [4.95, 6.45]
Osborne 2017 (4 rounds)	1133	45	17.8%	3.90 [2.76, 5.04]
Schreuders 2019 (4 rounds)	4998	275	21.0%	5.50 [4.87, 6.13]
Total (95% CI)	**174295**	**6676**	**100.0%**	**4.52 [3.50, 5.54]**

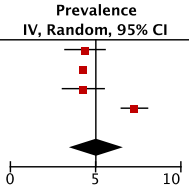
Heterogeneity: $Tau^2 = 1.19$; $Chi^2 = 56.89$, df = 4 ($P < 0.00001$); $I^2 = 93\%$
Test for overall effect: $Z = 8.66$ ($P < 0.00001$)



D

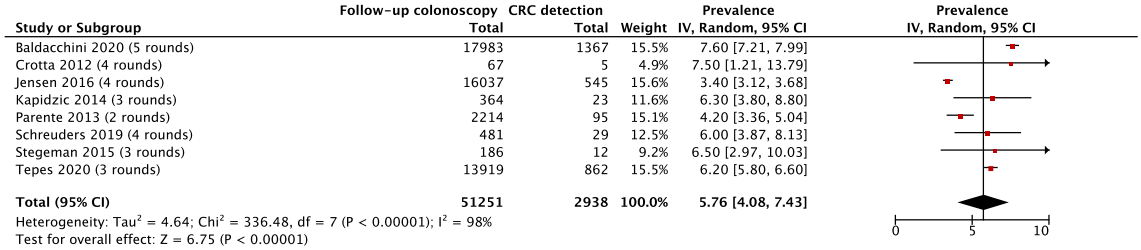
Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
Crotta 2012 (4 rounds)	1165	51	23.7%	4.40 [3.22, 5.58]
Jensen 2016 (4 rounds)	145452	6211	27.3%	4.30 [4.20, 4.40]
Osborne 2017 (4 rounds)	1061	46	23.4%	4.30 [3.08, 5.52]
Schreuders 2019 (4 rounds)	4385	322	25.6%	7.30 [6.53, 8.07]
Total (95% CI)	**152063**	**6630**	**100.0%**	**5.09 [3.52, 6.66]**

Heterogeneity: $Tau^2 = 2.35$; $Chi^2 = 57.24$, df = 3 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 6.36$ ($P < 0.00001$)

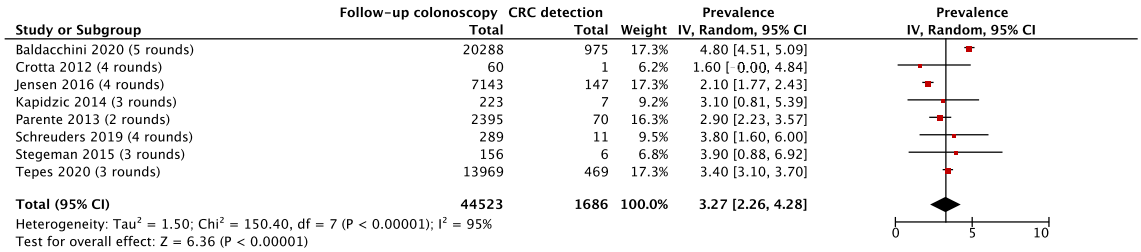


Supplementary figure 15. CRC PPV for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

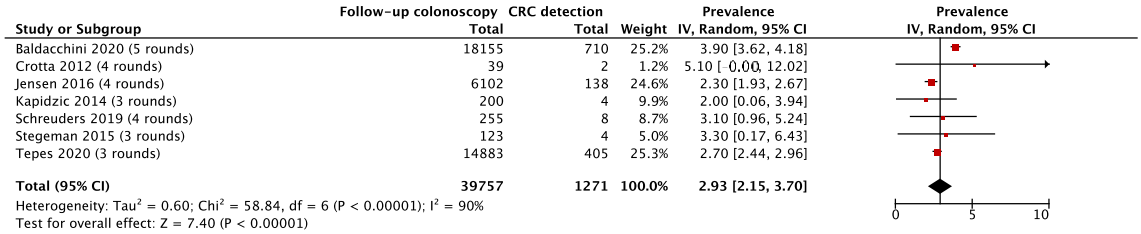
A



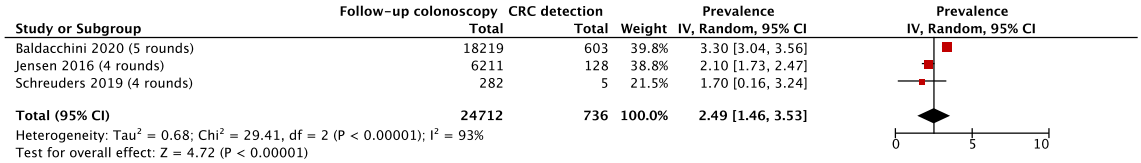
B



C

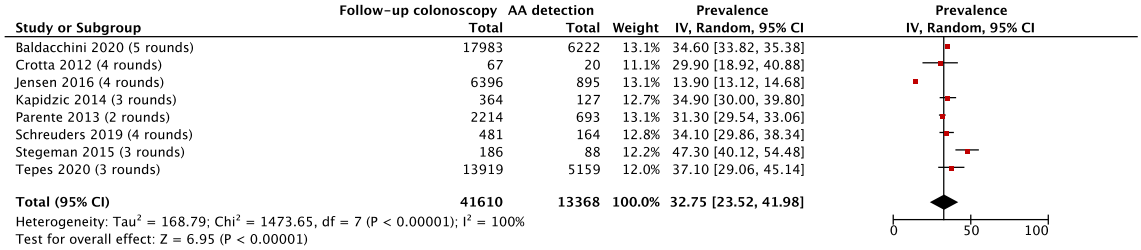


D

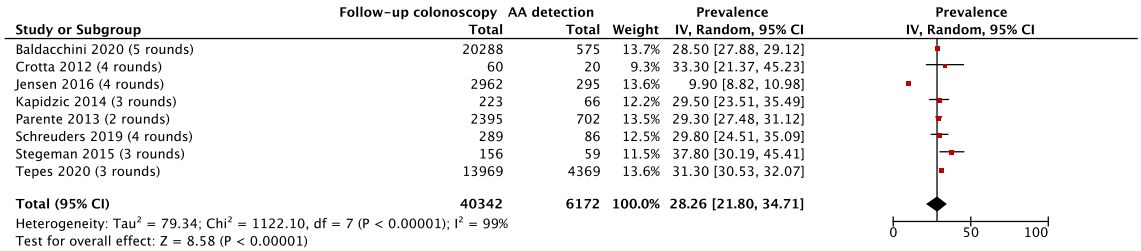


Supplementary figure 16. AA PPV for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

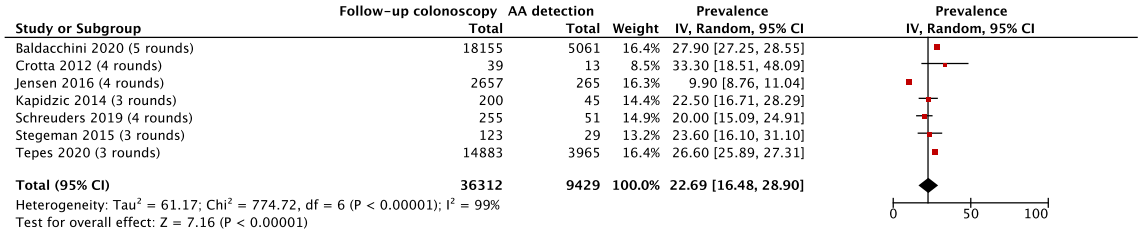
A



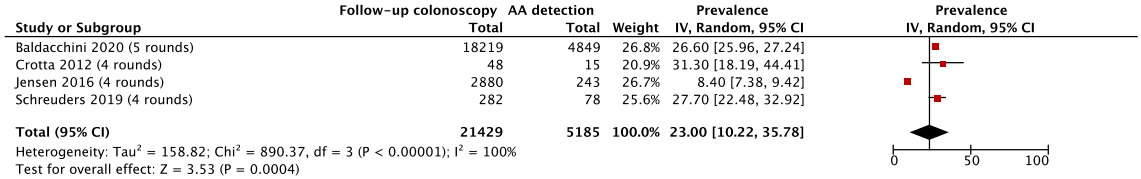
B



C

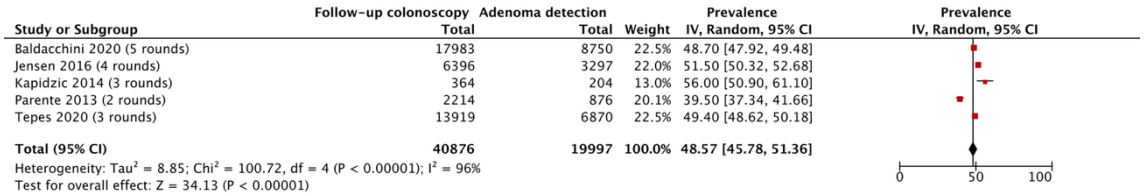


D

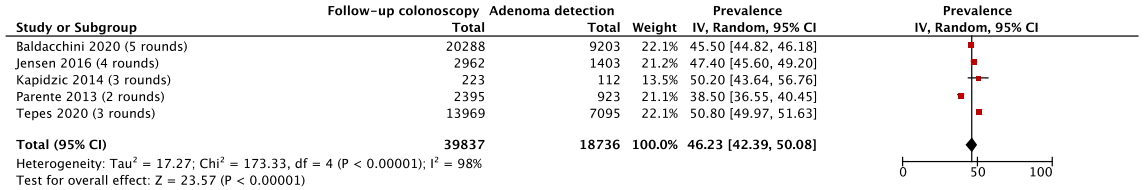


Supplementary figure 17. Adenoma PPV for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

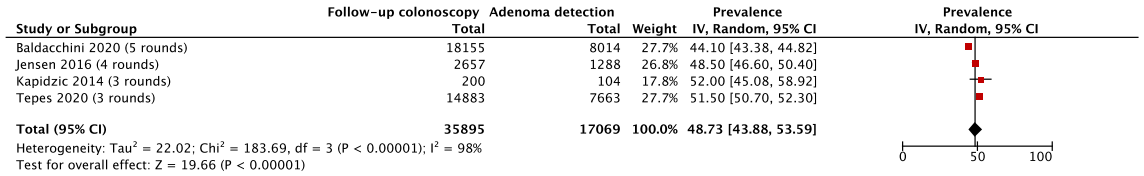
A



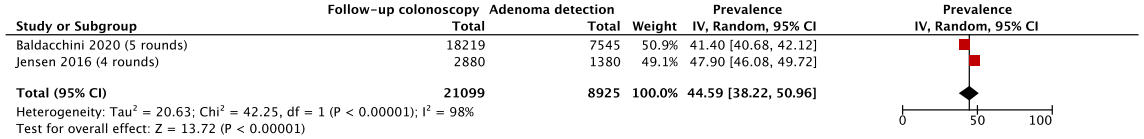
B



C



D

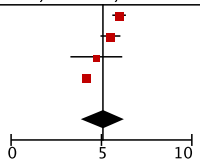


Supplementary figure 18. Positivity rate (A) and PPV for CRC (B), AA (C) and adenoma (D) of FIT per round in the overall screening population participating in biennial FIT-based organized screening programs with a cut-off of 20 µg Hb/g feces.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
1st round (4 studies)	685827	40870	26.8%	5.91 [5.58, 6.24]
2nd round (4 studies)	794686	43677	26.0%	5.43 [4.92, 5.94]
3rd round (3 studies)	797209	38413	19.8%	4.65 [3.26, 6.04]
4rd round (2 studies)	558186	22696	27.4%	4.10 [4.04, 4.16]
Total (95% CI)	**2835908**	**145656**	**100.0%**	**5.04 [3.86, 6.22]**

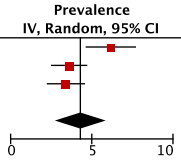
Heterogeneity: $Tau^2 = 1.32$; $Chi^2 = 134.14$, df = 3 ($P < 0.00001$); $I^2 = 98\%$
Test for overall effect: $Z = 8.38$ ($P < 0.00001$)



B

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (4 studies)	34183	2329	30.2%	6.12 [4.61, 7.63]
2nd round (4 studies)	36712	1515	35.2%	3.55 [2.47, 4.63]
3rd round (3 studies)	33077	1117	34.5%	3.35 [2.21, 4.49]
Total (95% CI)	**103972**	**4961**	**100.0%**	**4.26 [2.71, 5.80]**

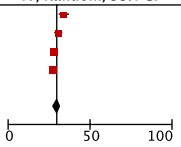
Heterogeneity: $Tau^2 = 1.46$; $Chi^2 = 9.50$, df = 2 ($P = 0.009$); $I^2 = 79\%$
Test for overall effect: $Z = 5.40$ ($P < 0.00001$)



C

Study or Subgroup	Follow-up colonoscopy Total	AA detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (4 studies)	34183	12094	20.1%	33.27 [30.51, 36.03]
2nd round (4 studies)	36712	5666	23.6%	29.81 [27.77, 31.85]
3rd round (3 studies)	33077	9039	27.2%	27.30 [26.07, 28.53]
4rd round (2 studies)	18267	4864	29.1%	26.61 [25.96, 27.26]
Total (95% CI)	**122239**	**31663**	**100.0%**	**28.89 [26.73, 31.05]**

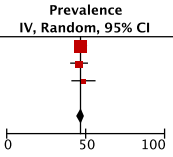
Heterogeneity: $Tau^2 = 4.08$; $Chi^2 = 28.05$, df = 3 ($P < 0.00001$); $I^2 = 89\%$
Test for overall effect: $Z = 26.17$ ($P < 0.00001$)



D

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (3 studies)	34116	15803	58.7%	46.08 [42.51, 49.65]
2nd round (3 studies)	36652	17221	27.1%	45.01 [39.76, 50.26]
3rd round (2 studies)	33038	15677	14.2%	47.80 [40.55, 55.05]
Total (95% CI)	**103806**	**48701**	**100.0%**	**46.03 [43.30, 48.77]**

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.37$, df = 2 ($P = 0.83$); $I^2 = 0\%$
Test for overall effect: $Z = 33.01$ ($P < 0.00001$)

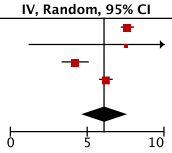


Supplementary figure 19. CRC PPV for the overall population participating in biennial FIT-based organized CRC screening programs with a cut-off of 20 µg Hb/g feces from A) 1st round, B) 2nd round, to C) 3rd round.

A

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	17983	1367	32.5%	7.60 [7.21, 7.99]
Crotta 2012 (4 rounds)	67	5	4.9%	7.50 [1.21, 13.79]
Parente 2013 (2 rounds)	2214	95	30.1%	4.20 [3.36, 5.04]
Tepes 2020 (3 rounds)	13919	862	32.5%	6.20 [5.80, 6.60]
Total (95% CI)	**34183**	**2329**	**100.0%**	**6.12 [4.61, 7.63]**

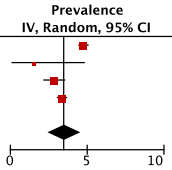
Heterogeneity: $Tau^2 = 1.79$; $Chi^2 = 60.22$, df = 3 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 7.94$ ($P < 0.00001$)



B

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	20288	975	31.5%	4.80 [4.51, 5.09]
Crotta 2012 (4 rounds)	60	1	8.3%	1.60 [0.00, 4.84]
Parente 2013 (2 rounds)	2395	70	28.7%	2.90 [2.23, 3.57]
Tepes 2020 (3 rounds)	13969	469	31.5%	3.40 [3.10, 3.70]
Total (95% CI)	**36712**	**1515**	**100.0%**	**3.55 [2.46, 4.63]**

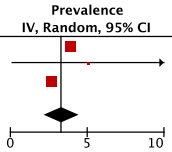
Heterogeneity: $Tau^2 = 0.95$; $Chi^2 = 56.36$, df = 3 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 6.40$ ($P < 0.00001$)



C

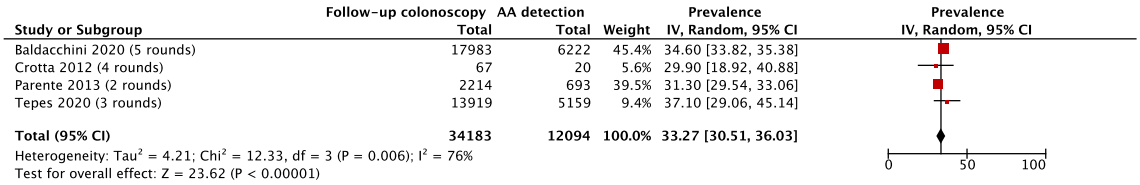
Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	18155	710	48.6%	3.90 [3.62, 4.18]
Crotta 2012 (4 rounds)	39	2	2.6%	5.10 [0.00, 12.02]
Tepes 2020 (3 rounds)	14883	405	48.8%	2.70 [2.44, 2.96]
Total (95% CI)	**33077**	**1117**	**100.0%**	**3.35 [2.20, 4.49]**

Heterogeneity: $Tau^2 = 0.69$; $Chi^2 = 37.75$, df = 2 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 5.71$ ($P < 0.00001$)

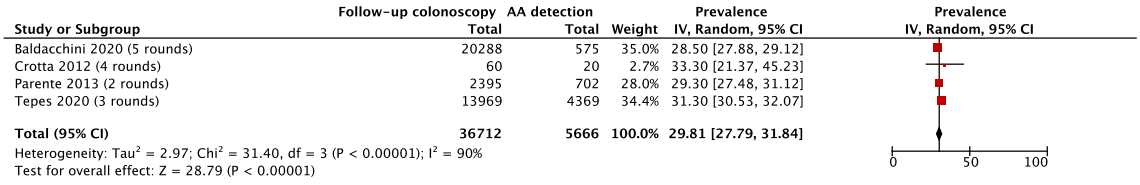


Supplementary figure 20. AA PPV for the overall population participating in biennial FIT-based organized CRC screening programs with a cut-off of 20 µg Hb/g feces from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

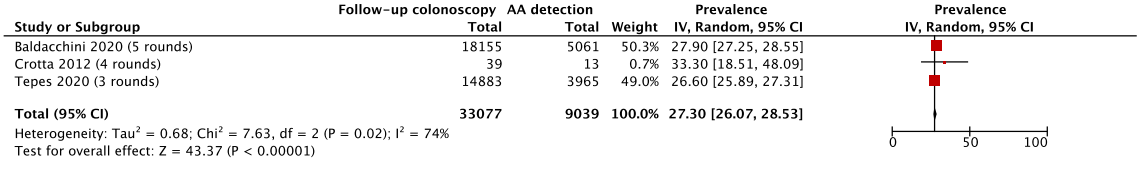
A



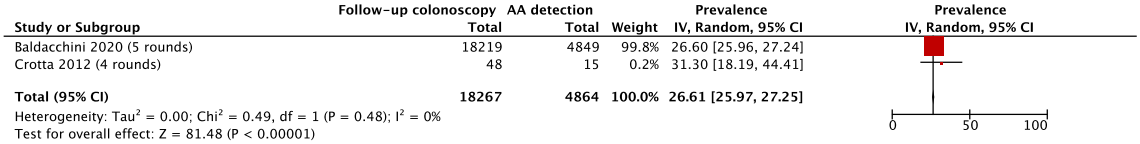
B



C

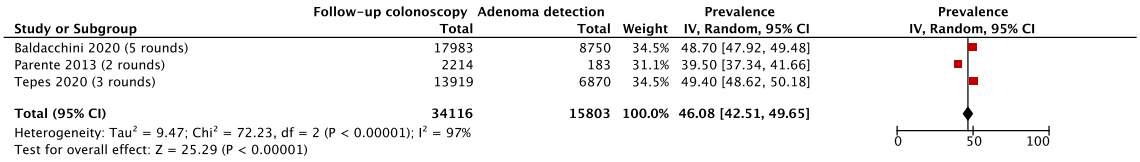


D

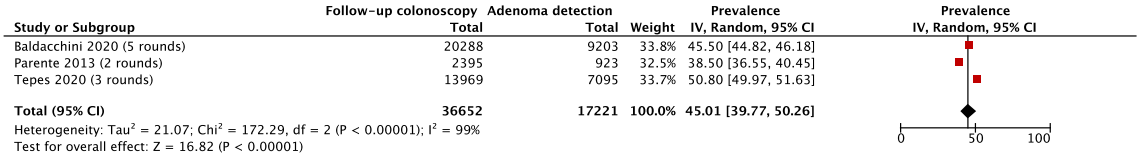


Supplementary figure 21. Adenoma PPV for the overall population participating in biennial FIT-based organized CRC screening programs with a cut-off of 20 µg Hb/g feces from A) 1st round, B) 2nd round, to C) 3rd round.

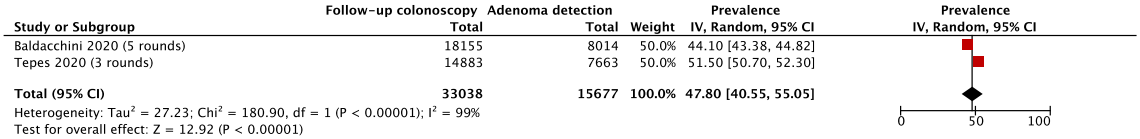
A



B



C

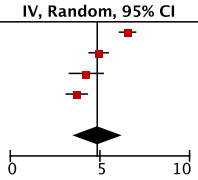


Supplementary figure 22. Positivity rate (A) and PPV for CRC (B), AA (C) and adenoma (D) of FIT per round for the consistent screeners participating in FIT-based organized screening programs.

A

Study or Subgroup	Participants Total	Positive Total	Weight	Prevalence IV, Random, 95% CI
1st round (5 studies)	3298506	1211047	25.8%	6.43 [5.98, 6.88]
2nd round (5 studies)	1332368	60422	25.5%	4.82 [4.29, 5.35]
3rd round (3 studies)	356460	13942	23.5%	4.13 [3.19, 5.07]
4rd round (3 studies)	233464	8695	25.3%	3.60 [3.01, 4.19]
Total (95% CI)	**5220798**	**1294106**	**100.0%**	**4.76 [3.41, 6.12]**

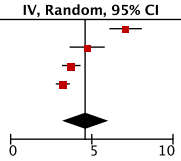
Heterogeneity: Tau² = 1.79; Chi² = 63.59, df = 3 (P < 0.00001); I² = 95%
Test for overall effect: Z = 6.91 (P < 0.00001)



B

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (5 studies)	204346	15856	24.0%	6.94 [5.98, 7.90]
2nd round (5 studies)	57628	2897	23.5%	4.57 [3.53, 5.61]
3rd round (3 studies)	12615	463	26.0%	3.59 [3.06, 4.12]
4rd round (2 studies)	7863	243	26.5%	3.07 [2.68, 3.46]
Total (95% CI)	**282452**	**19459**	**100.0%**	**4.49 [3.09, 5.89]**

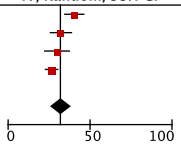
Heterogeneity: Tau² = 1.89; Chi² = 56.50, df = 3 (P < 0.00001); I² = 95%
Test for overall effect: Z = 6.28 (P < 0.00001)



C

Study or Subgroup	Follow-up colonoscopy Total	AA detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (4 studies)	55217	21421	25.2%	38.74 [32.94, 44.54]
2nd round (4 studies)	24602	7304	24.1%	30.62 [24.21, 37.03]
3rd round (3 studies)	12615	3328	22.2%	28.24 [20.73, 35.75]
4rd round (2 studies)	7863	1897	28.6%	24.91 [21.19, 28.63]
Total (95% CI)	**100297**	**33950**	**100.0%**	**30.51 [24.02, 36.99]**

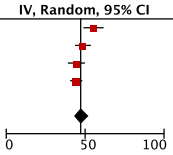
Heterogeneity: Tau² = 34.77; Chi² = 15.74, df = 3 (P = 0.001); I² = 81%
Test for overall effect: Z = 9.21 (P < 0.00001)



D

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
1st round (3 studies)	30819	15259	21.6%	53.41 [47.47, 59.35]
2nd round (3 studies)	14478	6211	25.5%	46.72 [42.15, 51.29]
3rd round (2 studies)	9844	4075	24.2%	42.58 [37.58, 47.58]
4rd round (2 studies)	7863	3280	28.8%	42.43 [39.00, 45.86]
Total (95% CI)	**63004**	**28825**	**100.0%**	**45.93 [41.43, 50.42]**

Heterogeneity: Tau² = 15.22; Chi² = 11.31, df = 3 (P = 0.01); I² = 73%
Test for overall effect: Z = 20.03 (P < 0.00001)

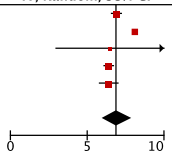


Supplementary figure 23. CRC PPV for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	24214	1682	23.8%	6.90 [6.58, 7.22]
Breekveldt 2022 (2 rounds)	149129	12183	24.3%	8.10 [7.96, 8.24]
Denters 2012 (2 rounds)	186	12	5.7%	6.50 [2.97, 10.03]
Portillo 2018 (3 rounds)	24398	1566	23.9%	6.40 [6.09, 6.71]
Zorzi 2018 (6 rounds)	6419	413	22.3%	6.40 [5.80, 7.00]
Total (95% CI)	**204346**	**15856**	**100.0%**	**6.94 [5.98, 7.90]**

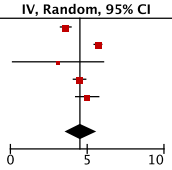
Heterogeneity: $Tau^2 = 0.99$; $Chi^2 = 142.59$, df = 4 ($P < 0.00001$); $I^2 = 97\%$
Test for overall effect: $Z = 14.12$ ($P < 0.00001$)



B

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	11121	402	23.5%	3.60 [3.25, 3.95]
Breekveldt 2022 (2 rounds)	33026	1874	23.8%	5.70 [5.45, 5.95]
Denters 2012 (2 rounds)	130	4	8.2%	3.10 [0.13, 6.07]
Portillo 2018 (3 rounds)	10124	455	23.2%	4.50 [4.10, 4.90]
Zorzi 2018 (6 rounds)	3227	162	21.4%	5.00 [4.25, 5.75]
Total (95% CI)	**57628**	**2897**	**100.0%**	**4.57 [3.53, 5.61]**

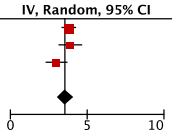
Heterogeneity: $Tau^2 = 1.17$; $Chi^2 = 98.86$, df = 4 ($P < 0.00001$); $I^2 = 96\%$
Test for overall effect: $Z = 8.59$ ($P < 0.00001$)



C

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	7359	278	42.3%	3.80 [3.36, 4.24]
Portillo 2018 (3 rounds)	2771	110	27.7%	3.90 [3.17, 4.63]
Zorzi 2018 (6 rounds)	2485	75	30.0%	3.00 [2.33, 3.67]
Total (95% CI)	**12615**	**463**	**100.0%**	**3.59 [3.07, 4.11]**

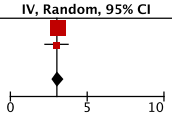
Heterogeneity: $Tau^2 = 0.12$; $Chi^2 = 4.49$, df = 2 ($P = 0.11$); $I^2 = 55\%$
Test for overall effect: $Z = 13.50$ ($P < 0.00001$)



D

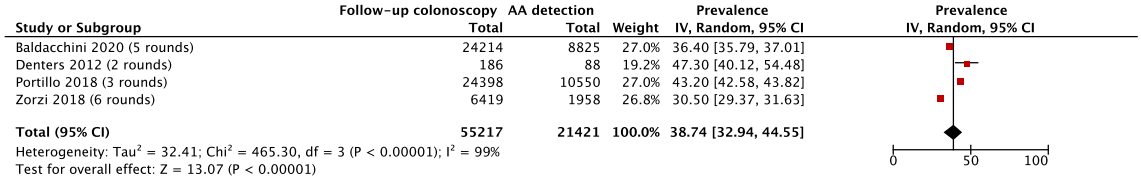
Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	5859	182	74.3%	3.10 [2.66, 3.54]
Zorzi 2018 (6 rounds)	2004	61	25.7%	3.00 [2.25, 3.75]
Total (95% CI)	**7863**	**243**	**100.0%**	**3.07 [2.69, 3.46]**

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.05$, df = 1 ($P = 0.82$); $I^2 = 0\%$
Test for overall effect: $Z = 15.78$ ($P < 0.00001$)

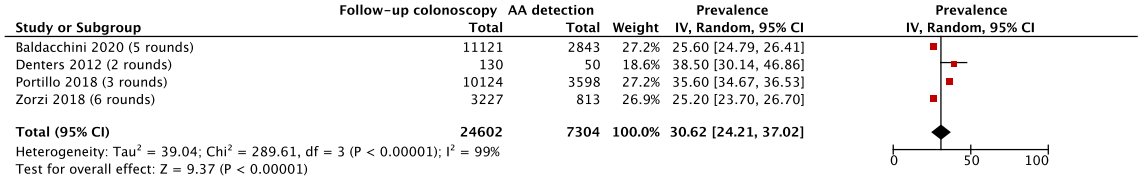


Supplementary figure 24. AA PPV for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

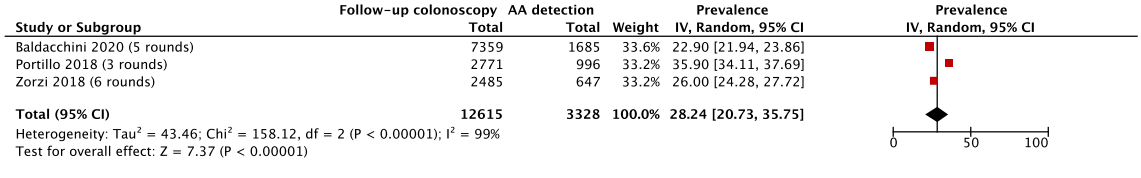
A



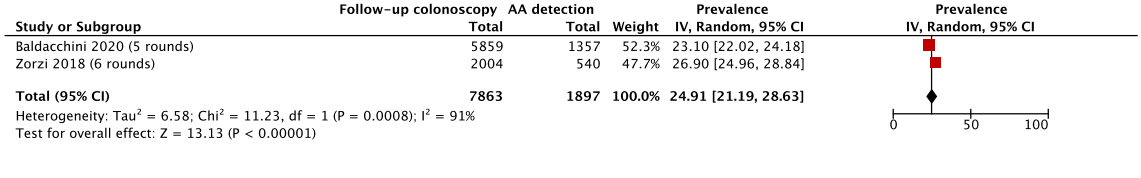
B



C



D

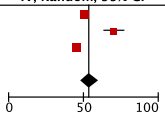


Supplementary figure 25. Adenoma PPV for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	24214	12262	37.4%	50.60 [49.97, 51.23]
Denters 2012 (2 rounds)	186	120	25.7%	69.90 [63.31, 76.49]
Zorzi 2018 (6 rounds)	6419	2877	37.0%	44.80 [43.58, 46.02]
Total (95% CI)	**30819**	**15259**	**100.0%**	**53.41 [47.47, 59.35]**

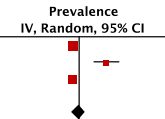
Heterogeneity: $Tau^2 = 24.49$; $Chi^2 = 105.81$, df = 2 ($P < 0.00001$); $I^2 = 98\%$
Test for overall effect: $Z = 17.62$ ($P < 0.00001$)



B

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	11121	4753	41.8%	42.70 [41.78, 43.62]
Denters 2012 (2 rounds)	130	85	18.0%	65.40 [57.22, 73.58]
Zorzi 2018 (6 rounds)	3227	1373	40.2%	42.50 [40.79, 44.21]
Total (95% CI)	**14478**	**6211**	**100.0%**	**46.72 [42.14, 51.29]**

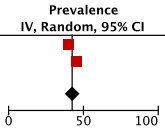
Heterogeneity: $Tau^2 = 12.83$; $Chi^2 = 29.48$, df = 2 ($P < 0.00001$); $I^2 = 93\%$
Test for overall effect: $Z = 20.00$ ($P < 0.00001$)



C

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	7359	2953	51.3%	40.10 [38.98, 41.22]
Zorzi 2018 (6 rounds)	2485	1122	48.7%	45.20 [43.24, 47.16]
Total (95% CI)	**9844**	**4075**	**100.0%**	**42.58 [37.59, 47.58]**

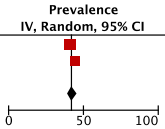
Heterogeneity: $Tau^2 = 12.34$; $Chi^2 = 19.67$, df = 1 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 16.71$ ($P < 0.00001$)



D

Study or Subgroup	Follow-up colonoscopy Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	5859	2392	53.3%	40.80 [39.54, 42.06]
Zorzi 2018 (6 rounds)	2004	888	46.7%	44.30 [42.13, 46.47]
Total (95% CI)	**7863**	**3280**	**100.0%**	**42.43 [39.01, 45.86]**

Heterogeneity: $Tau^2 = 5.30$; $Chi^2 = 7.46$, df = 1 ($P = 0.006$); $I^2 = 87\%$
Test for overall effect: $Z = 24.30$ ($P < 0.00001$)

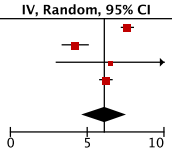


Supplementary figure 26. CRC PPV for the overall population participating in FIT-based organized CRC screening programs within open cohorts from A) 1st round, B) 2nd round, to C) 3rd round.

A

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	17983	1367	30.5%	7.60 [7.21, 7.99]
Parente 2013 (2 rounds)	2214	95	28.2%	4.20 [3.36, 5.04]
Stegeman 2015 (3 rounds)	186	12	11.0%	6.50 [2.97, 10.03]
Tepes 2020 (3 rounds)	13919	862	30.4%	6.20 [5.80, 6.60]
Total (95% CI)	**34302**	**2336**	**100.0%**	**6.10 [4.64, 7.55]**

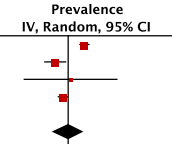
Heterogeneity: $Tau^2 = 1.77$; $Chi^2 = 60.16$, df = 3 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 8.22$ ($P < 0.00001$)



B

Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	20288	975	31.3%	4.80 [4.51, 5.09]
Parente 2013 (2 rounds)	2395	70	28.4%	2.90 [2.23, 3.57]
Stegeman 2015 (3 rounds)	156	6	8.9%	3.90 [0.88, 6.92]
Tepes 2020 (3 rounds)	13969	469	31.3%	3.40 [3.10, 3.70]
Total (95% CI)	**36808**	**1520**	**100.0%**	**3.74 [2.68, 4.80]**

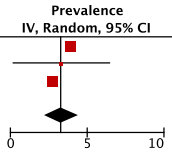
Heterogeneity: $Tau^2 = 0.91$; $Chi^2 = 54.25$, df = 3 ($P < 0.00001$); $I^2 = 94\%$
Test for overall effect: $Z = 6.90$ ($P < 0.00001$)



C

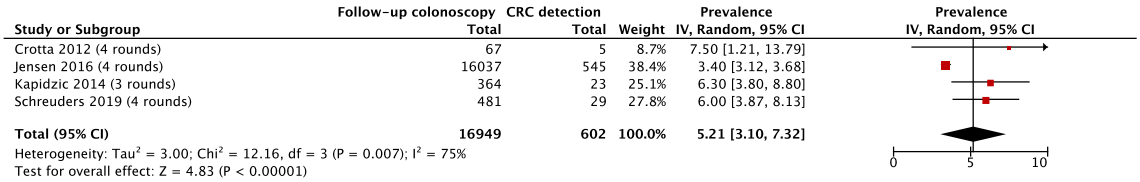
Study or Subgroup	Follow-up colonoscopy Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	18155	710	45.1%	3.90 [3.62, 4.18]
Stegeman 2015 (3 rounds)	123	4	9.7%	3.30 [0.17, 6.43]
Tepes 2020 (3 rounds)	14883	405	45.3%	2.70 [2.44, 2.96]
Total (95% CI)	**33161**	**1119**	**100.0%**	**3.30 [2.20, 4.40]**

Heterogeneity: $Tau^2 = 0.67$; $Chi^2 = 37.48$, df = 2 ($P < 0.00001$); $I^2 = 95\%$
Test for overall effect: $Z = 5.90$ ($P < 0.00001$)

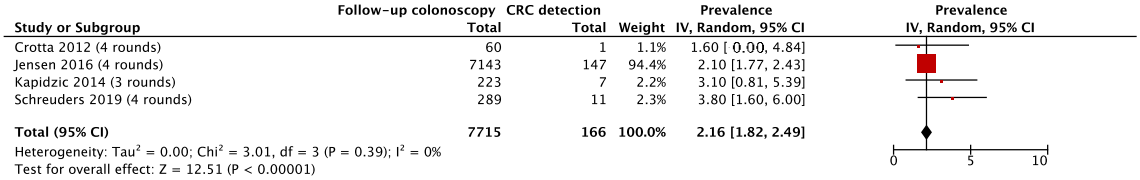


Supplementary figure 27. CRC PPV for the overall population participating in FIT-based organized CRC screening programs within closed cohorts from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

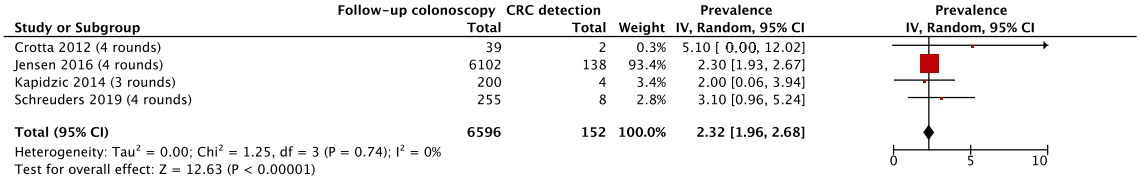
A



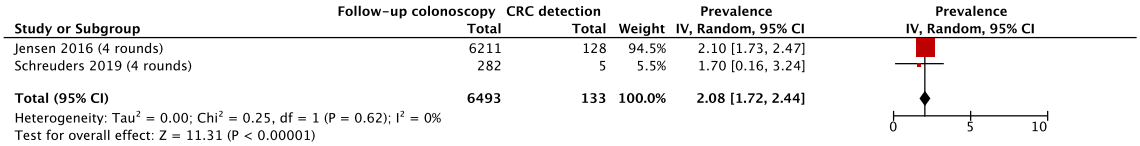
B



C

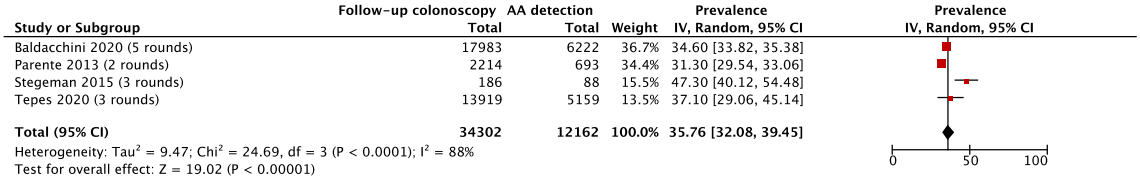


D

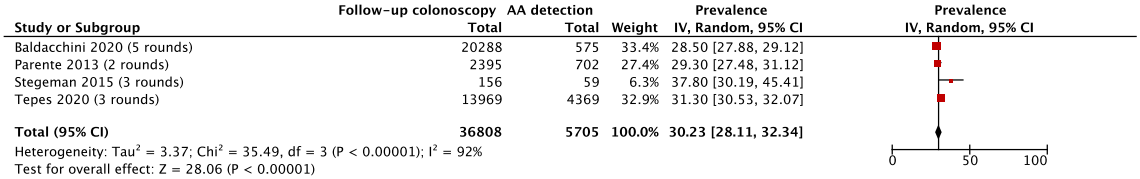


Supplementary figure 28. AA PPV for the overall population participating in FIT-based organized CRC screening programs within open cohorts from A) 1st round, B) 2nd round, to C) 3rd round.

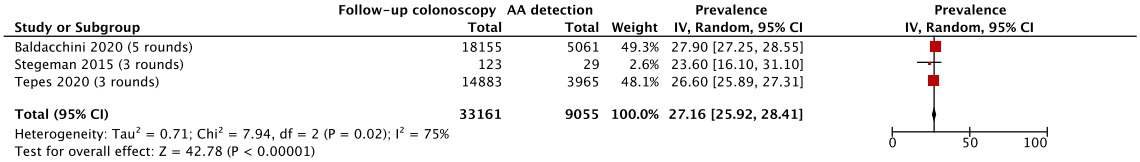
A



B

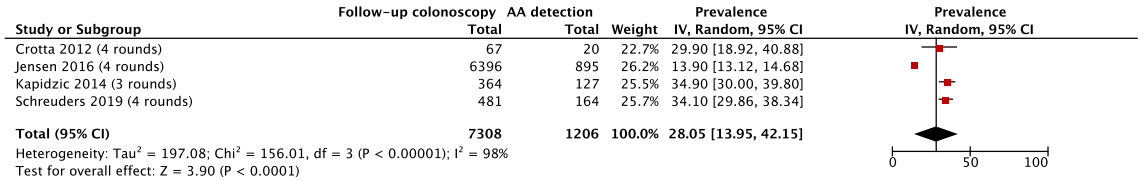


C

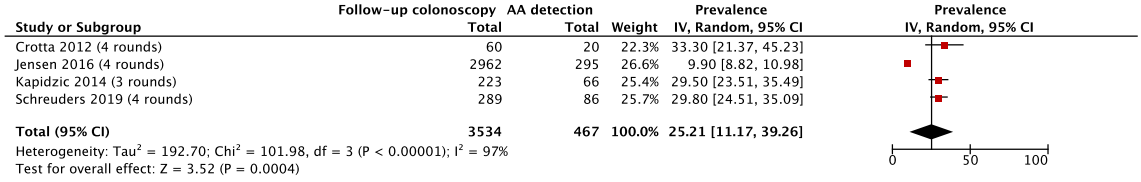


Supplementary figure 29. AA PPV for the overall population participating in FIT-based organized CRC screening programs within closed cohorts from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

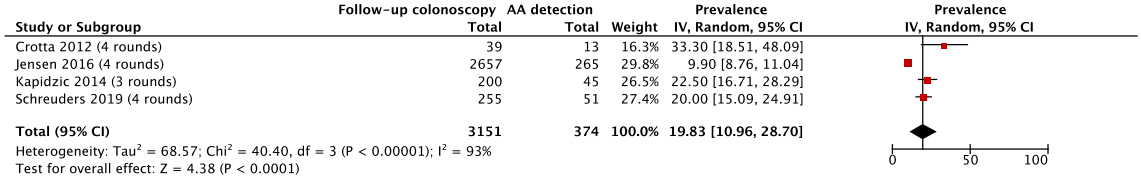
A



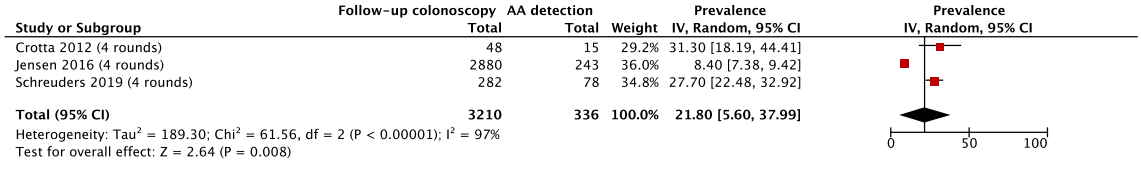
B



C

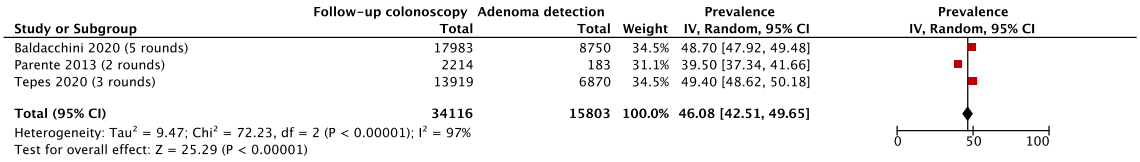


D

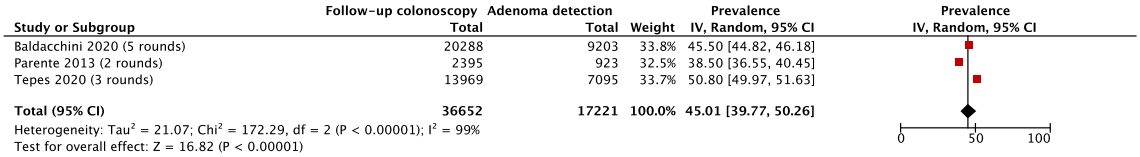


Supplementary figure 30. Adenoma PPV for the overall population participating in FIT-based organized CRC screening programs within open cohorts from A) 1st round, B) 2nd round, to C) 3rd round.

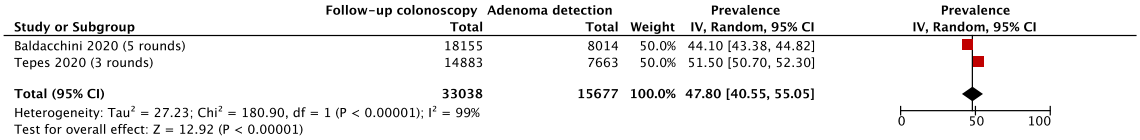
A



B

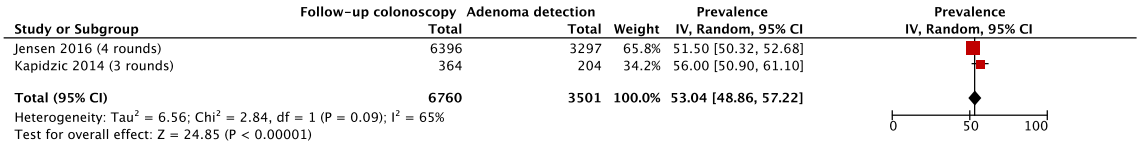


C

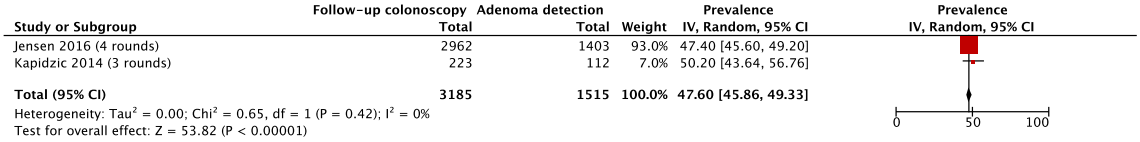


Supplementary figure 31. Adenoma PPV for the overall population participating in FIT-based organized CRC screening programs within closed cohorts from A) 1st round, B) 2nd round, to C) 3rd round.

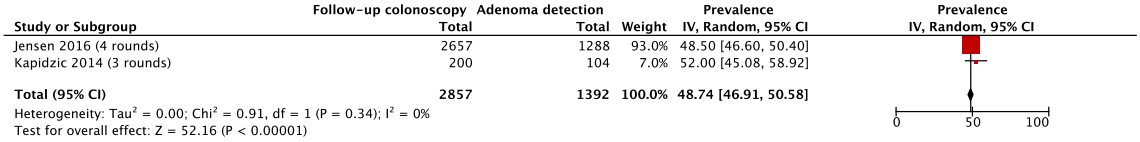
A



B

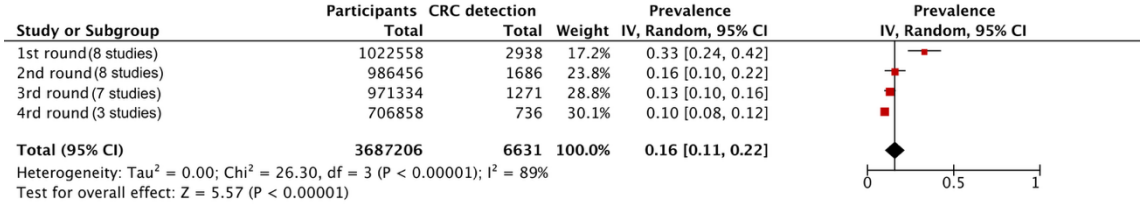


C

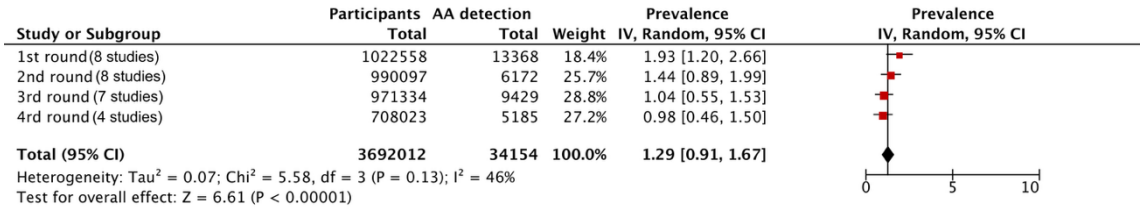


Supplementary figure 32. DR for CRC (A), AA (B) and adenoma (C) of FIT per round in the overall screening population participating in FIT-based organized screening programs.

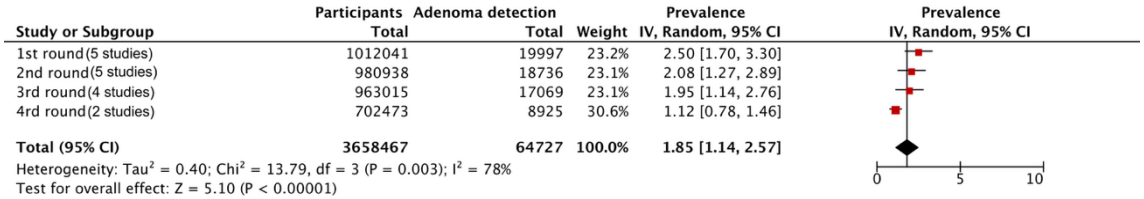
A



B

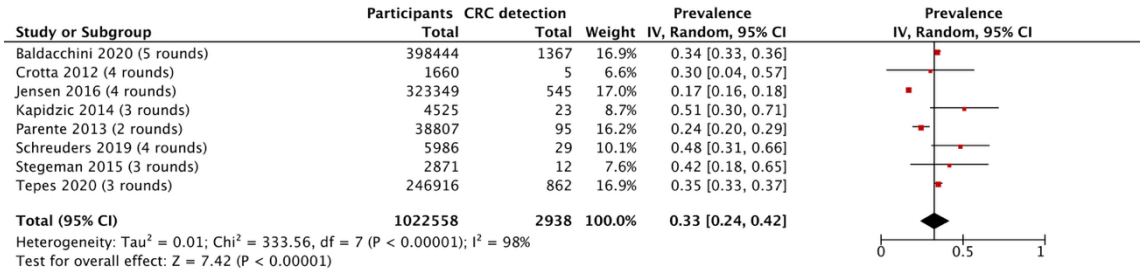


C

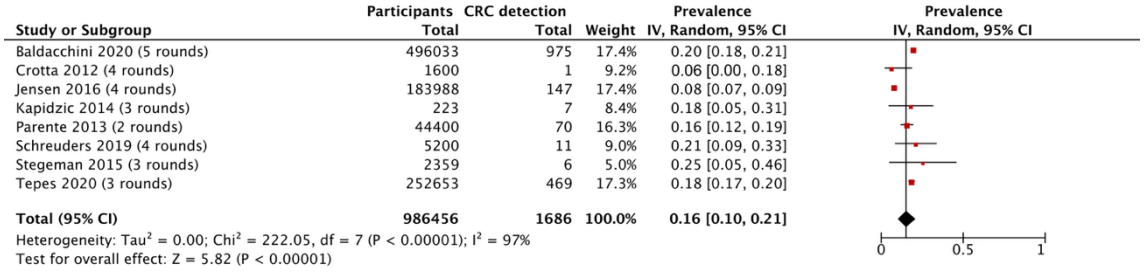


Supplementary figure 33. CRC DR for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

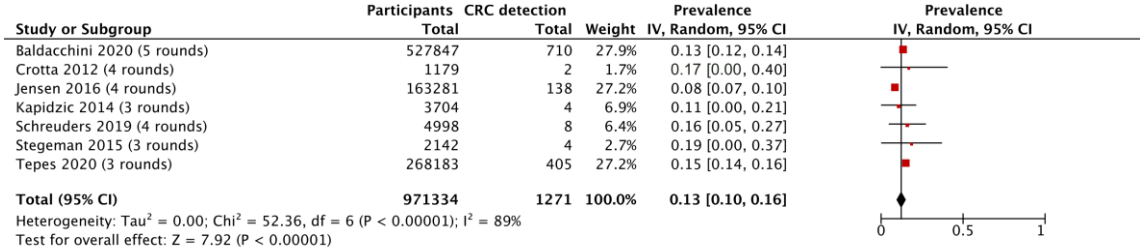
A



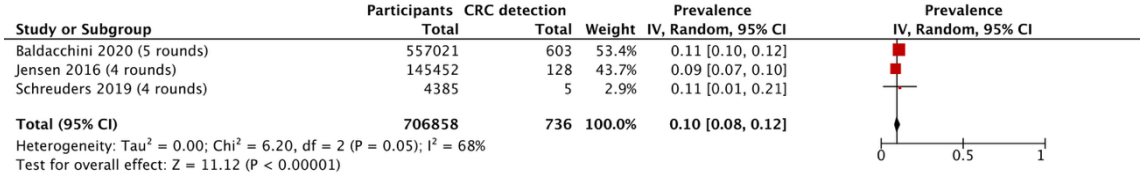
B



C

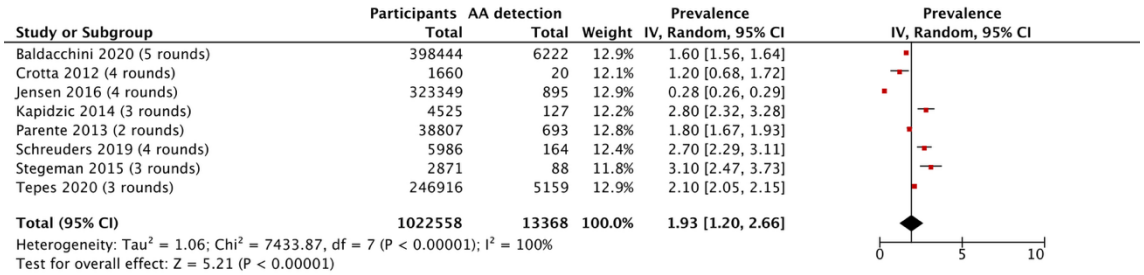


D

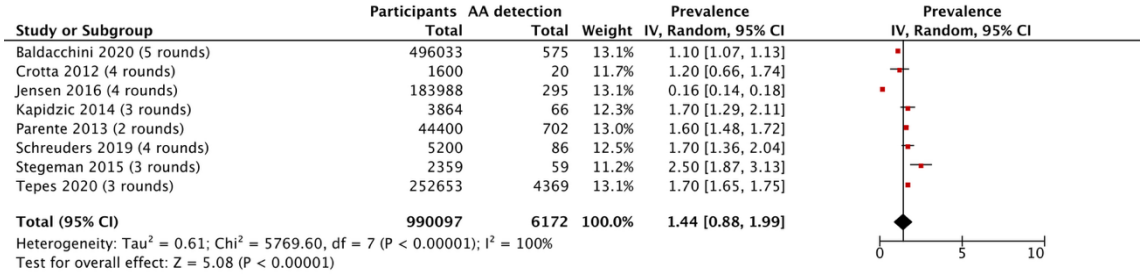


Supplementary figure 34. AA DR for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

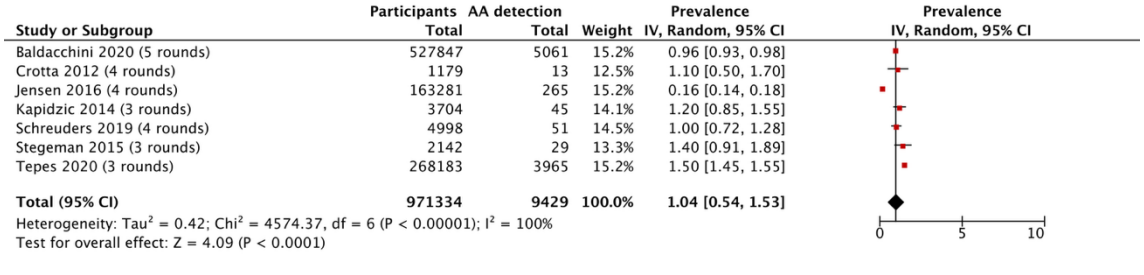
A



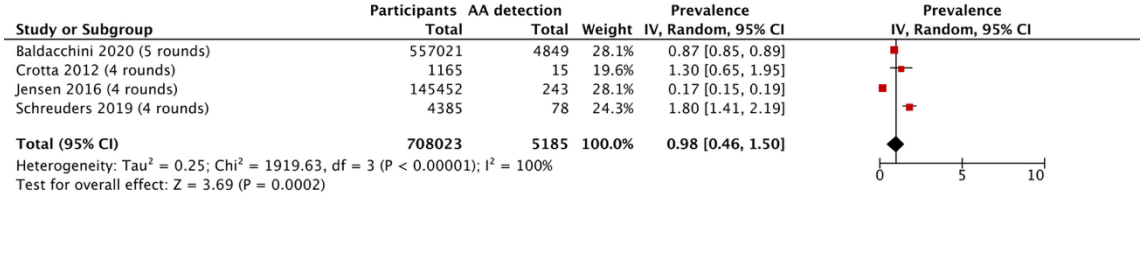
B



C



D

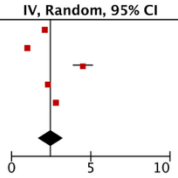


Supplementary figure 35. Adenoma DR for the overall population participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	398444	8750	20.4%	2.10 [2.05, 2.15]
Jensen 2016 (4 rounds)	323349	3297	20.5%	1.00 [0.97, 1.03]
Kapidzic 2014 (3 rounds)	4525	204	18.3%	4.50 [3.90, 5.10]
Parente 2013 (2 rounds)	38807	876	20.3%	2.30 [2.15, 2.45]
Tepes 2020 (3 rounds)	246916	6870	20.4%	2.80 [2.74, 2.86]
Total (95% CI)	**1012041**	**19997**	**100.0%**	**2.50 [1.70, 3.30]**

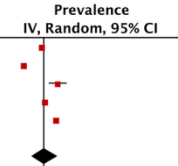
Heterogeneity: $Tau^2 = 0.82$; $Chi^2 = 3213.88$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 6.10$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	496033	9203	20.3%	1.90 [1.86, 1.94]
Jensen 2016 (4 rounds)	183988	1403	20.3%	0.76 [0.72, 0.80]
Kapidzic 2014 (3 rounds)	3864	112	18.7%	2.90 [2.37, 3.43]
Parente 2013 (2 rounds)	44400	923	20.2%	2.10 [1.97, 2.23]
Tepes 2020 (3 rounds)	252653	7095	20.3%	2.80 [2.74, 2.86]
Total (95% CI)	**980938**	**18736**	**100.0%**	**2.08 [1.27, 2.89]**

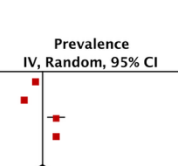
Heterogeneity: $Tau^2 = 0.83$; $Chi^2 = 3485.78$, df = 4 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 5.05$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	527847	8014	25.6%	1.50 [1.47, 1.53]
Jensen 2016 (4 rounds)	163281	1288	25.6%	0.79 [0.75, 0.83]
Kapidzic 2014 (3 rounds)	3704	104	23.1%	2.80 [2.27, 3.33]
Tepes 2020 (3 rounds)	268183	7663	25.6%	2.80 [2.74, 2.86]
Total (95% CI)	**963015**	**17069**	**100.0%**	**1.95 [1.14, 2.76]**

Heterogeneity: $Tau^2 = 0.67$; $Chi^2 = 2807.98$, df = 3 ($P < 0.00001$); $I^2 = 100\%$
Test for overall effect: $Z = 4.70$ ($P < 0.00001$)



D

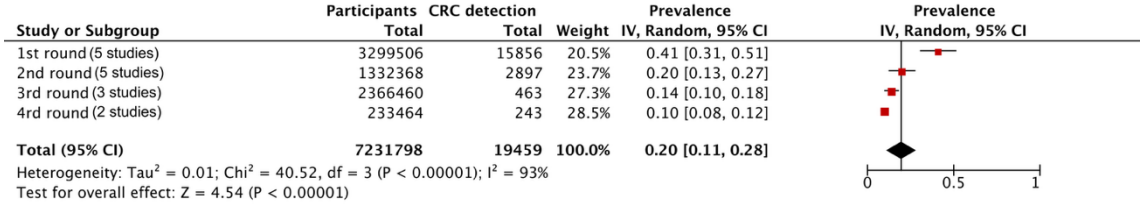
Study or Subgroup	Participants Total	Adenoma detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	557021	7545	50.2%	1.30 [1.27, 1.33]
Jensen 2016 (4 rounds)	145452	1380	49.8%	0.95 [0.90, 1.00]
Total (95% CI)	**702473**	**8925**	**100.0%**	**1.12 [0.78, 1.47]**

Heterogeneity: $Tau^2 = 0.06$; $Chi^2 = 145.77$, df = 1 ($P < 0.00001$); $I^2 = 99\%$
Test for overall effect: $Z = 6.39$ ($P < 0.00001$)

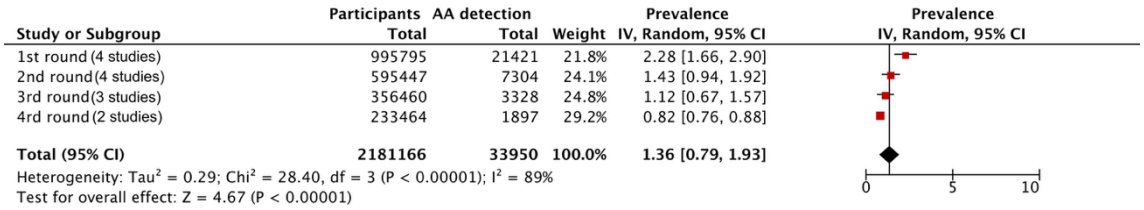


Supplementary figure 36. DR for CRC (A), AA (B) and adenoma (C) of FIT per round in the consistent screeners participating in FIT-based organized screening programs.

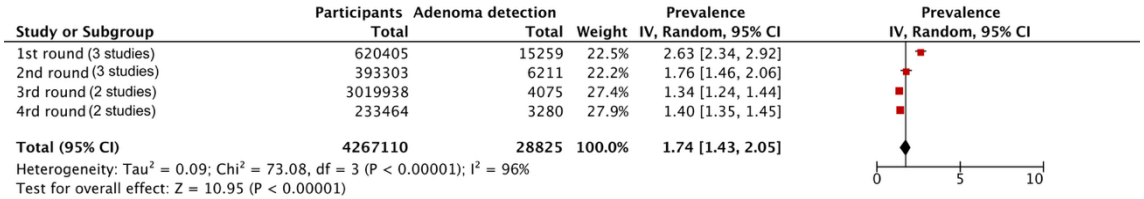
A



B



C

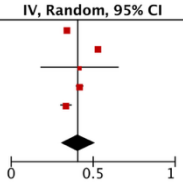


Supplementary figure 37. CRC DR for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

A

Study or Subgroup	Participants Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	494187	1682	22.5%	0.34 [0.32, 0.36]
Breekveldt 2022 (2 rounds)	2302711	12183	22.5%	0.53 [0.52, 0.54]
Denters 2012 (2 rounds)	2871	12	10.5%	0.42 [0.18, 0.65]
Portillo 2018 (3 rounds)	376390	1566	22.4%	0.42 [0.40, 0.44]
Zorzi 2018 (6 rounds)	123347	413	22.1%	0.33 [0.30, 0.37]
Total (95% CI)	**3299506**	**15856**	**100.0%**	**0.41 [0.30, 0.51]**

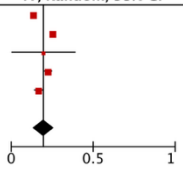
Heterogeneity: $Tau^2 = 0.01$; $Chi^2 = 578.36$, df = 4 ($P < 0.00001$); $I^2 = 99\%$
Test for overall effect: $Z = 7.64$ ($P < 0.00001$)



B

Study or Subgroup	Participants Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	295152	402	23.4%	0.14 [0.12, 0.15]
Breekveldt 2022 (2 rounds)	736921	1874	23.4%	0.25 [0.24, 0.26]
Denters 2012 (2 rounds)	2022	4	7.6%	0.20 [0.00, 0.39]
Portillo 2018 (3 rounds)	202144	455	22.9%	0.23 [0.20, 0.25]
Zorzi 2018 (6 rounds)	96129	162	22.7%	0.17 [0.14, 0.19]
Total (95% CI)	**1332368**	**2897**	**100.0%**	**0.20 [0.13, 0.26]**

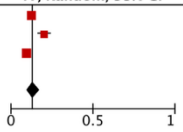
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 239.70$, df = 4 ($P < 0.00001$); $I^2 = 98\%$
Test for overall effect: $Z = 5.99$ ($P < 0.00001$)



C

Study or Subgroup	Participants Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	2233168	278	35.4%	0.13 [0.11, 0.14]
Portillo 2018 (3 rounds)	54522	110	27.5%	0.20 [0.16, 0.24]
Zorzi 2018 (6 rounds)	78770	75	37.0%	0.10 [0.09, 0.10]
Total (95% CI)	**2366460**	**463**	**100.0%**	**0.14 [0.10, 0.17]**

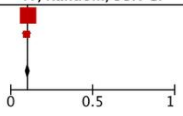
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 49.28$, df = 2 ($P < 0.00001$); $I^2 = 96\%$
Test for overall effect: $Z = 6.88$ ($P < 0.00001$)



D

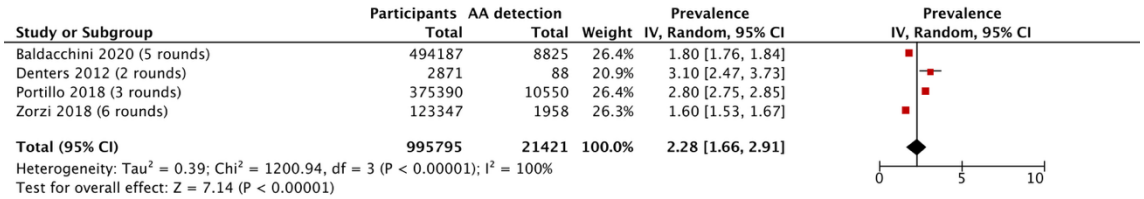
Study or Subgroup	Participants Total	CRC detection Total	Weight	Prevalence IV, Random, 95% CI
Baldacchini 2020 (5 rounds)	170395	182	74.6%	0.11 [0.09, 0.12]
Zorzi 2018 (6 rounds)	63069	61	25.4%	0.10 [0.07, 0.12]
Total (95% CI)	**233464**	**243**	**100.0%**	**0.10 [0.09, 0.12]**

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.52$, df = 1 ($P = 0.47$); $I^2 = 0\%$
Test for overall effect: $Z = 17.11$ ($P < 0.00001$)

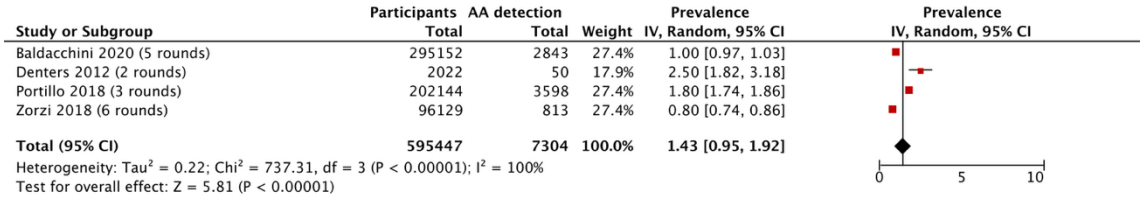


Supplementary figure 38. AA DR for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

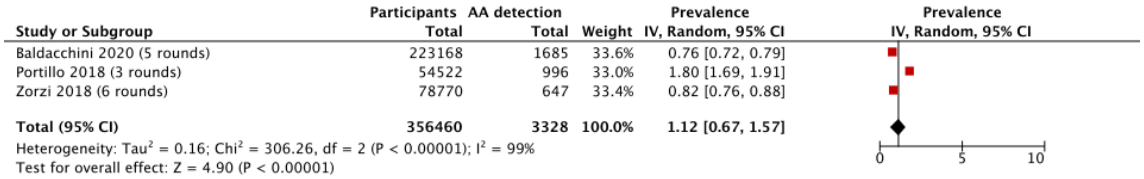
A



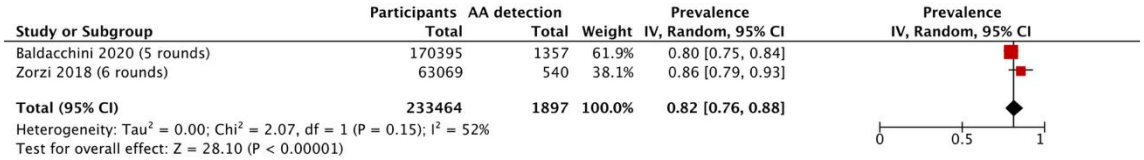
B



C

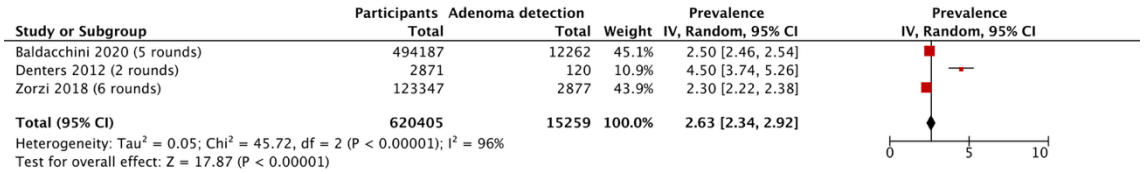


D

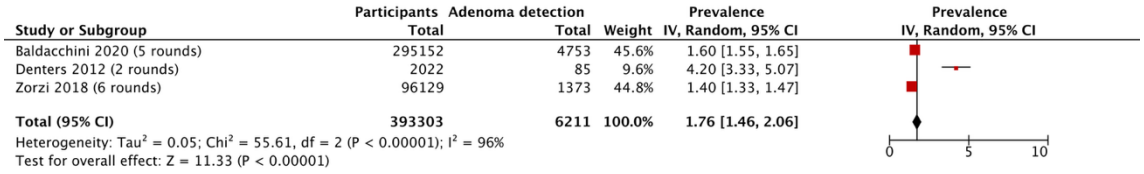


Supplementary figure 39. Adenoma DR for the consistent screeners participating in FIT-based organized CRC screening programs from A) 1st round, B) 2nd round, C) 3rd round, to D) 4th round.

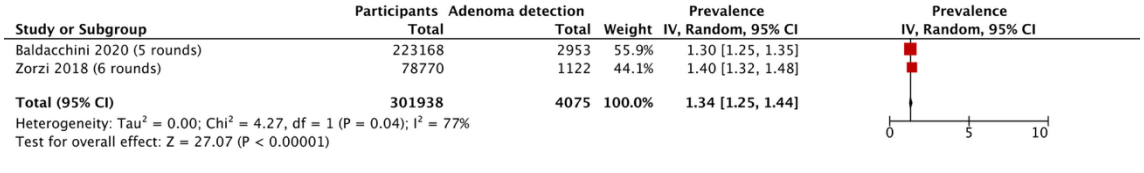
A



B



C



D

